ELECTROTECHNICAL

Workshop

User’s Guide

Famic Technologies Inc.

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Electrotechnical Workshops User's Guide for Automation Studio™.

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1 Introduction

This *Electrotechnical Workshop User's Guide* provides the information required to use and work efficiently with the Electrotechnical workshop of Automation Studio™. This guide includes technical specifications for components, procedures to define properties, instructions for building and simulating circuits, and provides examples of applications.

Automation Studio™ is an innovative system design, simulation, and documentation software solution for the design and support of automation, fluid power and electrical systems. It is intended to be used by engineers, maintenance and support personnel in a wide variety of related fields.

The software consists of several modules (also called "Workshops") and libraries that can be added according to the users' specific needs and requirements. Each library contains hundreds of symbols that are compliant with ISO, IEC, JIC or NEMA standards. Users may simply select the appropriate components and drag and drop them onto workspace to quickly create virtually any type of system. Systems can consist of one single technology (hydraulics, pneumatics, electrotechnical, etc.) or of a combination of multiple technologies that will interact with each other during simulation as they would in the reality.

Automation Studio™ supports the following functions: editing, simulation, file and diagram management, printing, and display.

The electrotechnical workshop is a graphic editing tool that manipulates and simulates electrotechnical diagrams. The designation used complies with international standards IEC 61082-1 to 6, IEC 61346-1, IEC 60617-2 to 11, IEC 61666, IEEE 315 and NEMA ICS 19-2002.

1.1 Links to Other Workshops

The editor of the electrotechnical workshop is not graphically compatible with the other Automation Studio™ editors.

Only components from the electrotechnical workshop can be used in an electrotechnical diagram.

Components from the electrotechnical workshop cannot be used on other types of diagrams.

However, variables from other workshops can be associated with components from the electrotechnical workshop without any problem. In the same manner, variables from the

electrotechnical workshop can be associated with components from other workshops.

Automation Studio™ components are either emitters or receivers.

An emitting component transmits its state or value to receiving components that are associated with it.

A receiver component receives its state, value and “Tag Name” from the emitter which it is associated with.

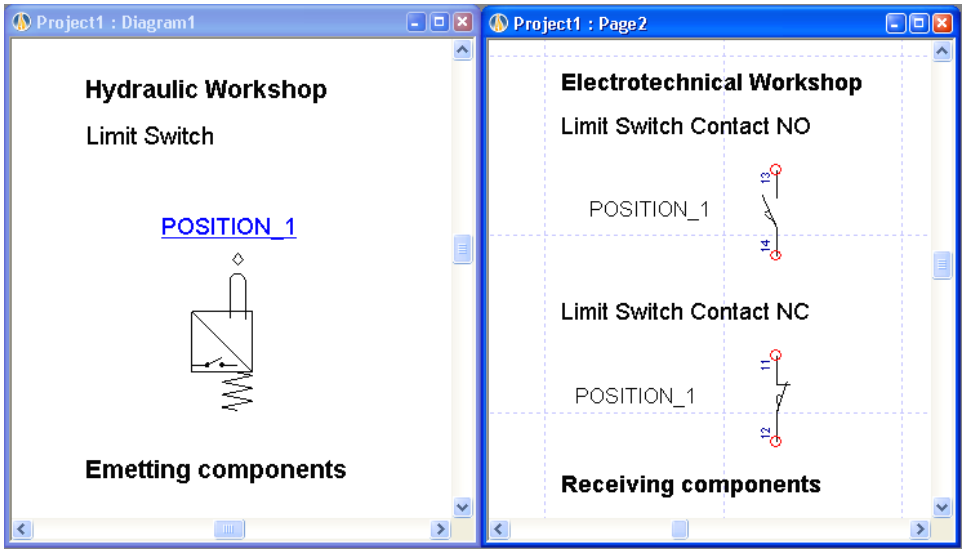


Figure 1-1: Compatibility between Workshops

2 Electrotechnical Diagram Editor Menus

This chapter describes the electrotechnical diagram editor menus that are not covered in the *Automation Studio™ User's Guide*. For a description of items common to all workshops, please refer to *Automation Studio™ User's Guide*.

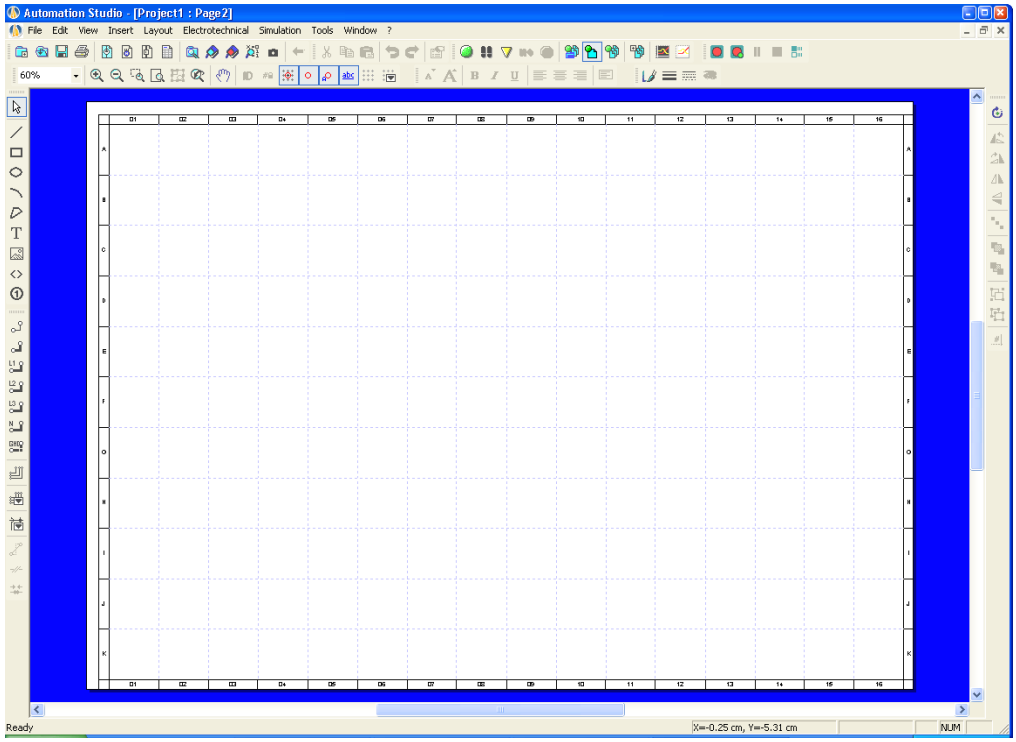


Figure 2-1 : Main Electrotechnical Diagram Editor Window

☞ To access the main window of the electrotechnical diagram editor, you must create an electrotechnical diagram.

2.1 File Menu

The "File" menu is the same as in the standard editor.

2.2 Edit Menu

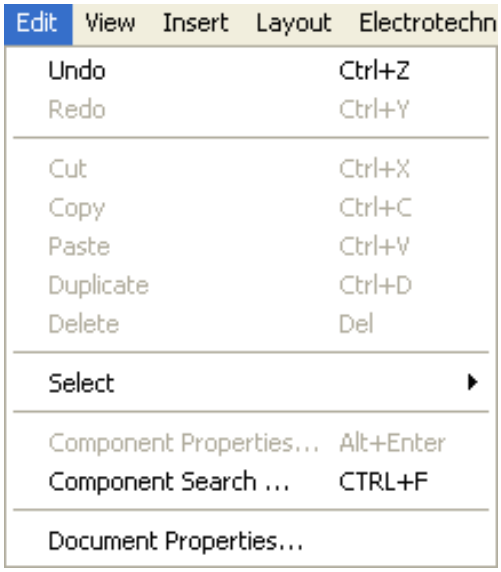


Figure 2-2: Edit Menu

The options that are specific to the Electrotechnical Editor are the following:

Select

This option provides specialized selection command.

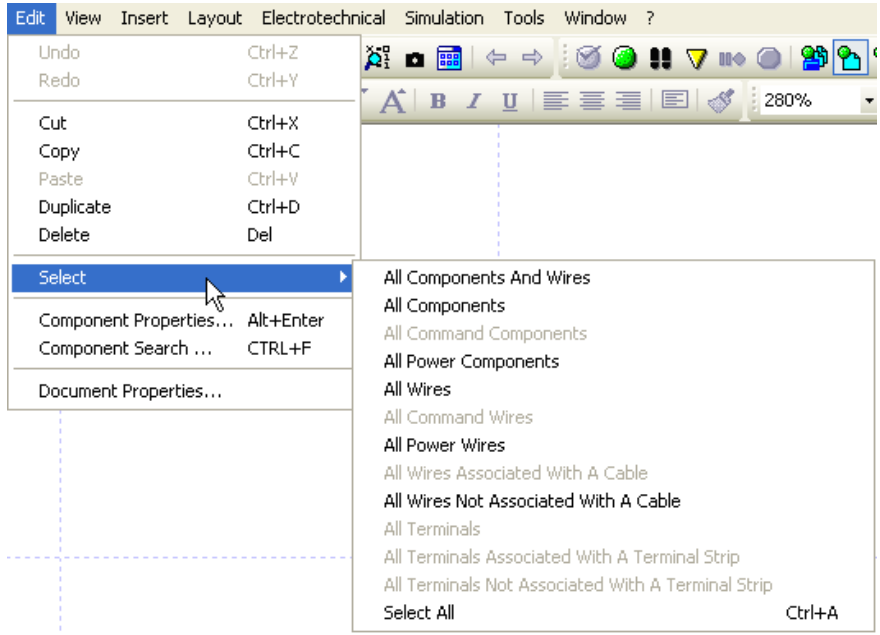


Figure 2-3 Edit Menu

Component Properties . . .

This option gives you access to the "Selected Component" properties dialog box. The properties of electrotechnical components are described in the section "Creating your first electrotechnical diagram" and in the various specialized sections.

Component Search

This command allows you to search components associated to specific essential subdivision and location. Refer to *Automation Studio User's Guide* for details on this feature.

Document Properties . . .

This option gives you access to the document properties: Page Setup, Map Locator, Summary Info, Electrotechnical Diagram and History.

2.2.2 Document Properties Command in the Edit Menu

This command gives you access to the "Document Properties" dialog box. Properties of an electrotechnical document are grouped into five branches:

1. Page Setup;

2. Map Locator;

3. Summary Information;

4. Electrotechnical Diagram;

5. History;

6. References.

“Page Setup”, “Summary Information”, “History” and “References” display similar information as for all documents generated using Automation Studio™. Refer to *Automation Studio™ User’s Guide* for a complete description of these branches.

When this dialog box opens, the electrotechnical diagram branch is selected by default.

2.2.3 Electrotechnical Diagram Branch, Part 1

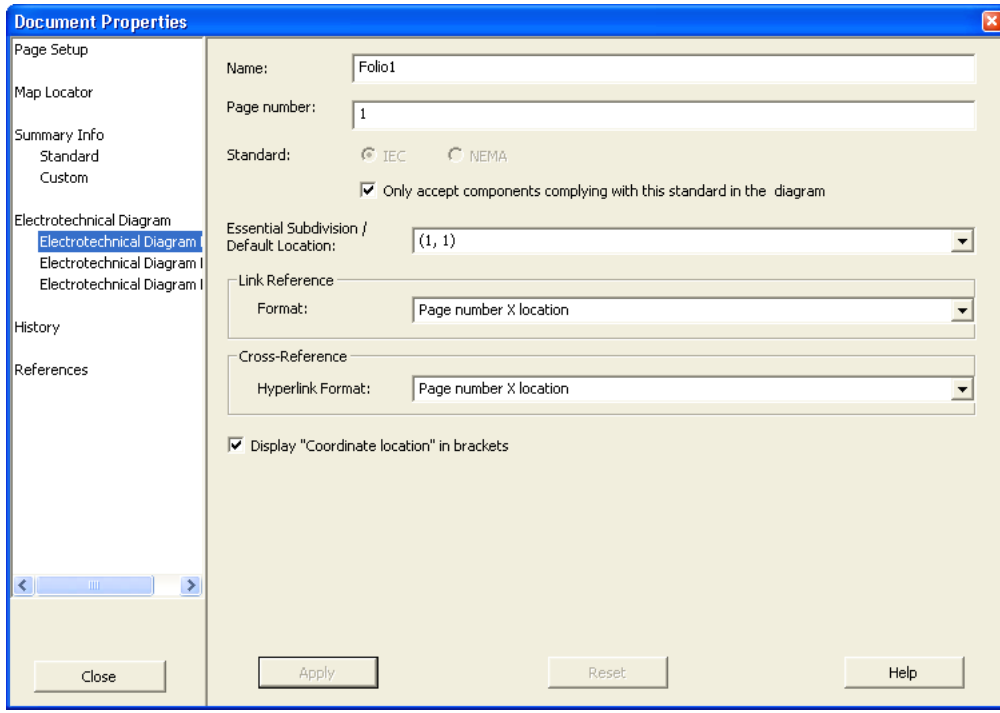


Figure 2-4 : Document Properties Dialog Box, Electrotechnical Branch Part 1



In order to produce easily understandable diagrams, it is advisable to use the same wire identification and numbering format for all diagrams included in the same project.

The options available from the Electrotechnical Diagram dialog box are as follows:

Section	Description
Name	Diagram name as it appears in the Project Explorer.
Page Number	Diagram number used for links and to locate components.
Standard	Selection of the IEC or NEMA standard. Only the components of the selected standard will be allowed on the diagram.

Section	Description
Only accept components complying with this standard in the diagram.	When checked, only the components of the selected standard are authorized on the diagram.
Essential Subdivision	This alphanumeric code is assigned to a component inserted on the diagram. Changing this code will not affect the component already on the diagram. The "Essential subdivision" combined with the "Location" allows you to cross-reference a component to a physical location, such as a plant grid number.
Location	This alphanumeric code is assigned to a component inserted on the diagram. Changing this code will not affect the component already on the diagram. The "Location" combined with the "Essential subdivision" allows you to cross-reference a component in a physical location, such as a plant grid number.
Link Reference - Cross Reference	Provides a variety of reference format structure.
Display "Coordinate Location" in Brackets	If checked, this displays the coordinate location in parentheses.

2.2.4 Electrotechnical Diagram Branch, Part 2

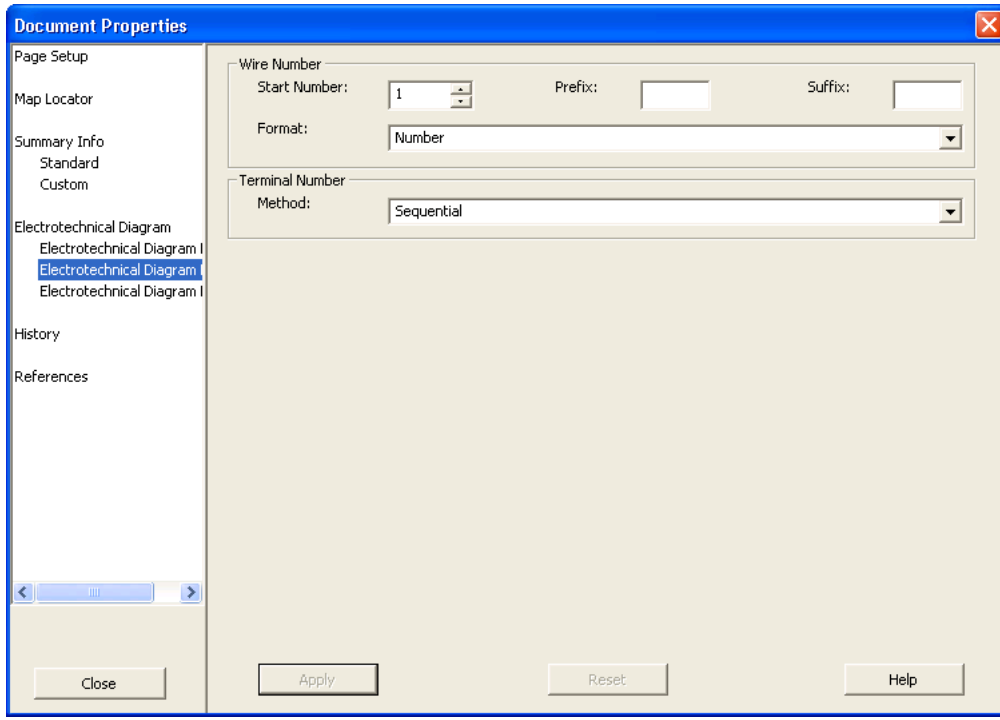


Figure 2-5 : Document Properties Dialog Box, Electrotechnical Branch Part 2

Section	Description
Command wire number	Provides fields to define the command wire number format.
Start Number	Defines the starting number for wire numbering on the current diagram.
Format	Allows the selection of the wire identifier format for the current diagram.
Prefix	Defines the wire identifier prefix for the current diagram.
Suffix	Defines the wire identifier suffix for the current diagram.

Section	Description
Terminal Number	Display of the terminal number.
Method	Display mode for the terminal number.

Display Format

The IEC standard component identifier display format is as follows:

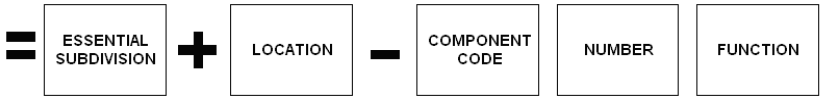


Figure 2-6 : IEC Standard Identification Component Format

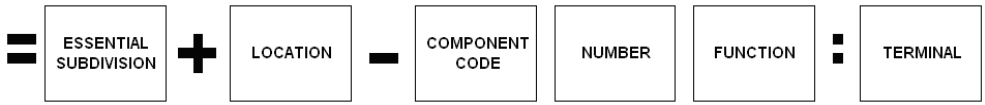


Figure 2-7 : IEC Standard Identification Terminal Format

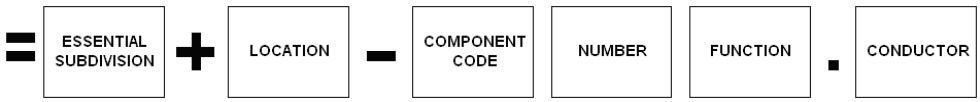


Figure 2-8 : IEC Standard Identification Wire Format

Example:

=R016+3A2-Q1:2

R016: Essential subdivision; room 016,

3A2: location; component location 3A2

Q1: designation, number; breaker 1

2: terminal; terminal 2

Decoding: Terminal 2, breaker 1 of the component at location 3A2 in room 016

= 1T5-M3

IT5: essential subdivision; pump 1 of the cooling turbine 5

M3: designation, number: motor 3

Decoding: Motor 3, pump 1 of the cooling turbine 5

= -K1

K1: designation, number; Breaker 1

Decoding; Breaker 1

The NEMA standard component identifier display format is as follows:

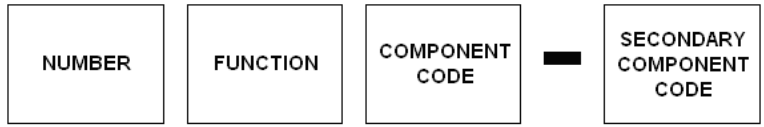


Figure 2-9 : NEMA Standard ID Component Format

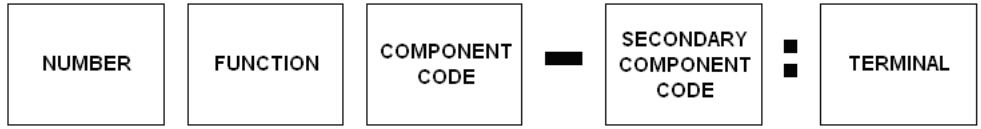


Figure 2-10 : NEMA Standard ID Terminal Format

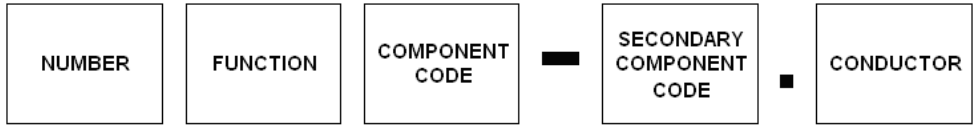


Figure 2-11 : NEMA Standard ID Wire Format

Examples:

1JCR

1: component "number" in this cabinet

J: function; testing

CR designation; control relay

Decoding: First control relay for testing

26CR-TDC

26: number of the component,

CR: designation; control relay

TDC: secondary designation; delay switch NO

Decoding: Delay NO switch controlled by relay 26

2.2.5 Electrotechnical Diagram Branch, Part 3

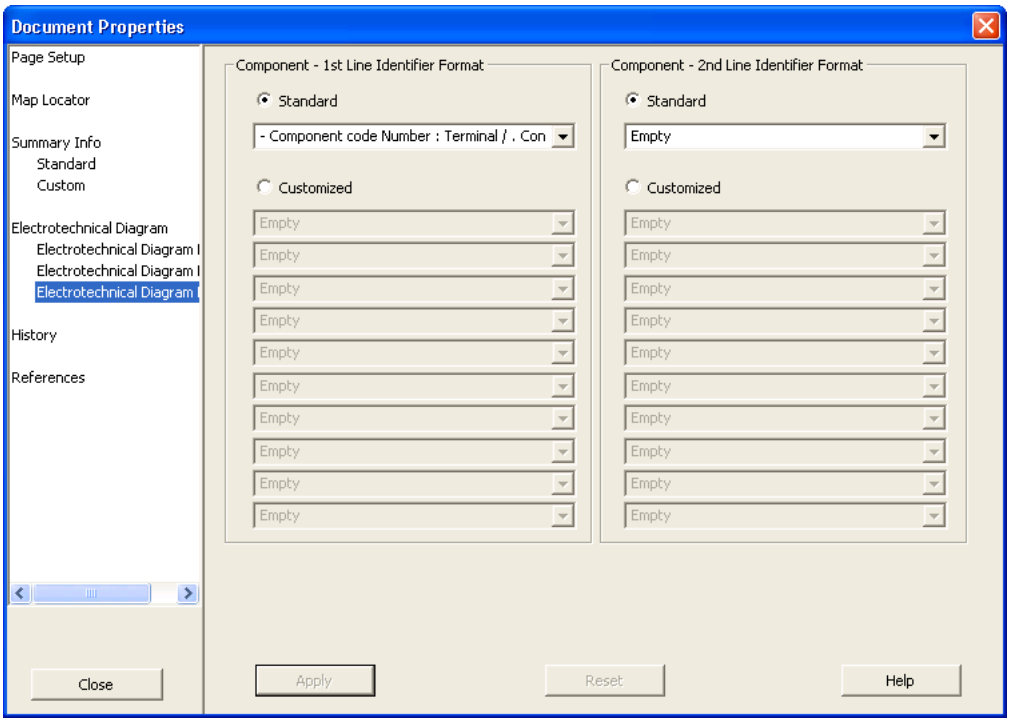


Figure 2-12 : Document Properties Dialog Box, Electrotechnical Branch Part 3

Section	Description
Component – 1st / 2nd	Specific areas used to define the first line and the second line of

Section	Description
Line Identifier Format	the component identifier.
Standard	Permit to select how the identifier will be defined by using a list of pre-defined formats.
Customized	Permit to build the identifier by adding several elements to the identifier. Each element is selectable trough a list.

2.3 View Menu

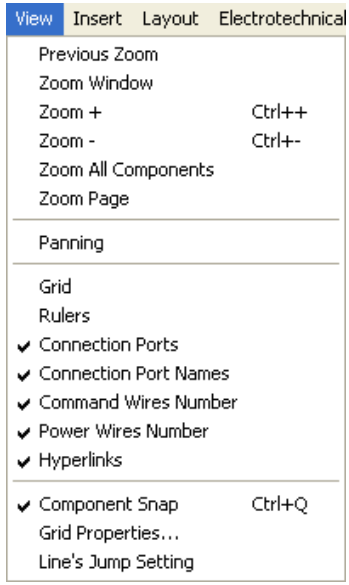


Figure 2-13 : View Menu

The View Menu commands which are specific to the electrotechnical diagram are:

Command Wire Number

If selected, this option displays the wire numbers

Hyperlinks

If selected, this option displays the hyperlinks

2.4 Insert Menu



Figure 2-14 : Insert Menu

The Insert Menu commands which are specific to the electrotechnical diagram are:

Command Wire

This option identifies the connections between components as being command wires. All future connections will also be command wires. The command wire's characteristics can be set from the Electrotechnical Editor properties available from the menu "Tools → Option"

The wire format can be modified using the selected wire pop-up menu.

Power Wire

It identifies the connections between components as being power wires. All future connections will also be a power wire. The power wire's characteristics can be set from the Electrotechnical Editor properties available from the menu "Tools → Option"

The wire format can be modified using the selected wire pop-up menu.

2.5 Layout Menu

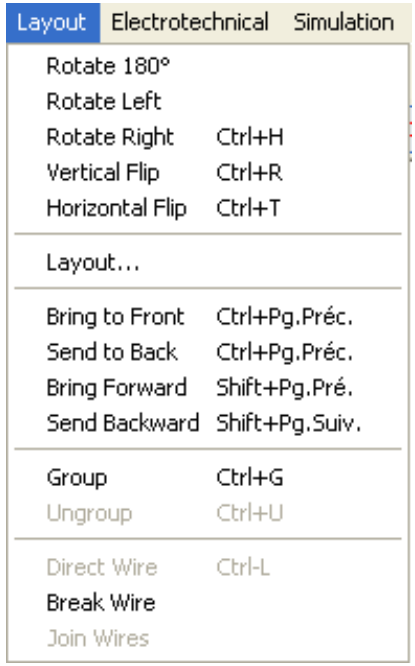


Figure 2-15 Layout Menu

The Layout Menu commands specific to the electrotechnical diagram are:

Direct Wire

This option redraws the wire as a direct diagonal or straight line between two points.

Break Wire

This command allows you to insert a connection point.

Direct wires can not be broken.

Join Wires

This is a tool which removes the connection point between two wires. The joined wire is then

re-numbered following the automatic numbering rule.

2.6 Electrotechnical Menu

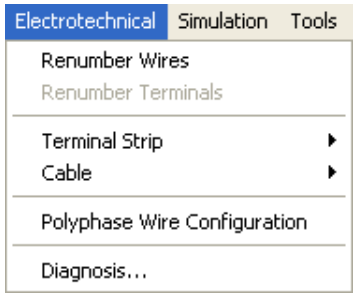


Figure 2-16 : Electrotechnical Menu

The commands available from the Electrotechnical menu are as follows:

Renumber Wires

This option allows the renumbering the diagram's wires. The renumbering follows the automatic numbering rules.

Renumber Terminals

This option allowas the renumbering of the diagram's terminals. See section 3.5.2 Terminal Numbering.

Terminal Strip

Gives access to the functions that are applicable to the terminal strips For further information, refer to the section 3.6 Management of Terminal Strips

Cable

This option provides tools for cable management. For further information, refer to chapter 4 Cable Management in this guide.

Polyphase Wire Configuration

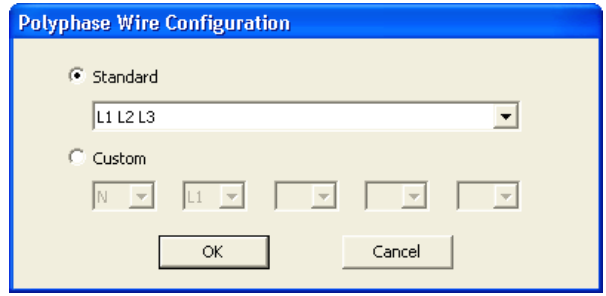


Figure 2-17 : Polyphase Wire Configuration Dialog Box

This tool allows for the configuration of polyphase wires. The configuration can follow one of the provided standard configurations, as well as a custom standard. The user can then define the number and sequence of wires in the polyphase wire.

Diagnosis

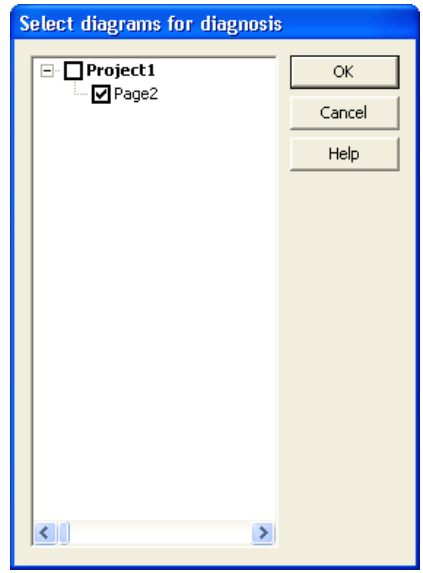


Figure 2-18 : Select Diagrams for Diagnosis Dialog Box

A diagnosis will be executed only on the diagrams selected in the “Select Diagram for Diagnosis” dialog box.

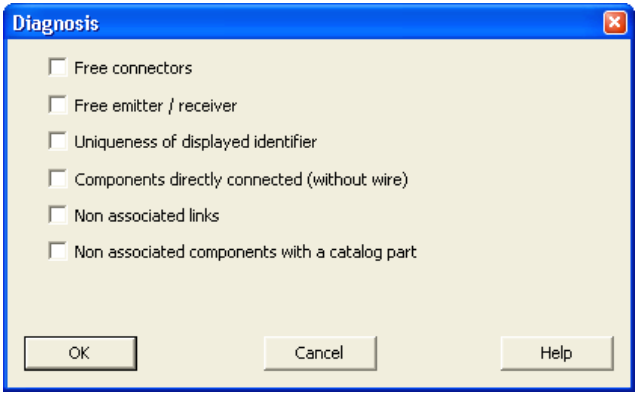


Figure 2-19 : Diagnosis Dialog Box

The actions available from the "Diagnosis…" command are as follows:

Sections	Description
Free Connectors:	Allows the user to detect free connectors in the current diagram.
Free emitter / receiver:	Allows the user to detect non-referenced emitter components and receiver components not linked to an emitter in the current diagram.
Uniqueness of displayed identifiers:	Allows the user to verify whether each component identifier in the current diagram is unique.
Components directly connected (without wire):	Allows the user to detect components of the current diagram that are directly linked to each other without a connecting wire.
Non associated links	Allows the detection of input links that are not connected to output links (or vice versa) in the current diagram.
Non associated components with a catalog part	Allows the detection of components that do not have a part number assigned in their "Component Properties" → "Catalog Information" dialog box.

You can select one or several diagnoses. The results are displayed in the message window. The components with an error or warning message appear in a specific color which indicates a

fault on the diagram. (see Automation Studio™ User's Guide.)

It is possible to simulate a project even though there are free connections; a warning message is displayed in the message window. The simulation night not work correctly if a receiver component such as a contact is not linked to an emitter component such as a relay. A warning message is displayed in the message window. The uniqueness of displayed identifiers does not affect the simulation. The presence of directly connected components without wire does not affect the simulation. (Normally, components are connected by a wire.).

2.7 Simulation Menu

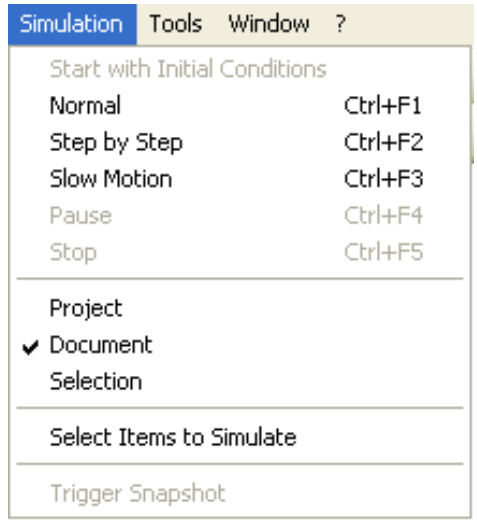


Figure 2-20 : Simulation Menu

This menu is identical to the standard editor.

2.8 Tools Menu

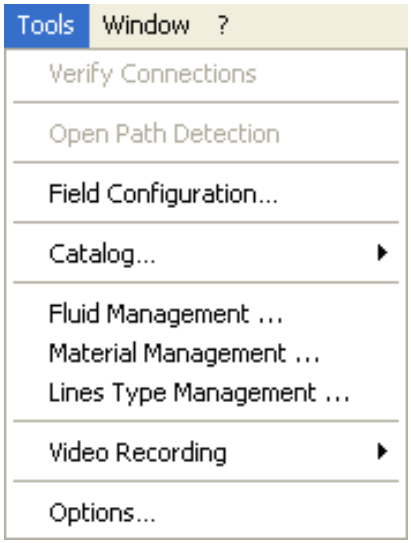


Figure 2-21 : Tools Menu

The options of the “Tools” menu give access to the options of the electrotechnical workshop.

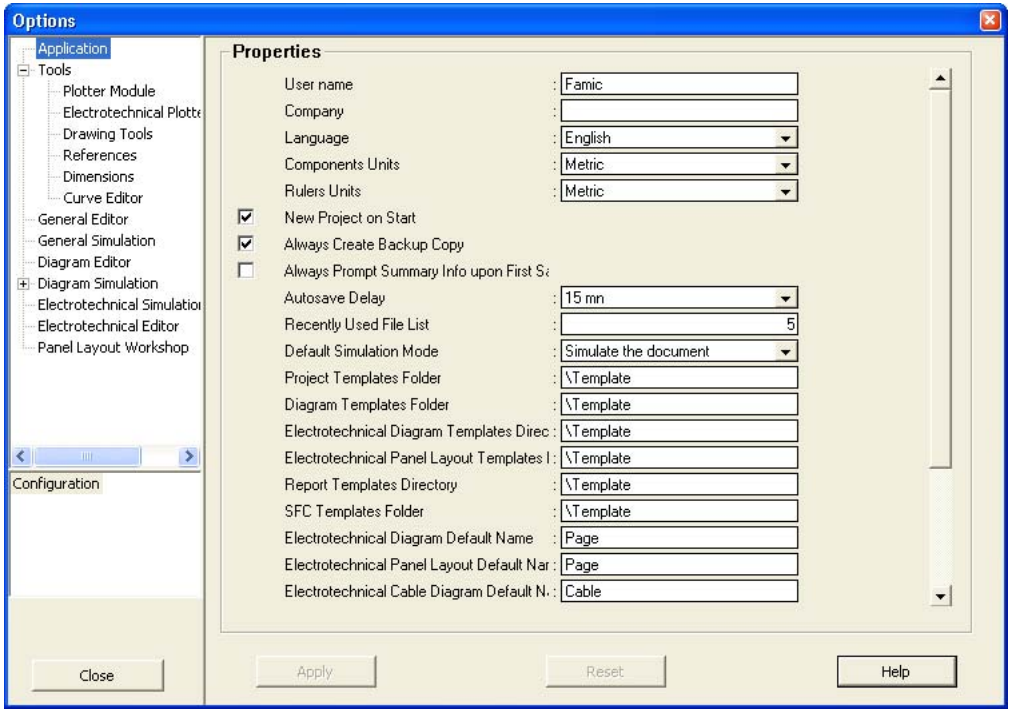


Figure 2-22 Options Dialog Box

These options are divided into several branches:

- Electrotechnical Plotter Module;
- Electrotechnical Editor;
- Electrotechnical Simulation;
- Panel Layout Editor;
- Panel Layout Workshops;
- Dimensions.

2.8.1 Electrotechnical Plotter Module

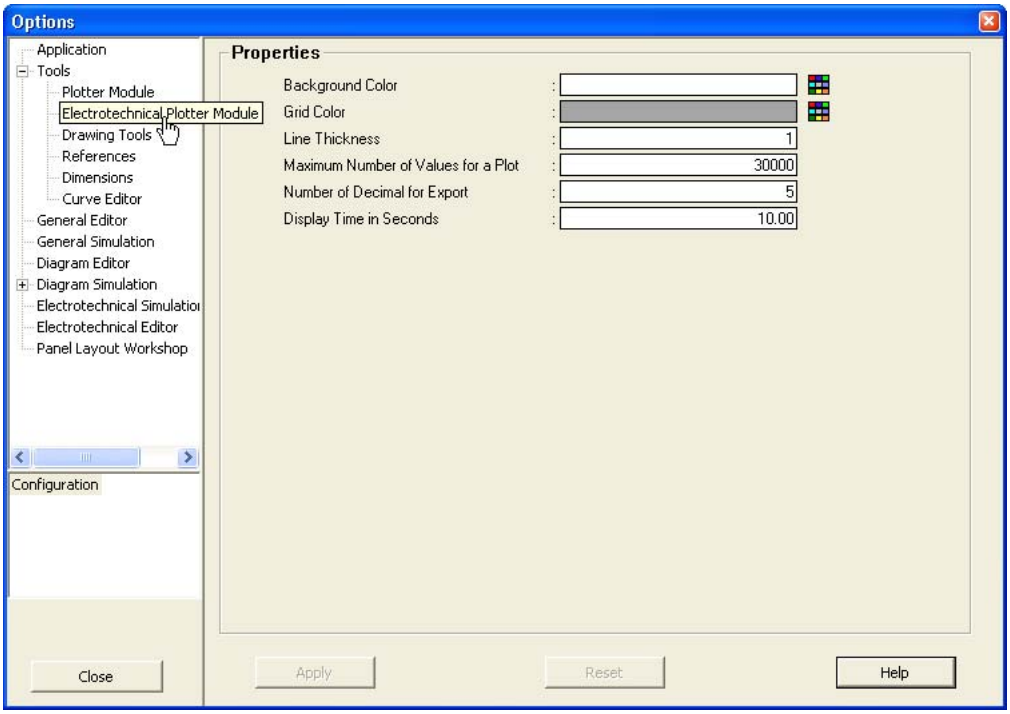


Figure 2-23 : Electrotechnical Plotter Module Option

This electrotechnical plotter module option operates exactly as does the standard plotter module.

2.8.2 Electrotechnical Editor Options

The options of the electrotechnical editor allow the definition of the editor's default properties for the workshop. They apply to every new diagram.

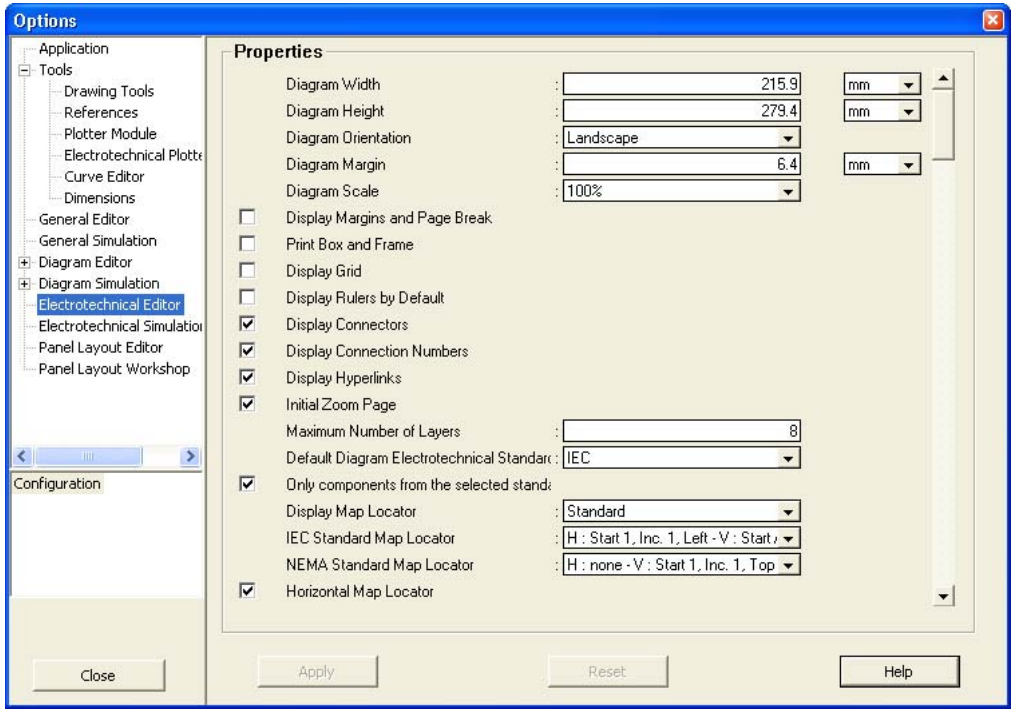


Figure 2-24: Electrotechnical Editor Options

The default properties of the electrotechnical editor are as follows:

Properties	**Description**
Diagram Width	Allows the user to specify the width of the diagram page.
Diagram Height	Allows the user to specify the height of the diagram page.
Diagram Orientation	Allows the user to specify the orientation of the diagram page.
Diagram Margin	Allows the user to specify the margins of the diagram page.
Diagram Scale	Allows the user to specify the scale of the diagram page.
Display Margins and Page	Allows the user to display or hide the margins and page

Properties	Description
Breaks	breaks.
Print Box and Frame	Allows the user to print the title block and borders around the page.
Display Grid	Allows the user to display the grid.
Display Rulers by Default	Allows the user to display the ruler.
Display Connectors	Allows the user to display the connectors.
Display Connection Numbers	Allows the user to display the connection numbers.
Display Hyperlinks	Allows the user to display the hyperlink standard font.
Initial Zoom Page	Allows the user to set the initial zoom to the diagram page.
Maximum Number of Layers	Allows the user to define the number of maximum layers.
Default Diagram Electrotechnical Standard	Allows the user to determine the default standard to be used for new diagrams.
Only Components from the Selected Standard are Accepted by Default	If unchecked, the next diagram will accept components from both IEC and NEMA standards.
Display Map Locator	Allows the user to display the map locator and to choose its format.
IEC Standard Map Locator	Allows the user to select the default map locator format for new IEC diagrams.
NEMA Standard Map Locator	Allows the user to select the default map locator format for new NEMA diagrams.

Properties	Description
Horizontal Map Locator	For a non-standard map locator, allows the user to display the horizontal map locator.
Page Number Horizontal Map Locator	For a non-standard map locator, this allows the user to display the page number as a prefix to the horizontal map locator number.
Default Start Symbol Horizontal Map Locator	For a non-standard map locator, this allows the user to enter the start number for the horizontal map locator numbering.
Default Increment Horizontal Map Locator	For a non-standard map locator, this allows the user to enter the increment value for the horizontal map locator numbering.
Default Origin Horizontal Map Locator	For a non-standard map locator, this allows the user to enter an origin for the horizontal map locator numbering.
Default Width Horizontal Map Locator	For a non-standard map locator, this allows the user to enter the unit and value for the width of the horizontal map locator.
Default Vertical Map Locator	For a non-standard map locator, this allows the user to display the vertical map locator.
Page Number Vertical Map Locator	For a non-standard map locator, this allows the user to display the page number as a prefix to the vertical map locator number.
Default Start Symbol Vertical Map Locator	For a non-standard map locator, this allows the user to enter the start number for the vertical map locator numbering.
Default Increment Vertical Map Locator	For a non-standard map locator, this allows the user to enter increment value for the vertical map locator numbering.
Default Origin Vertical Map Locator	For a non-standard map locator, this allows the user to enter an origin for the vertical map locator numbering.
Default Width Vertical	For a non-standard map locator, this allows the user to enter

Properties	Description
Map Locator	the unit and value for the width of the vertical map locator.
Default Areas Map Locator	Predetermines the number of zones used when selecting a non-standard map locator.
Default Automatic Adjustment Map Locator	When selected, the last horizontal and vertical map locators are adjusted so that all locators have equal width.
Display Grid Map Locator.	This allows the user to define whether the map locator should be displayed when a new electrotechnical diagram is created.
Default (IEC or NEMA) 1st Line Type	When the standard is used, it allows the user to define, for new diagrams, the display format for the first line of the component identifier.
Default (IEC or NEMA) 1st Line Comp. Id. Format	When the IEC standard is used, it allows the user to define, for new diagrams, the display format for the second line of the component identifier.
Default (IEC or NEMA) first customized list of first line	When the customized formatting way is used, allows the user to define a part of the identifier on the first line. All parts combined together will form the default component identifier for new diagrams.
Default (IEC or NEMA) 2nd Line Type	When selected, the identifier format for the second line is chosen trough a pre-defined list, else it is customized.
Default (IEC or NEMA) 2nd Line Comp. Id. Format	When the standard formatting way is used, allows the user to define, for new diagrams, the display format for the second line of the component identifier by using a pre-defined list.
Default (IEC or NEMA) first customized list of second line	When the customized formatting way is used, allows the user to define a part of the identifier on the second line. All parts combined together will form the default component identifier for new diagrams.
Disregard the Essential Subdivision and Location	Allows disregarding the use of the “Essential Subdivision”

Properties	Description
for components Identification by Default	and "Location" component properties in the identifier.
Wire Numbering Start Number	This allows the user to select the start number for the wire numbering.
Default Wire Number Prefix	This allows the user to define the prefix value for the wire numbers.
Default Wire Number Suffix	This allows the user to define the suffix value for the wire numbers.
Default (IEC or NEMA) Wire Number Format	This allows the user to select the display format for the wire numbers when using this standard.
Power Wire Numbering Identical to Command Wire Numbering by Default	This allows the use of the same wire numbering rule for command and power wires.
Power phase (1, 2, 3, neutral), default wire numbering start number (IEC & NEMA)	Allows the user to configure the wire numbering starting number for each phase.
Power phase (1, 2, 3, neutral), default wire number prefix	Allows the user to configure the wire number prefix for each phase.
Power phase (1, 2, 3, neutral), default wire number suffix	Allows the user to configure the wire number suffix for each phase.
Power phase (1, 2, 3, neutral), default wire number format (IEC & NEMA)	Allows the user to configure the wire number format for each phase.

Properties	Description
Wire Number Identical to I/O Address by Default	Allows the setting of the wire numbers connected to a PLC port with the PLC function address number.
Disregard the Essential Subdivision and Location for Wire Numbering by Default	Allows you to disregard the use of the "Essential Subdivision" and "Location" component properties in wire numbering.
Display Command Wire Number by Default	Allows the user to display the command wire numbers.
Display Power Wire Number	Allows the user to display the power wire numbers.
Terminal Numbering Method	Allows the user to select the default method for terminal numbering.
Default (IEC & NEMA) Reference Id. Format	Allows the user to select the display format for the link identifiers standard.
(IEC & NEMA) Localization Hyperlink Format	Defines the default format for hyperlinks when using a standard.
"Coordinate Location" displayed by default	Allows the user to display the default coordinate locations. The coordinates are displayed according to the selected map locator format.
External Type Component and Wire Color	Allows the user to determine the default colors of components and of external wires. (outside the cabinet)
Clip Jumper Type Wire Color	Allows the user to determine the wire color of clip jumpers linking two terminals of a terminal strip by default.
(IEC & NEMA)	Allows the user to modify the editing format (tracing format,

Properties	Description
Command Wire Appearance	wire color), for command wires. This option is only taken into consideration when using a standard.
(IEC & NEMA) Power Wire Appearance	Allows the user to modify the editing format (tracing format, wire color) for power wires. This option is only taken into consideration when using a standard.
(IEC & NEMA) Phase (1, 2, 3, neutral) Wire Appearance	Allows the user to modify the editing format (tracing format, wire color) for phased power wires. This option is only taken into consideration when using a standard.
(IEC & NEMA) Ground Wire Appearance	Allows the user to modify the editing format (tracing format, wire color) for ground power wires. This option is only taken into consideration when using a standard
Default Wire Number Font Height	Defines the default font size for the wire numbers.
Default Wire Number color	Defines the default font color for the wire numbers.

2.8.3 Electrotechnical Simulation

The electrotechnical workshop simulation options determine the colors and properties of wires and components during simulation. They apply to all electrotechnical simulated diagrams in the same project.

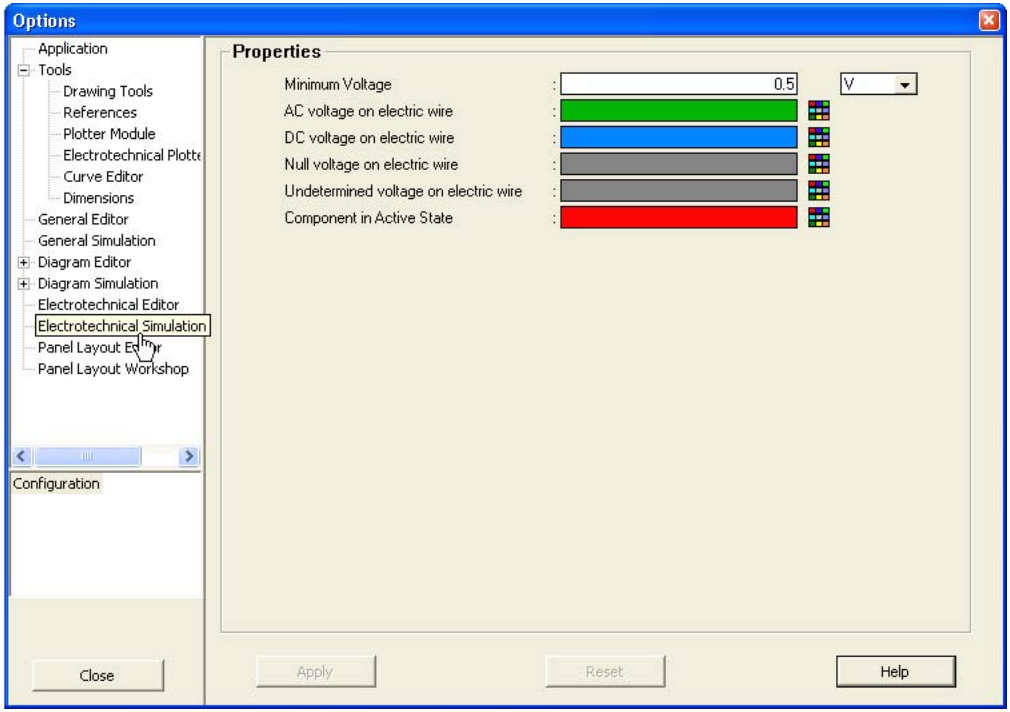


Figure 2-25 : Electrotechnical Simulation Options

The default properties of the electrotechnical editor in Figure 2-25 are as follows:

Properties	Description
Minimum Voltage	Allows the user to determine the unit and value of the minimum voltage of any wire.
AC voltage on electric wire	Allows the user to determine the color of wires with alternating current during simulation.
DC voltage on electric wire	Allows the user to determine the color of wires with direct current during simulation.
Null voltage on electric wire	Allows the user to determine the color of the wires with no current during simulation.

Properties	Description
Undetermined voltage on electric wire	Allows the user to determine the color of the wires with undetermined current during simulation.
Component in Active State	Allows the user to determine the color of active components during simulation.

2.8.4 Panel Layout Editor Options

The panel layout editor options determine the properties for the diagram page settings. Those settings apply to all new panel layout diagrams that are created.

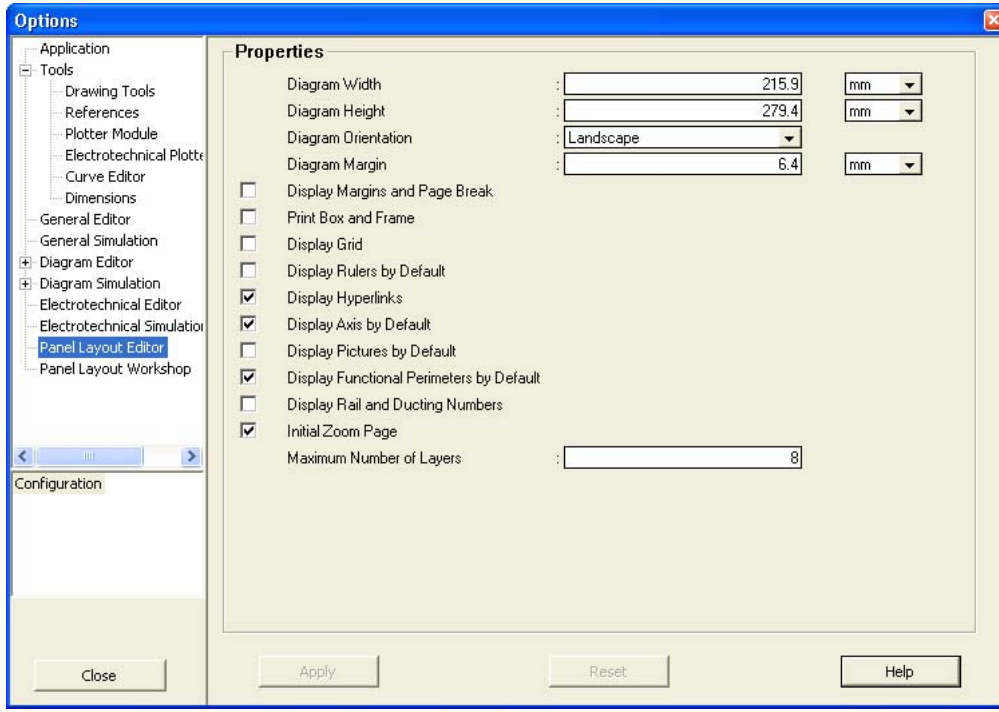


Figure 2-26 : Panel Layout Editor Options

The default properties of the panel layout editor are as follows:

Properties	Description
Diagram Width	Defines the default width of the diagram.
Diagram Height	Defines the default height of the diagram.
Diagram Orientation	Defines the default orientation of the diagram.
Diagram Margin	Defines the default margins of the diagram.
Display Margins and Page Break	Allows the showing the margins and the page break in the diagram view in AS.
Print Box and Frame	Allows the printing of the default title box and the diagrams borders.
Display Grid	Allows the display of the grid over the AS diagrams.
Display Rulers by Default	Allows the display of the rulers in AS.
Display Hyperlinks	Allows the display of the hyperlinks in the AS diagrams.
Display Axis by Default	Allows the display of the default panel layout axis.
Display Pictures by Default	Allows the display of the component images in AS diagrams.
Display Functional Perimeters by Default	Allows the display of functional component perimeters in AS diagrams.
Display Rail and Ducting Numbers	Allows the display of rail and ducting identification numbers in AS diagrams.
Initial Zoom Page	Allows the page to be zoomed when opened for the first time.
Maximum Number of layers	Defines how many layers may be used for each diagram.

2.8.5 Panel Layout Workshop Options

The panel layout workshop options determine the properties used for the views on the diagram. Those settings apply to all new panel layout diagrams that are created.

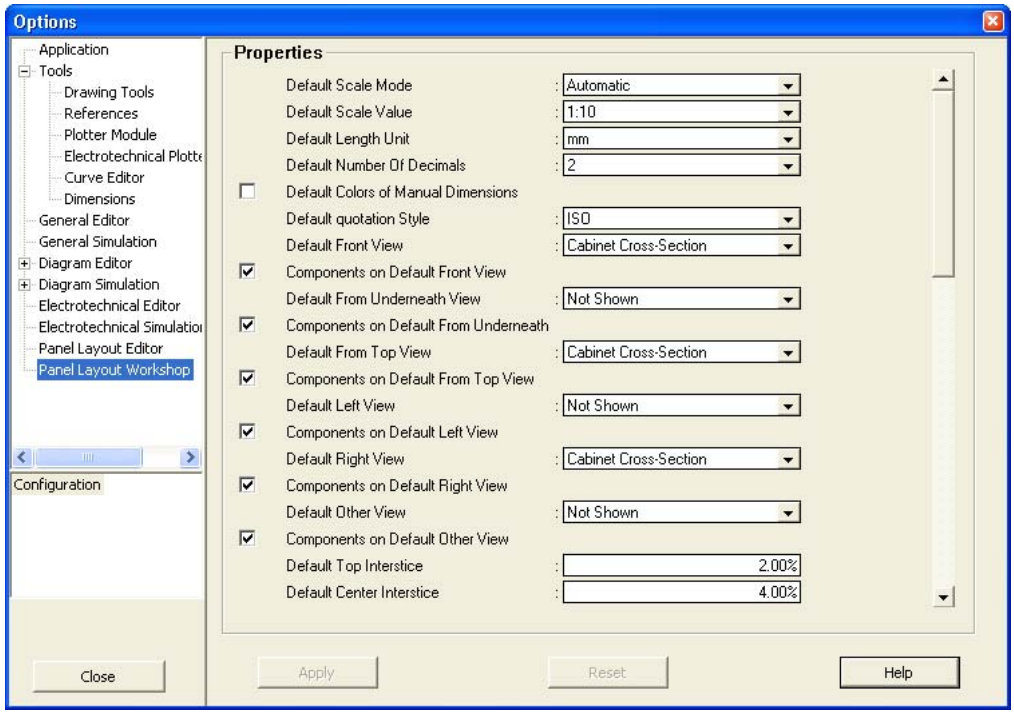


Figure 2-27 : Panel Layout Workshop Options

The default properties of the panel layout workshop are as follows:

Properties	Description
Default Scale Mode	Allows the automatic adjustment of the zoom level depending on the cabinet or mounting panel dimensions.
Default Scale Value	Defines the default scale when in manual scale mode.
Default Length Unit	Defines the default unit used in the control panel workshop.

Properties	Description
Default Number of Decimals	Defines the default quantity of numbers for quotations.
Default Colors of Manual Quotation	Allows the selection of the default color for quotations.
Default Quotation Style	Allows the selection of the default style for quotations.
Default Views	Defines each view used in the default diagram. The possibilities are: ● Not Shown (the front view must be shown); ● Closed Cabinet; ● Cabinet Cross-Section; ● Mounting Panel; ● Inside Door (used for the front view only).
Components on Default Views	Allows the display of the component box or its axes with its functional perimeters for each view.
Default Other View	Defines another view used in the default diagram. The possibilities are: ● Not Shown; ● Closed Cabinet Front View; ● Cabinet Back View; ● Mounting Panel Front View; ● Mounting Panel Back View; ● Inside Door Front Picture.
Default Interstices	Defines spacing between views.
Default Placement	Defines either ISO or “North American” view as default.
Colors	Defines the color of several drawing elements.

Properties	Description
Line Width	Defines the line width for several drawing elements.
Label Properties	The labels are not currently used in Automation Studio™.
Functional Perimeter Line Width	Defines the width of the line around the component functional perimeters.
Hyperlink Text Height	Defines the hyperlink font size used.
Frame View	Allows the display of view borders on the diagram.
Cabinet Thickness	Defines the thickness of the cabinet walls. AS must be closed and reopened to activate this setting.
Mounting Panel Thickness	Defines the thickness of the mounting panel. AS must be closed and reopened to activate this setting.

2.8.6 Dimensions Options

The dimensions options dialog box permits the configuration of the quotations default properties on the diagram.

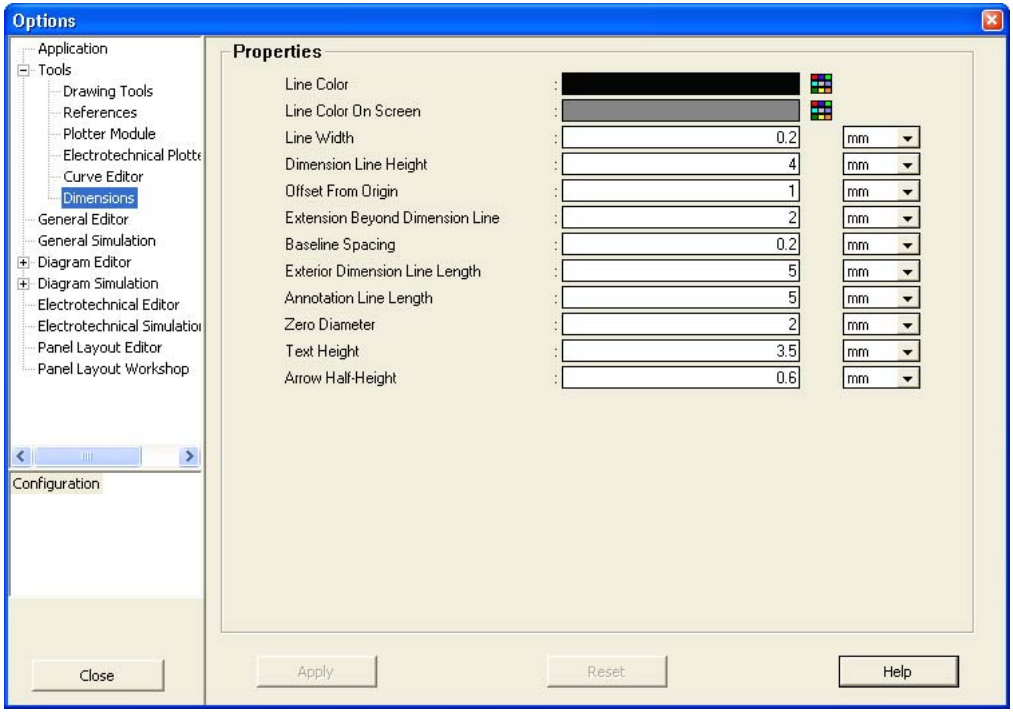


Figure 2-28 : Dimensions Options

The default properties for the quotations are as follows:

Properties	Description
Line Color	Defines the line color of the quotations when printed on paper.
Line Color On Screen	Defines the line color of the quotations on the screen.
Line Width	Defines quotation line default width.
Dimension Line Height	Defines the default height of the dimension line compared to its tied point.
Offset From Origin	Defines the offset between the quotation arrow and the dimension line of the quote diameter.

Properties	Description
Extension Beyond Dimension Line	Defines the length of the line over the quotation dimension line.
Baseline Spacing	Defines the spacing between the dimension line and the displayed value.
Exterior Dimension Line Length	Defines the length of the line between the arrow and the dimension value of the linear quote.
Annotation Line Length	Defines the length of the line between the arrow and the dimension value of the diameter quote.
Zero Diameter	Defines the zero symbol diameter used for stacked quotations.
Text Height	Defines the font size of the quotation value.
Arrow Half-Height	Defines the quotation arrow size.

2.9 Window Menu

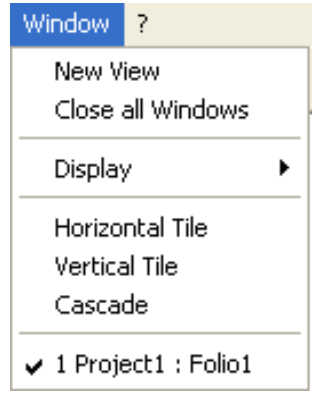


Figure 2-29 : Window Menu

The “Window” menu is identical to the one found in the standard editor.

2.10 Help Menu

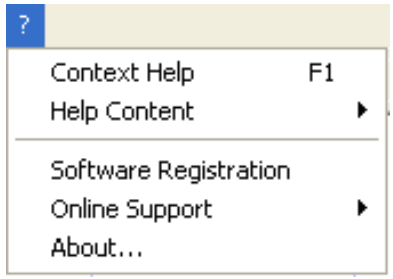


Figure 2-30 : Help Menu

The “?” menu is identical to the one found in the standard editor.

3 Electrotechnical Workshop Capability

3.1 Introduction

This chapter presents the new capabilities introduced by the Electrotechnical workshop in detail. It covers the following points:

- Component management;
- Cross-reference management;
- Wire management;
- Cable management;
- Terminal strip management.

3.2 Component Management

Electrotechnical components are managed in the same way as components of other workshops.

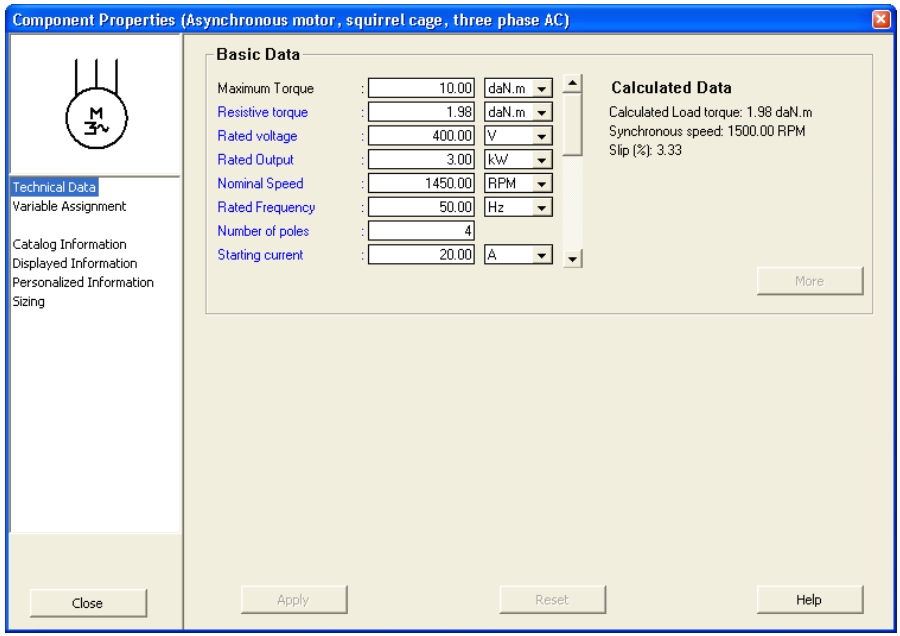


Figure 3-1 : Technical Data Branch of Motors

The Component Properties command of the Edit menu and the component pop-up menu give you access to the properties of the selected component. The Technical Data or the Configuration branch in the component properties dialog box allow you to define the behavior of the component.

The component Help files describe each of the properties used.

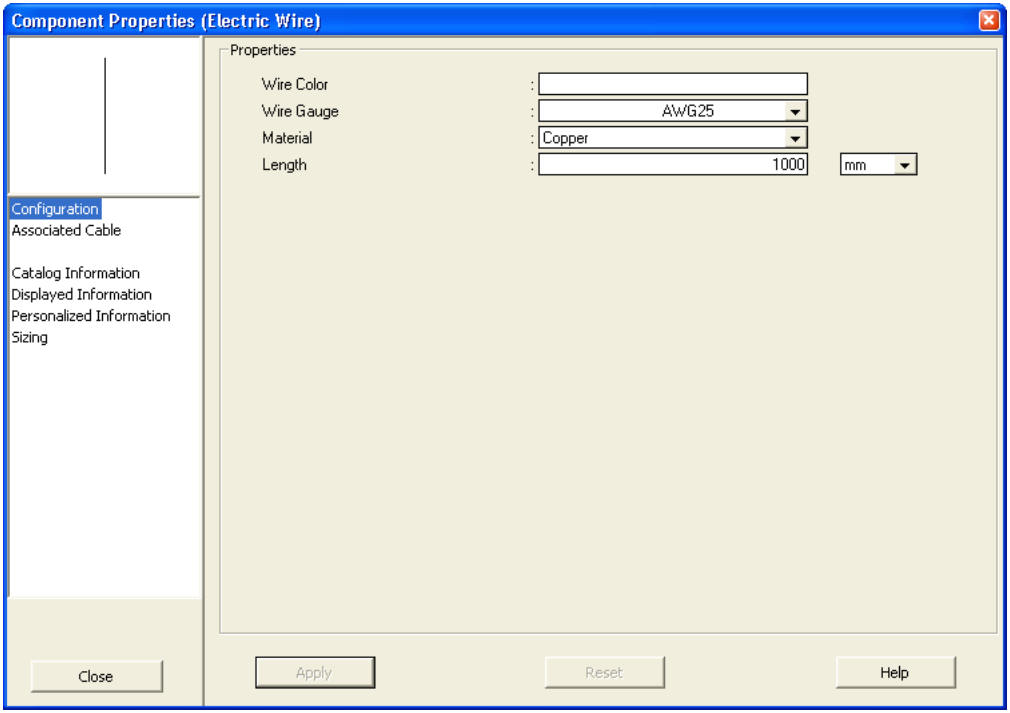


Figure 3-2 : Configuration Branch of an Electric Wire

The component pop-up menu allows you to define a component as inside or outside a cabinet. This information is required to define from what a cable is connected, and what it is connected to. By default, components are considered internal.

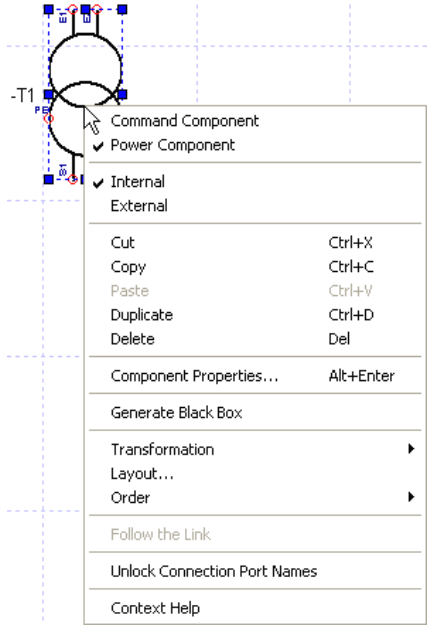


Figure 3-3 : Component Pop-Up Menu

3.3 Wire Management

3.3.1 Wire Types

The Electrotechnical workshop offers seven wire types:

- Command wire, to connect command elements, represented by a thin line;
- Power wire, to connect power elements, represented by a thick line;
- Power wire phase 1, phase 2, phase 3, neutral;
- Ground wire.

You can access these two types of wire from the Insert toolbar and from the Insert menu. By default, components are linked by command wires.

The wire pop-up menu allows switching from one wire type to another. For an existing project, the active wire type is the last one used in the project.

3.3.2 Wire Simulation

The wire resistivity may be calculated in simulation mode. The equivalent impedance of each wire in the circuit is evaluated by using their individual data, which are:

- The wire gauge;
- The material the wire is made of;
- The wire length.

☞ Enabling the evaluation of the wire impedances may considerably slow down the simulation speed. It is recommended to simulate the wires only when it is necessary to study their behavior..

The steps to activate wire simulations are :

1. Uncheck the “Perfect Wires” box in the “Project Properties” → “Options” → “Electrotechnical Simulation” dialog box.

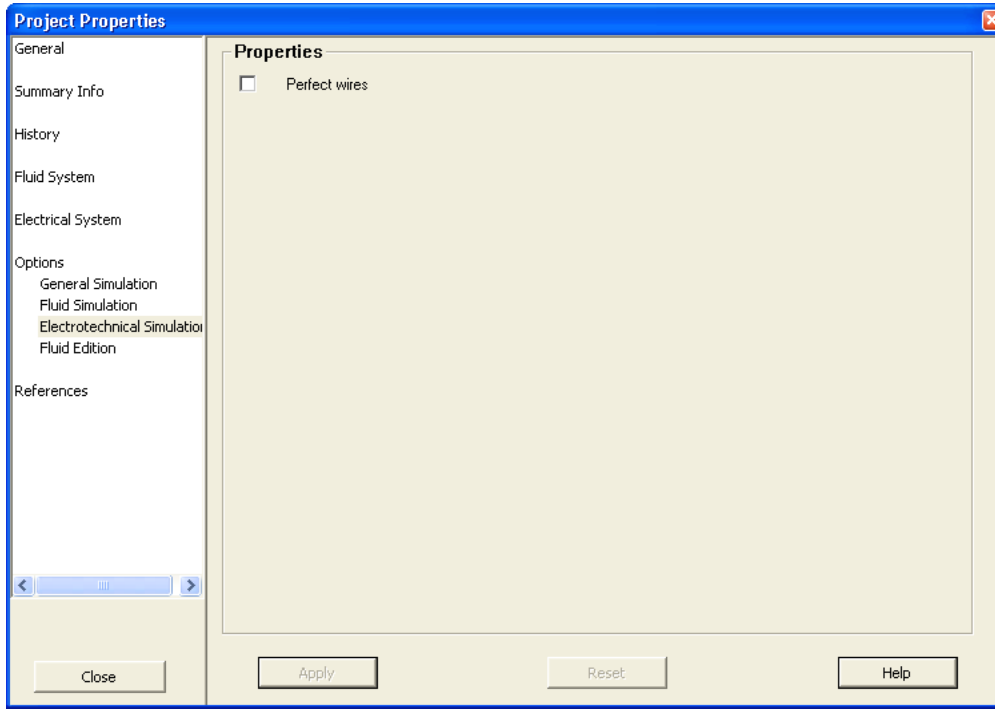


Figure 3-4 : "Perfect Wires" option for the project

2. Define each wires that must be simulated as "Internal' or "external" type by using the wire context menu. Refer to section *3.3.3 Wire Definition* for more details.

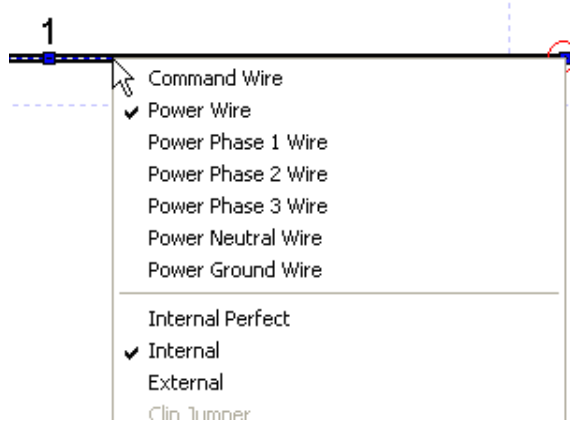


Figure 3-5 : Change to an "Internal" type of wire

It is possible to select several or even all the wires in the page and modify their type in one action. Refer to the select all wires function in the "Edition" → "Select" menu.



The « Internal Perfect » wires won't show the "Configuration" and "Sizing" branches in the "Component Properties dialog box". So, it is not possible to configure those wires properties.

3. Enter the data, like the gauge, material and length, to correctly describe the wires in their respective "Component Properties" → "Configuration" dialog boxes.

3.3.3 Wire Definition

A wire can be defined in four ways:

1. "Internal Perfect" indicates a wire without any properties. It's impedance will not be evaluated in simulation mode. The "Configuration" and "Sizing" branches won't appear in its "Component Properties" dialog box.

   If the perfect wire is connected to a terminal, it will be considered to be on the "From" side of the terminal.

   The "Perfect" wires do not appear in the project reports.

2. "Internal" indicates a connection wire on the back panel of a cabinet. If the perfect wire is connected to a terminal, it will be considered to be on the "From" side.

   The technical properties of the wire will be calculated in simulation mode if the "Perfect Wires" option in the "Project Properties" → "Options" → "Electrotechnical Simulation" dialog box is unchecked.

3. "External" indicates a connection wire off the back panel of a cabinet, to connect components in front of a cabinet, or to connect an external power source to a cabinet. An external wire is normally connected to a terminal. It will appear on the "To" side;

   The technical properties of the wire will be calculated in simulation mode if the "Perfect Wires" option in the "Project Properties" → "Options" → "Electrotechnical Simulation" dialog box is unchecked.

4. "Clip Jumper" indicates a clip jumper mounted on the terminal strip.

   This information is required to define the terminal strip connections. You can define the

type of wire through its pop-up menu. By default, wires are defined as internal and will appear on the "From" side of the terminal.

The resistance of a clip jumper is negligible. It is considered as a perfect wire.

☞ A wire can only be a clip jumper wire if it is between two terminals. Once a clip jumper wire is disconnected at either end, such as by deleting a terminal, etc., the wire automatically changes into an internal wire.

3.3.4 Wire Numbering

The wires are automatically numbered according to the following rules:

- The wires are numbered in their order of creation;
- Links spread the wire numbers from the emitter to the receiver;
- Connected wires have the same number;
- A wire that does not have its number locked which is connected to a locked numbered wire takes the number, the prefix, and the suffix of the locked number. Note that the number may include letters;
- Two wires connected to the same terminal carry the same number;
- Two wires connected to an interrupt terminal (contact, fuse, etc.) do not have the same number;
- When using the IEC standard, wires are numbered from left to right and from top to bottom, independently of the map locator direction;
- When using the NEMA standard, wires are numbered from top to bottom and from left to right, independently of the map locator direction;
- There can be two identical wire numbers if the essential subdivisions or the locations are different;
- You can lock a wire number from the pop-up menu. Locked numbers are not modified during an automatic renumbering. A locked number is represented by an asterisk displayed on the right of the number;
- The display format of the wire identifier is defined in the document properties; it applies

to all existing and future wires of the document;

- You can manually number a wire using the Displayed Information pop-up menu of the component;

- The Renumber Wires command in the Electrotechnical menu, allows you to renumber all the wires of the active document which are not locked. Renumbering is performed following these rules;

- The name of the terminal to which a wire is connected can be used as a prefix to the wire number. This can be done manually through the standard component properties dialog box. The Displayed Information branch gives access to the "prefix" property, which can be modified. You can also assign an automatic prefix from the pop-up menu.

3.3.5 Wire Properties

Wire properties is accessed in the same way as any other component.

Configuration Branch

The Configuration branch of the wire properties dialog box allows you to define the wire color and gauge. This branch is not shown if the wire is "Internal Perfect".

Associated Cable Branch

The Associated Cable branch of the wire properties dialog box allows you to define a wire as a conductor of a cable.

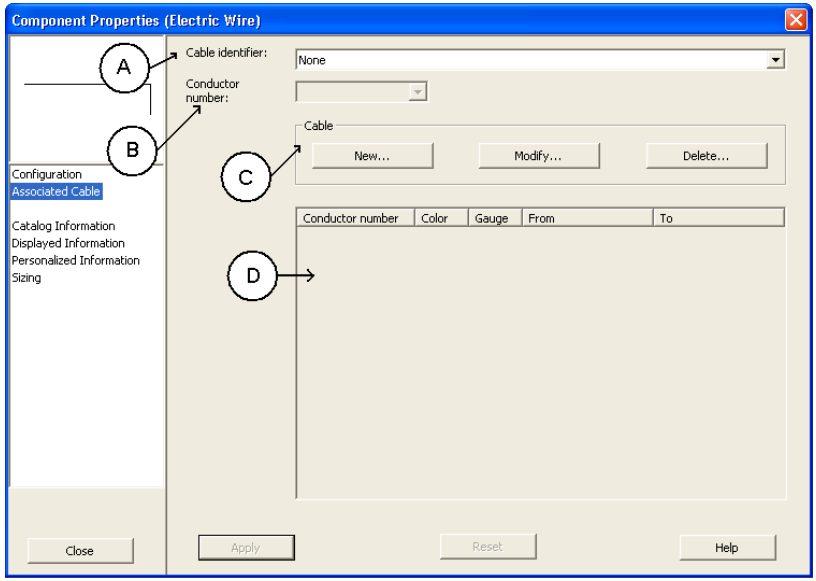


Figure 3-6: Associated Cable branch of wires

The properties of this dialog box are:

Properties		Description
A	Cable Identifier	Allows to select among existing cables, the cable which should be associated with the wire.
B	Conductor Number	Allows to select a free conductor from a drop-down menu. Only free connectors appear in the list.
C	Cable	Allows cable management. Allows you to create, modify or delete a cable (see chapter 4 Cable Management).
D	Assignment table	This table lists the conductors used for the selected cable. This table cannot be edited.

To associate a wire with a cable:

1. Select the cable identifier.
2. Select the number of the conductor.
3. Click on the Apply button.
4. Close the dialog box.

☞ A NEMA type wire is compatible with an IEC type wire and vice versa.

To dissociate a wire from a cable:

1. Select “None” from the cable identifier.
2. Click on the Apply button.
3. Close the dialog box.

To create, modify or delete a cable from this dialog box, see chapter 4 Cable Management

The software manages the display of the wire identifiers of cables as follows:

- The cable identifier followed by the conductor number is displayed under the wire.
- When a wire is linked to a cable, the Location and the Essential Subdivision properties become those of the cable and can no longer be modified by the user.
- When a wire is disconnected from a cable, the cable identifier disappears and another wire can be associated as a conductor with the cable. The Location, Essential Subdivision, Displayed Component Code, Tag Name and Comment Properties take their initial values.
- Moving a wire within the same project does not affect its association with its cable.
- If a wire is moved to another project, its association with its cable is lost.
- When a wire is pasted in another project, it is disconnected if the new project does not contain a cable with the same Tag Name.
- When a wire is pasted (after having been copied) in the same project, there are various

possible outcomes:

a) There is a free conductor in the cable from which the copied wire is linked:

The pasted wire keeps its link with the cable,

The wire occupies the first available conductor in the cable.

b) There are no free conductors in the cable from which the copied wire is linked:

The wire is disconnected.

The warning message “The wire X has been dissociated from cable Y, there are no more free connectors” appears in the message window.

Sizing Branch

The “Sizing” branch is not shown if the wire is “Internal Perfect”.

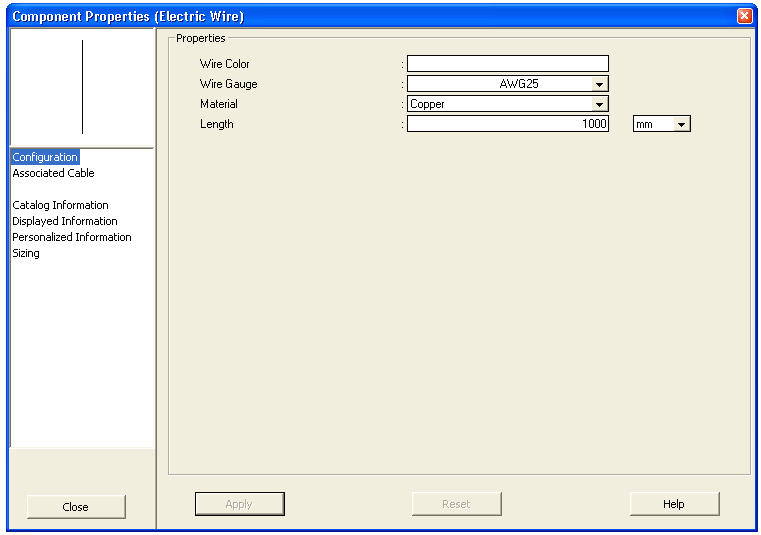


Figure 3-7 : Configuration Branch of Wire

Displayed Information branch

The “Displayed Information” branch of the Wire properties dialog box allows you to define

and/or modify the displayed information of the wire.

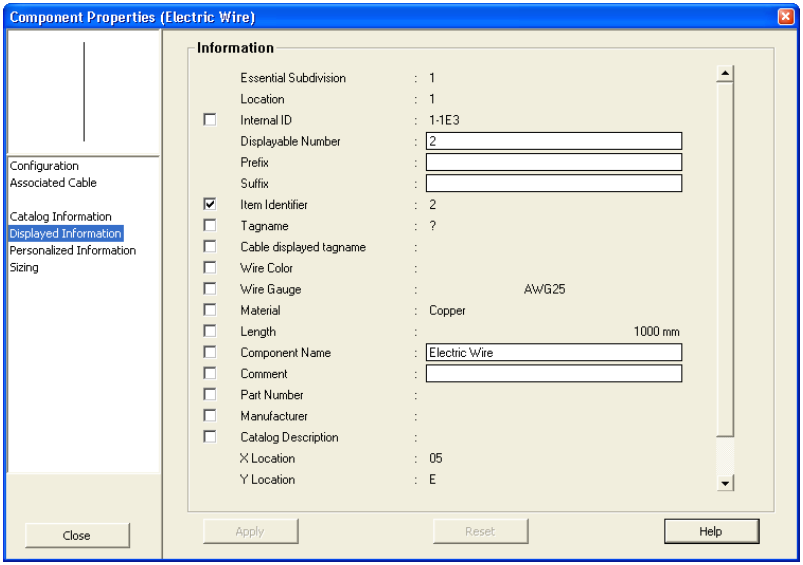


Figure 3-8: Displayed Information Branch of Wire Properties

For more information refer to section 5.2 Displayed Information***Error! Reference source not found.*** of this manual.

3.3.6 Polyphase Links

It is possible to connect 3-phase components using the polyphase wire tool. This tool allows the creation of 2 to 5 parallel wires in a single step.

This polyphase tool is a group of power wire that becomes independent once drawn on the diagram.

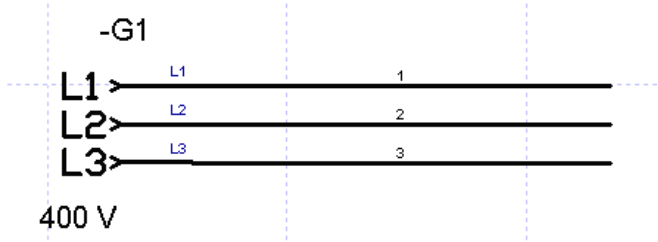


Figure 3-9 : Polyphase Link Example

You can define the polyphase wire tool by accessing the dialog box from the “Format” toolbar or from the “Electrotechnical” menu:



Figure 3-10 : Access to the Polyphase Wire Tool Configuration Dialog Box

The configuration dialog box is as follows:

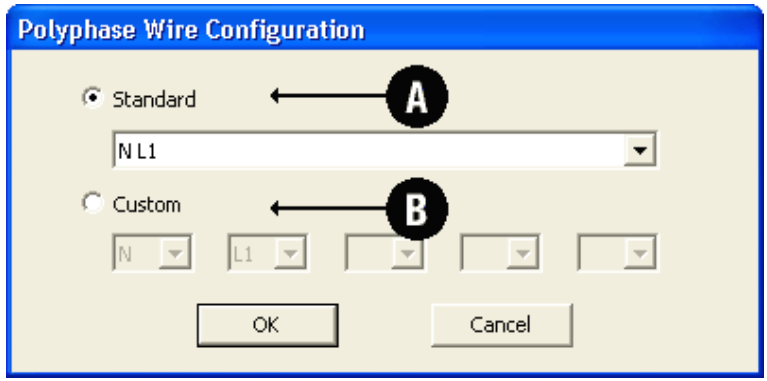


Figure 3-11 : Polyphase Wire Tool Configuration Dialog Box

You can choose from a list of standard configurations or personalize it.

Here is a list of the predefined standard wires:

Zone	Nom	Description
A	Standard	Allows selecting a configuration using the predefined list box. The standard configurations are: N L1 L1 L2 L1 L2 L3 N L1 L2 L3 N L1 PE L1 L2 PE L1 L2 L3 PE N L1 L2 L3 PE

Zone	Nom	Description
B	Custom	Allows you to choose each wire that composes the polyphase wire including the position from left to right or from top to bottom. Each drop-down list gives you a choice of electrotechnical wires. You can define from 2 to 5 wires in your personalized polyphase wire. The possible values are: L1 : phase 1 L2 : phase 2 L3 : phase 3 N : neutral PE : ground

You can modify the format (style and color) of each wire (L1, L2, L3, neutral or ground) by accessing the "Electrotechnical Editor "branch from the "Options" dialog box.

When using the polyphase wire tool, the cursor draws the master wire. The other wires are slave wires. The slave wires are placed on the right, if you start to draw vertically and at the bottom of the master wire if you start to draw horizontally. If you start to draw from a connector, the slave wires tend to position themselves on the nearest connector. The polyphase drawing tool search connectors from components on top, bottom, right and left. If connectors are found at a distance smaller than 2 grid steps, the tool makes the connection.

3.3.7 Wire Pop-Up Menu

Right-clicking on a selected wire opens the wire pop-up menu. The following figure shows the wire pop-up menu.

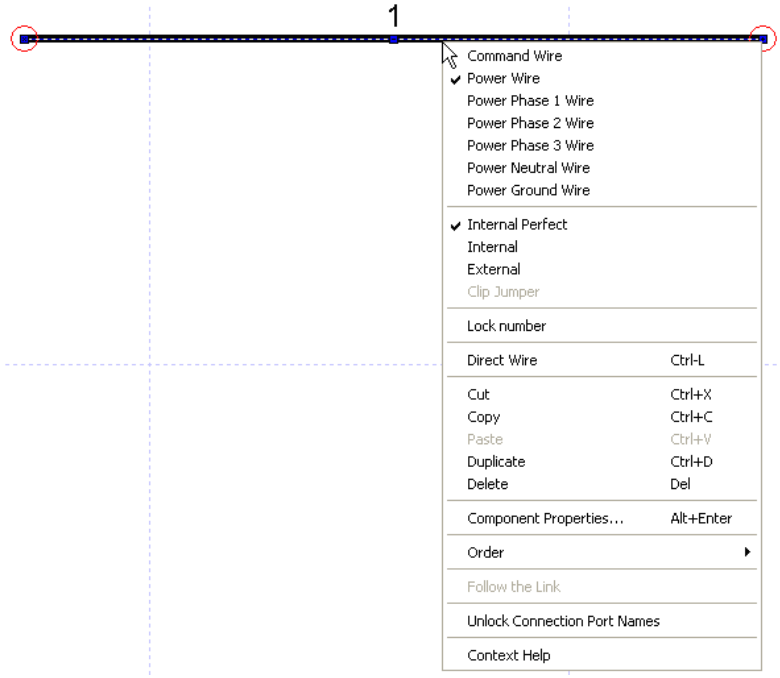


Figure 3-12 : Wire Pop-Up Menu

The pop-up menu commands for wires of the electrotechnical workshop are:

Command	**Description**
Command Wire	Allows the user to define a wire as a command wire, which is displayed as a thin line.
Power Wire	Allows the user to define a wire as a power wire, which is displayed as a thick line.
Power Phase 1 Wire	Allows the user to define a wire as a Phase 1 Power Wire, which is displayed as a thick line.
Power Phase 2 Wire	Allows the user to define a wire as a Phase 2 Power Wire which is displayed as a thick line.
Power Phase 3 Wire	Allows the user to define a wire as a phase 3 power wire.

Command	Description
	(displayed as a thick line)
Power Neutral Wire	Allows the user to define a wire as a neutral power wire. (displayed as a thick line)
Power Ground Wire	Allows the user to define a wire as a ground power wire. The wire is displayed as a thick line.
Internal Perfect	Allows the user to define a wire as being perfect. Its impedance is not taken into account in simulation mode. This type of wire won't appear on the reports.
Internal	Allows the user to define a wire as being inside a cabinet. This information is required to define connections to terminal strips. Internal wires are displayed on the "From" side of the terminal strip picture.
External	Allows the user to define a wire as being outside a cabinet. This information is required to define connections to terminal strips. External wires are displayed on the "To" side of the terminal strip picture.
Clip Jumper	Allows the user to define a wire as being a clip jumper connecting two terminals of a terminal strip. This information is required to define terminal strip connections. Clip jumpers are shown as a vertical line with connectors on terminal strip diagrams.
Set prefix at …	Allows the user to automatically assign the wire with the component terminal name, which is attached as a prefix to the wire number. When the wire is connected between two component terminals, you can select the one that is required. This can also be done manually using the standard dialog box of the component properties.
Lock Number	Allows the user to lock the wire number. Automatically renumbering wires does not affect locked numbers.
Direct Wire	Allows the user to draw a direct wire between two terminals by

Command	Description
	the shortest path.

3.3.8 Wire Number Alignment

The "Align Satellites Tool" allows to greatly simplifying the wire number alignment procedure. This tool is located in the "Layout" toolbar.



Figure 3-13 : Align Satellites Tool

To quickly align multiple wire numbers:

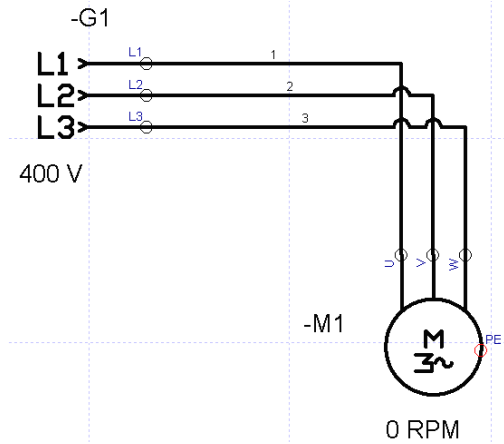


Figure 3-14 : Circuit with Unaligned Wire Numbers

1. Click on the "Align Satellites Tool"  button to activate the alignment mode;

2. Click and drag horizontally or vertically the alignment line over the wires at the location you want the wire numbers to be located;

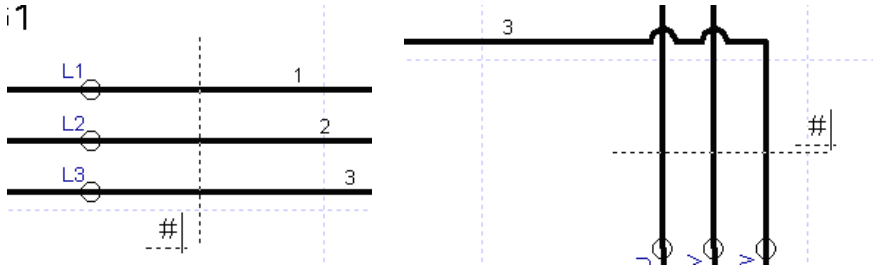


Figure 3-15 : Alignment Localization Line

3. Release the mouse button to permit the wire number to align them.

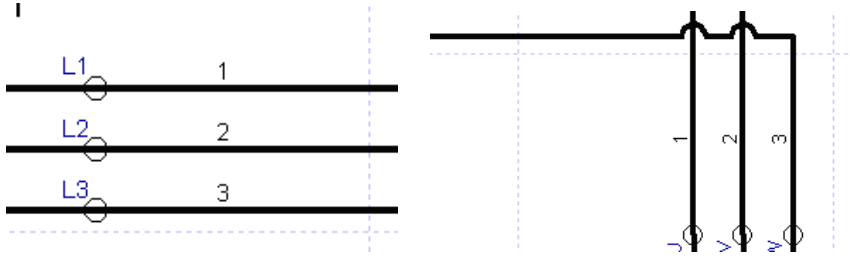


Figure 3-16 : Aligned Wire Number

3.4 Managing Essential Subdivisions and Locations

Subdivisions and locations are managed in the dialog box found under “Project Properties” → “Electrical System” → “Management”.

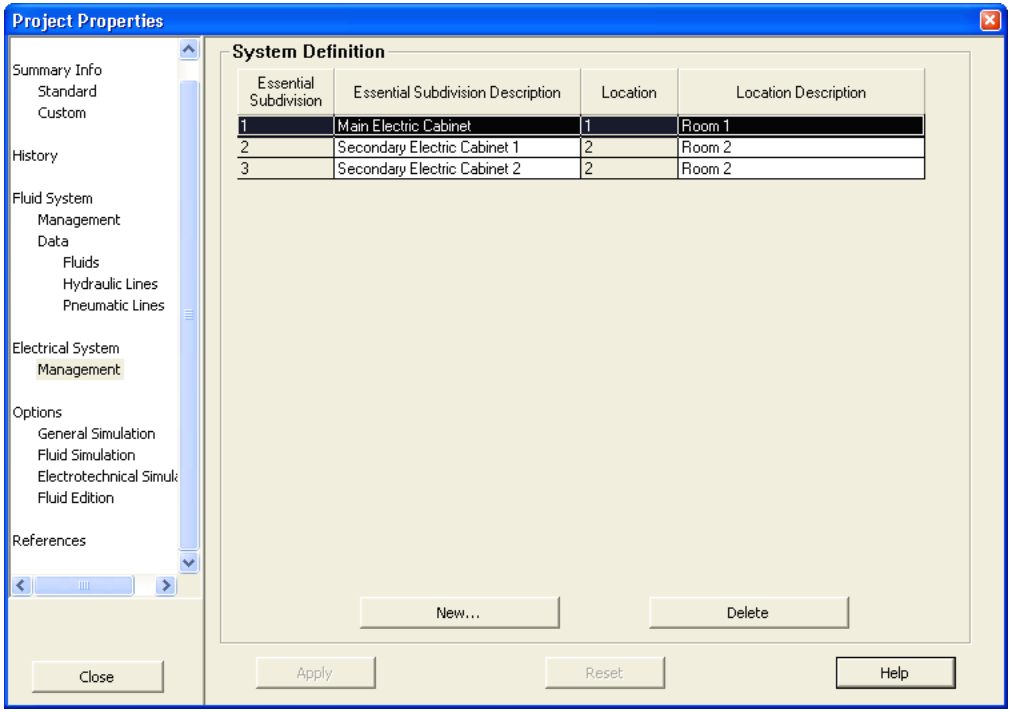


Figure 3-17: "Project Properties" → "Electrical System" → "Management" Dialog Box

Field	Description
Essential Subdivision	Assigns an identification to an essential subdivision. This subdivision number is used to identify a common group of components on the schematic.
Essential Subdivision Description	Describes an essential subdivision.
Location Number	Assigns an identification to a location. This number is normally used to identify where each circuits are physically located.
Location Description	Describes a location.

Field	Description
New…	Allows the creation of a new pair (Subdivision, Location), by adding a new line in the dialog box.
Delete	Allows the suppression of an existing pair (Subdivision, Location). This is possible only if the project has no component that belongs to this circuit.

☞ While names can be given to subdivisions and locations, it is preferable not to do so, the component variable names already use them.

(Essential Subdivision, Location) Pair Assignment

To assign a specific pair (subdivision and location) to a component, after you configured properly essential subdivisions in the "Project Properties" → "Electrical System" → "Management", there are different ways to proceed:

- When a component is inserted on a diagram, the selected subdivision in the "View" toolbar is automatically associated to the component;

- When a component is selected, it is possible to modify installation and circuit pair by choosing the desired pair in the "View" toolbar;



Figure 3-18: "View" Toolbar, (Installation, Circuit) Assignment

- When it is possible to configure a component, you can modify pair by "Component Properties" → "Builder" → "Identification" → "Essential Subdivision / Location", and directly choosing the desired pair.

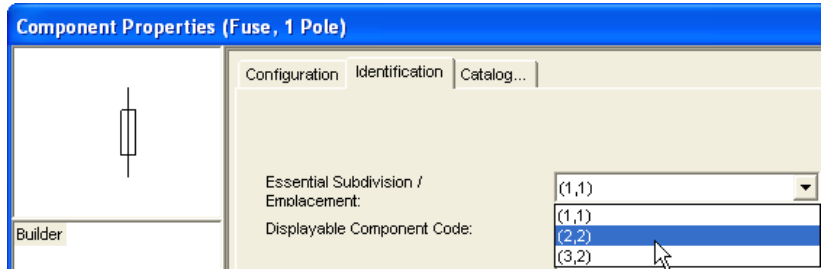


Figure 3-19: "Component Properties", (Subdivision, Location) Assignment

3.5 Management of Terminals

3.5.1 Inserting Terminals

Terminals are inserted on a diagram from the library as any other components. The insertion is done according to the following rules:

1. When a terminal is inserted on a wire, the wire is automatically cut at the insertion point. The two new wire sections carry the same number.

2. When a terminal is deleted, the two wire sections, which were connected to the terminal, are unified to form a single wire. The number of this wire is the same as the number of the two sections.

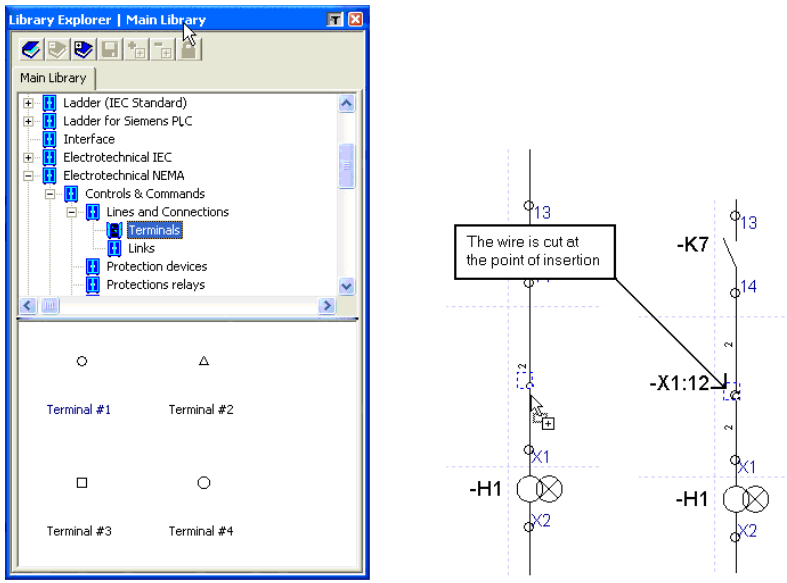


Figure 3-20: Inserting terminals

3.5.2 Terminal Numbering

The terminals are automatically numbered according to the following rules:

- Terminals are numbered progressively as they are created.

- Terminals not associated with a terminal strip are not automatically numbered. You can

manually modify the number of these terminals (Displayed Information branch of the component properties).

- Terminals are numbered sequentially by default.
- When using the IEC standard, terminals are numbered from left to right and from top to bottom independently of the direction of the map locators.
- When using NEMA standard, terminals are numbered from top to bottom and from left to right independently of the direction of the map locators.
- The terminal numbering method for the active document is defined in the document properties (see section 2.2 Edit Menu). They apply to all terminals of the document (existing and future).
- The terminal numbering method for the entire project is defined in the Terminal Numbering Method property of the Electrotechnical Editor branch of the Options command of the Tools menu.
- You can configure the software or the document to give the same name to terminals that are connected by clip jumpers.
- You can configure the software or the document to give terminals the same number as the wire which it is connected to.
- You can manually number a terminal in the displayed Information branch of its properties.
- You can lock the number of a terminal from its shortcut menu. Locked numbers are not modified when diagram terminals are automatically renumbered. Locked terminals are represented by an asterisk displayed to the right of its number.
- The “Renumber Terminals” command of the Electrotechnical menu allows you to renumber all the terminals of the active document which number is not locked. Renumbering is carried out according to the automatic numbering rules.



The numbering method for the active document is defined in the “document properties” (see Document Properties . . . in the section 2.2 Edit Menu). The numbering rules apply to all existing and future terminals.

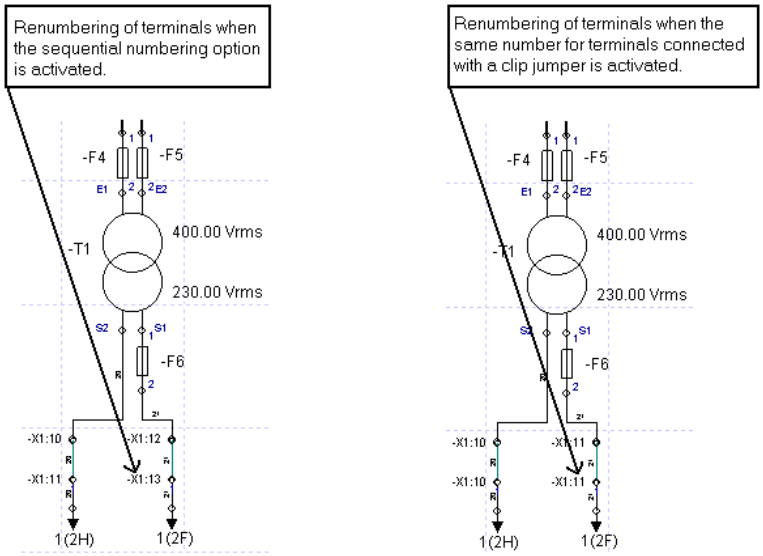


Figure 3-21: Terminal Numbering

3.5.3 Connecting Terminals With Clip Jumpers

The Electrotechnical workshop allows you to connect terminals of a terminal strip with a clip jumper.

- From the wire shortcut menu, you can define a wire as being a clip jumper between two terminals. It is not necessary that these terminals follow each other.
- By default, clip jumpers are of different color than wires so as to distinguish them (see section 2.8.2 Electrotechnical Editor Options).
- If the method Sequential - Same number for terminals connected by a jumper is selected, terminals linked by clip jumper have the same number.
- In the terminal connection diagram, a vertical line identifies clip jumpers. Black dots indicate connection points.

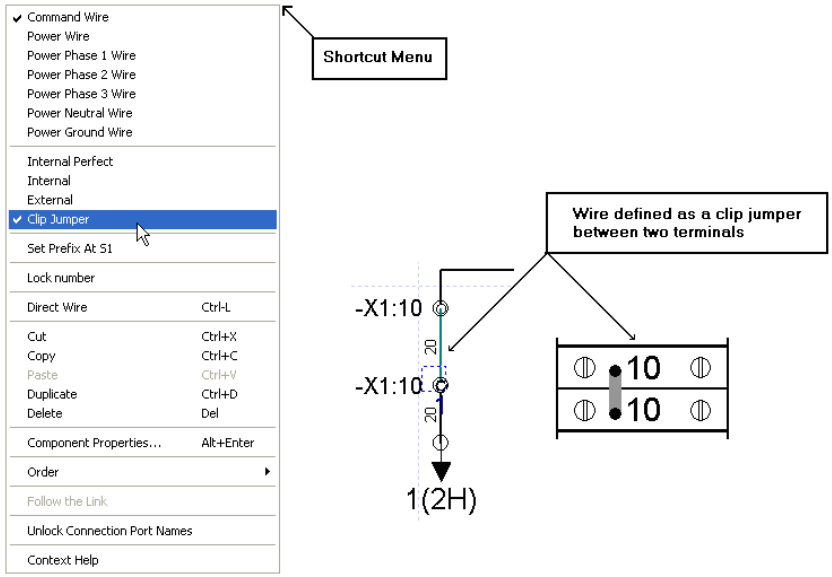


Figure 3-22: Terminals connected by clip jumpers

3.5.4 Terminal Properties

It is possible to access terminal properties just the same way as for any other component.

3.5.4.1 Associated Terminal Strip Branch

The Associated Terminal Strip branch of the terminal properties dialog box allows you to define a terminal as belonging to a terminal strip.

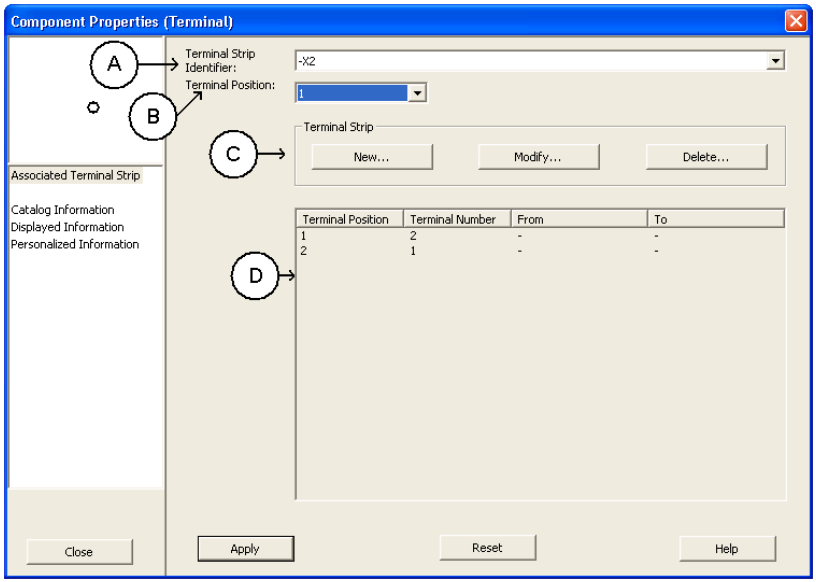


Figure 3-23: Terminal properties, Associated Terminal Strip branch

The properties of this dialog box are:

	Section	Description
A	Terminal Strip Identifier	Allows to select the desired terminal strip which the terminal should be associated with.
B	Terminal Position	Allows the selection of a free position of the terminal strip which the terminal should be associated with. Only free positions appear in the menu.
C	Terminal Strip	Gives access to the management of terminal strips. Allows the user to create, modify or delete a terminal strip. (see section 3.6 Management of Terminal Strips)

	Section	Description
D	Terminal association table	Displays the relevant information on the selected terminal strip.

To associate a terminal with a terminal strip:

1. Select the identifier of the terminal.
2. Select the number of positions.
3. Click on the Apply button.
4. Close the dialog box.

☞ A NEMA terminal is compatible with an IEC terminal strip and vice versa.

To dissociate a terminal from a terminal strip:

1. Select “None” as a terminal strip identifier.
2. Click on the Apply button.
3. Close the dialog box.

You can create, modify or delete a terminal strip from this dialog box, see section 3.6 Management of Terminal Strips.

The software manages the display of terminal and terminal strip identifiers as follows:

- The terminal strip identifier followed by the number of terminals is displayed near the terminal.
- When the terminal is linked to a terminal strip, the Essential Subdivision, Location, Displayable Component Code and Displayable Number properties are no longer modifiable by the user. They take the property values of the terminal strip.
- The Tag Name property takes the value of the terminal strip identifier.

- When a terminal is dissociated from a terminal strip (the link between the terminal and its terminal strip is deleted), the terminal strip identifier disappears from the terminal and the terminal strip position can be associated with another terminal. The "Essential Subdivision", Location, Displayable Component Code, Displayable Number, Tag Name and Comment show their initial values.

- Moving a terminal component within the same project does not affect its association with its terminal strip.

- When the terminal is moved into another project, its association with its terminal strip is lost.

- When a terminal is pasted into another project, it is dissociated if this other project does not contain a terminal strip with the same Tag Name. (The Tag Name displays the terminal strip identifier which the terminal is associated with).

- When a terminal is pasted (after having been copied) in the same project, there are various possible outcomes:

  a) There is a free position in the terminal strip from which the copied terminal is linked:

  The pasted terminal keeps its link with the terminal strip,

  The terminal takes the first free position of the terminal strip.

  b) There are no free positions in the terminal strip from which the copied terminal is linked to:

  The terminal is dissociated.

  The warning message "The terminal X has been dissociated from terminal strip Y, there are no more free positions" is displayed in the message window.

3.5.4.2 Displayed Information Branch

The Displayed Information branch of the terminal properties dialog box allows you to define and /or modify the terminal's displayed information. .

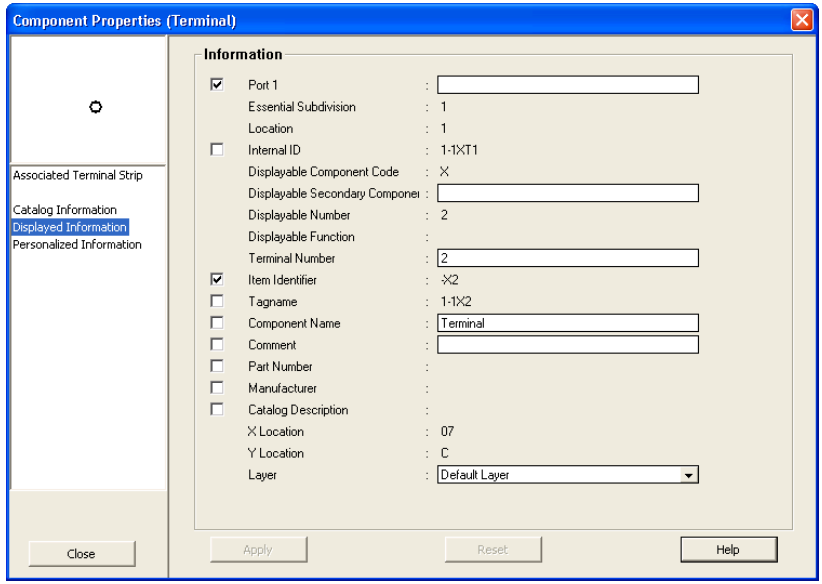


Figure 3-24: Terminal properties, Displayed Information branch

The displayed information options of a terminal are:

Properties	**Description**
Port 1	Allows the user to modify and display the name of the port.
Essential Subdivision	Allows the user to display the essential subdivision for the component identifier if the terminal is not associated with a terminal strip. The sequential component number is adjusted according to the number of similar components present with the same essential subdivision number.
Location	Allows the user to display the location for the identifier of the component if the terminal is not associated with a terminal strip. The sequential component number is adjusted according to the number of similar components present with the same location.

Properties	Description
Internal ID	Allows the user to display or not the internal identifier of the component.
Displayable Component Code	Allows the user to display and to modify the component code. The default code comes from the database.
Displayable Secondary Component	Allows the user to define a secondary designation for the component. This is used to display the identifiers according to the NEMA standard. (See section 2.2.3 Electrotechnical Diagram Branch, Part 1)
Displayable Number	Allows the user to modify the terminal number when it is not associated with a terminal strip.
Displayable Function	Allows the user to define a function for the component. This is used for the display of identifiers according to the IEC or NEMA standard (See section 2.2.3 Electrotechnical Diagram Branch, Part 1)
Terminal Number	Allows the user to display and to modify the terminal number in the terminal strip. A default number is assigned when the terminal is associated with the terminal strip.
Item Identifier	Allows the user to display the terminal identifier. This identifier changes when the terminal is associated with or dissociated from a terminal strip.
Tagname	Allows the user to display the terminal strip identifier which the terminal is associated with.

Properties	Description
Component Name	Allows the user to display and to modify the component name. The default name comes from the database.
Comment	Allows the user to display and to modify a comment concerning the component.
Part Number	Allows the user to display the part number from the catalog information.
Manufacturer	Allows the user to display the manufacturer from the catalog information.
Catalog Description	Allows the user to display the description of the component from the catalog information.
X Location	Information field. Displays the X location of the component.
Y Location	Information field. Displays the Y location of the component.
Layer	Allows the user to select the layer that the component belongs to.

3.5.5 Terminal Shortcut Menu

The terminal shortcut menu is accessible in the editor by clicking on the right button of the mouse when a terminal is selected. The following figure presents the terminal shortcut menu. .

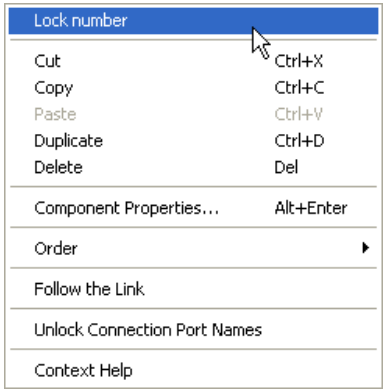


Figure 3-25: Terminal shortcut menu

The Lock Number command of the terminal shortcut menu in the electrotechnical workshop allows the user to lock the terminal number. A locked number is not modified during an automatic renumbering of the terminals.

3.6 Management of Terminal Strips

3.6.1 Creating a Terminal Strip

To create a terminal strip:

1. Open the New Terminal Strip dialog box by choosing Electrotechnical → Terminal → New or by clicking on the New button of the Associated Terminal Strip branch in the terminal properties dialog box.

The New Terminal Strip dialog box is displayed on the screen.

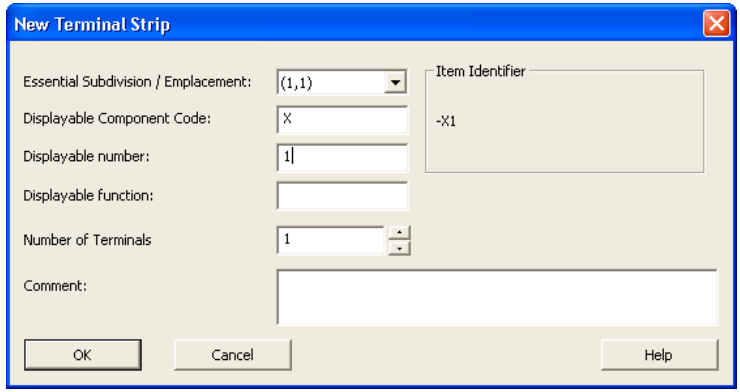


Figure 3-26: Creating a terminal strip

2. If required, modify the default information of the new terminal strip.

3. Enter the number of terminals of the terminal strip.

4. Edit a comment regarding the terminal strip if desired.

5. Click on the OK button to create the terminal strip.

You can now select this terminal strip to associate terminals with its connections from the terminal properties dialog box (see Associated Cable Branch in section 3.3.5 Wire Properties).

3.6.2 Modifying a Terminal Strip

To modify a terminal strip:

1. Open the Modify Terminal Strip dialog box by choosing Electrotechnical →Terminal → Modify or by clicking on the Modify button of the Associated Terminal Strip branch of the terminal properties dialog box.

2. Select the identifier of the terminal strip to be modified from the drop-down menu.

3. Modify the information of the terminal strip as needed.

4. Modify the number of terminals of the terminal strip if required.

5. Modify the comment entered for the terminal strip if desired.

6. Click on the OK button to accept the modifications.

The modified information is updated on all the diagrams in which the terminal strip is mentioned.

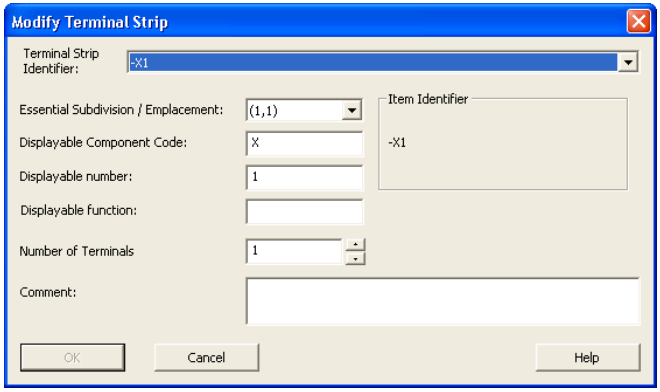


Figure 3-27: Modifying a terminal strip

3.6.3 Deleting a Terminal Strip

To delete a terminal strip:

1. Open the Delete Terminal Strip dialog box by choosing Electrotechnical → Terminal → Delete or by clicking on the Delete button of the Associated Terminal Strip branch in the terminal properties dialog box.

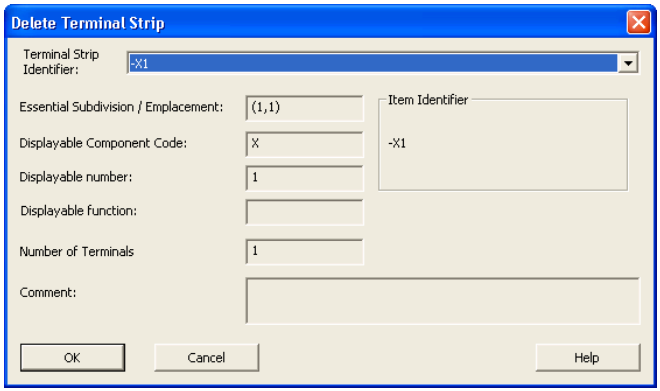


Figure 3-28: Deleting a terminal strip

2. From the drop-down menu, select the identifier of the terminal strip to be deleted.

3. Click on the OK button to confirm the deletion of the terminal strip.

The terminals that were associated with the terminal strip are freed. The content of the terminal strip diagram (if it exists) is deleted. The terminal information is updated on all diagrams in which this terminal strip was mentioned.

3.6.4 Creating a Terminal Strip Connection Diagram

To create a terminal strip connection diagram for an existing terminal strip: :

1. Choose File → New → Electrotechnical → Terminal Strip Diagram or from the project explorer contextual menu, choose New → Electrotechnical → Terminal Strip Diagram.

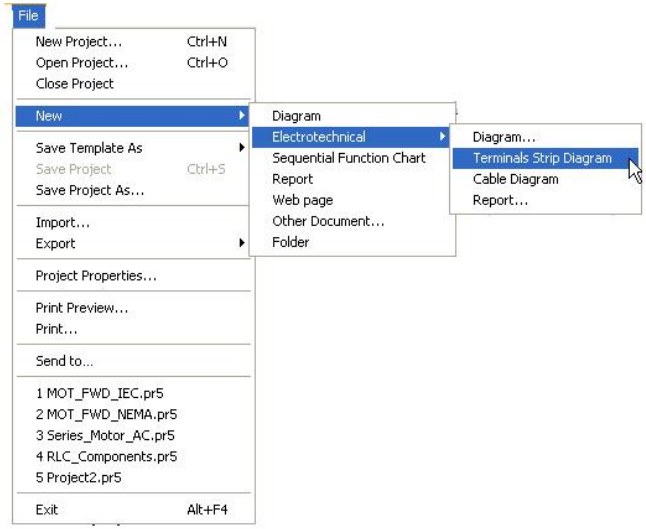


Figure 3-29: Creating a terminal strip connection diagram

The document properties dialog box opens on the screen.

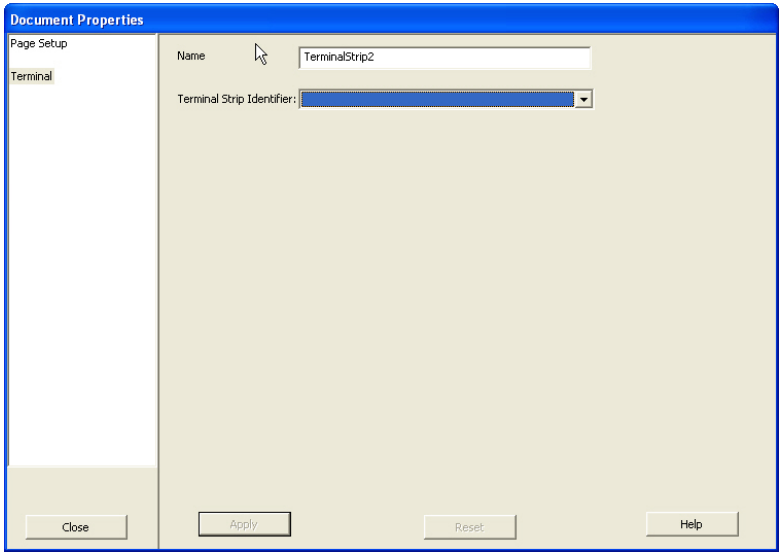


Figure 3-30: Dialog box, creating a terminal strip connection diagram

2. Modify the name of the diagram if desired. The name of the diagram can be modified later in the document properties dialog box.

3. Select the terminal strip identifier from the drop-down menu.

4. Click on the Apply button to validate the selection.

5. Click on the Close button to close the dialog box and to create the terminal strip connection diagram desired.

The terminal strip connection diagram is created and displayed.

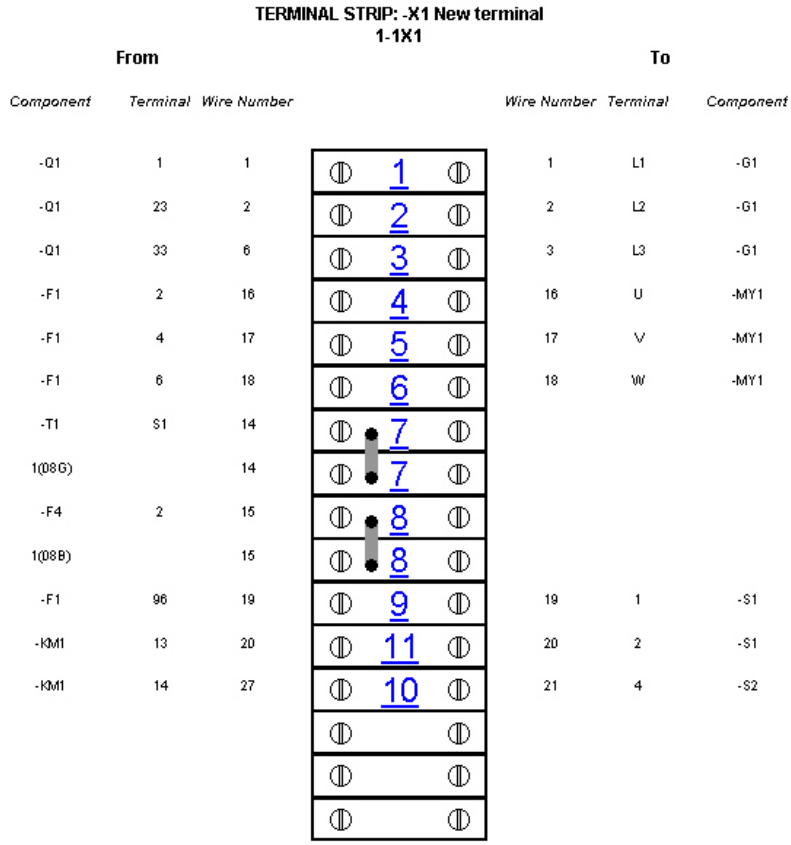


Figure 3-31: Terminal strip connection diagram

The information contained on the diagram is automatically updated when the terminal strip or its properties are modified. The internal wires are the internal connections to terminals and the external wires are the external connections to terminals.

For each wire on each terminal, the software displays the wire number, the component terminal and the component.

At the center of the diagram, the software displays the number of each terminal and the clip jumpers connecting various terminals. The information comes from the wires properties. Only terminals that are in use are detailed

The deletion of a terminal strip connection diagram has no effect on the terminals in the diagrams.

4 Cable Management

This section provides the knowledge and procedures for:

- Cable creation;
- Creating wiring drawings;
- Modify an existing cable.

4.1 Automation Studio™ Cables

4.1.1 Cables

A cable is an assembly of one or more conductors normally surrounded by a protective a sheath.

4.1.2 Cables in Automation Studio™

Attaching wires to a specific cable.

Cables are not shown on a drawing. Each wire is assigned to a cable and the cable number is displayed on the wire. An independent wiring drawing is available.

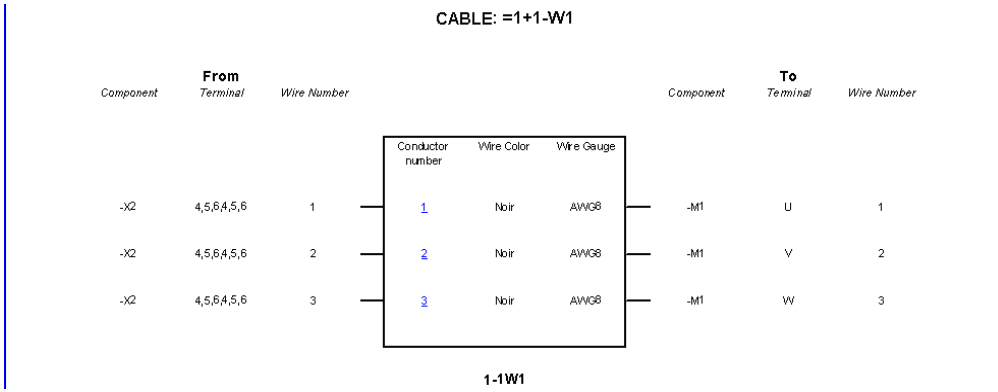


Figure 4-1 : Cable Drawing

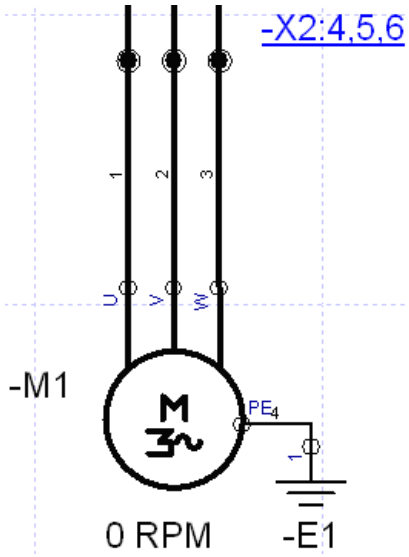


Figure 4-2 : Cable

4.1.3 Cables Drawing

Cables are not displayed on electrotechnical diagrams. The wires that are parts of a specific cable are tagged with the cable ID.

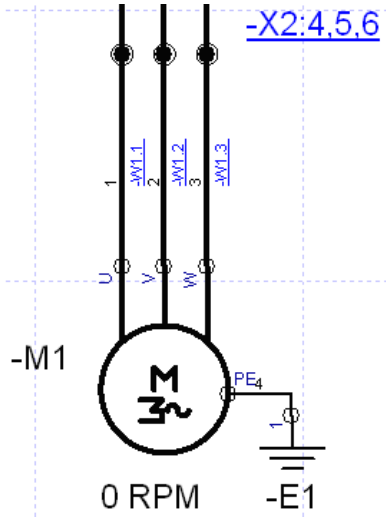


Figure 4-3 : Cable Wire Identification

Example:

-W1.1

Where: -W1 : Cable designation

1 : Wire designation in the cable

4.1.4 Wiring Diagram

A wiring diagram is the collated data of a specific cable.

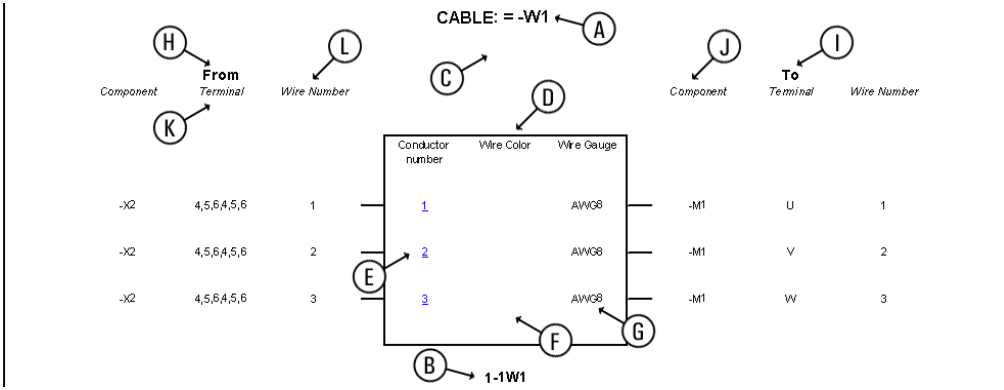


Figure 4-4 Wiring Drawing

Zone	Name	Description
A	Cable identification	The cable builder assigns the identification.
B	Displayed identification	The cable builder assigns the identification. Refer to the description after this table.
C	Comment	The cable builder assigns the comment.
D	Cable representation	The cable is represented by a rectangle. Each wire is displayed on both sides of the rectangle.
E	Conductor number	A cable wire number is assigned through the component.

Zone	Name	Description
		properties dialog box.
F	Wire color	Wire color is assigned using the component properties dialog box.
G	Wire size	Wire size defined in the AWG standard. It is assigned using the component properties dialog box.
H	From	The internal component to which the cable is attached.
I	To	The external component to which the cable is attached.
J	Component	Component found at the end of the cable.
K	Terminal	Component terminal to which the cable is attached.
L	Wire number	Wire number

The cable identification complies with this format:

=	Essential subdivision	+	Location	-	Designation	Number	Function

If the essential subdivision has not been entered, the = sign is not shown.

If the location has not been entered, the + sign is not shown.

4.2 Operation with Cables

Cable management is composed of:

- Creation of a virtual cable; this cable is not displayed on any drawing but has an ID and properties;
- Assigning wires to a cable;
- Cable modification;
- Cable elimination;
- Wiring diagram creation.

4.2.1 Cable Creation

To create a cable:

1. Open the New Cable dialog box by choosing “Electrotechnical” → “Cable” → “New”;

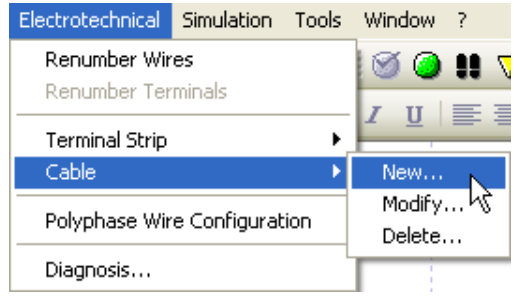


Figure 4-5 : Cable Creation Menu

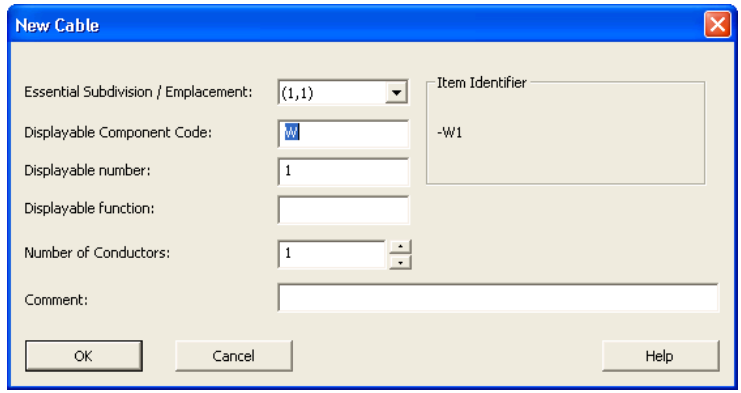


Figure 4-6 : New Cable Dialog Box

2. If needed modify the default entries;

   Make sure to provide the proper amount of conductor for the cable;

3. If needed, enter the displayable function and comment;

4. Click on the OK button to finish creating the new cable.

4.2.2 Assigning Wires to a Cable

To link a conductor to a cable:

1. Select the wire;

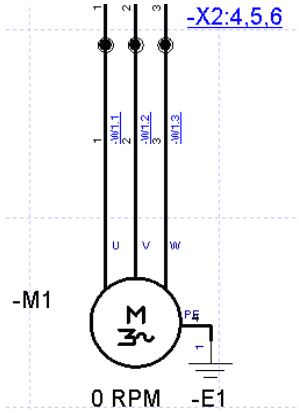


Figure 4-7 : Example - Motor Circuit with Terminal

2. Open the component properties dialog box for the wire;

3. Select the “Associated Cable” function;

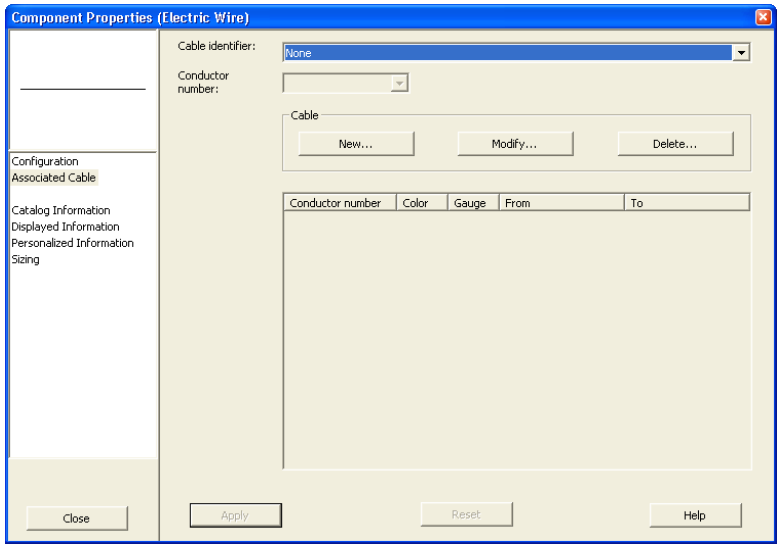


Figure 4-8 : Component Properties (Electric Wire)

4. Select the desired cable from the “Cable identifier” drop-down menu;

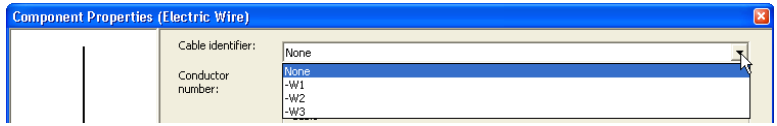


Figure 4-9 : Drop-Down Menu for Cable Identifier

5. Select a wire number in the cable from the "Conductor number" drop-down menu;

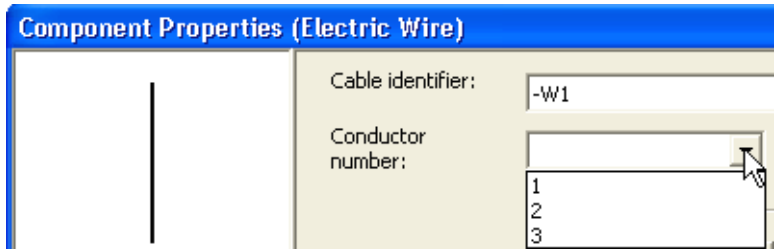


Figure 4-10 : Drop-Down Menu for Conductor Number

6. Click on " Apply ";

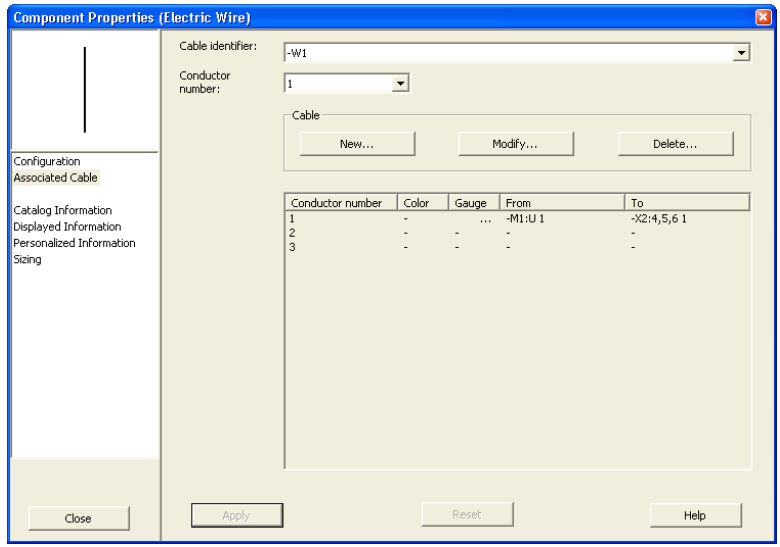


Figure 4-11 : Component Properties (Electric wire)

7. Click on "Close".

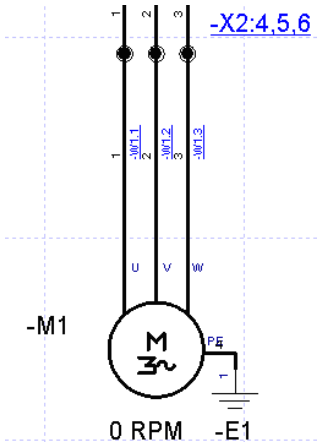


Figure 4-12 : Motor Circuit with Terminal

The conductor is now part of a cable and is identified accordingly. The identification tag is also a hyperlink to the associated wiring diagram if the tag was created.

4.2.3 Cable Modification

Cable properties can be modified as required. For example, the number of conductors the cable may contain can be increased or decreased.

To modify the cable properties or identification:

1. Open the "Cable Modify" dialog box, by clicking on the menu "Electrotechnical" → "Cable" → "Modify";

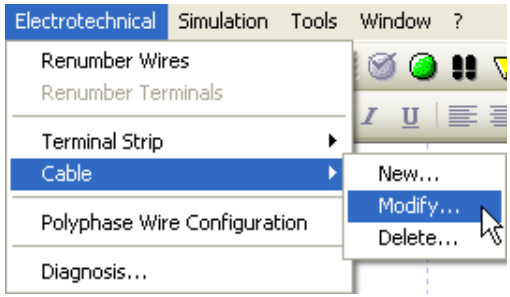


Figure 4-13 : Cable Modification Menu

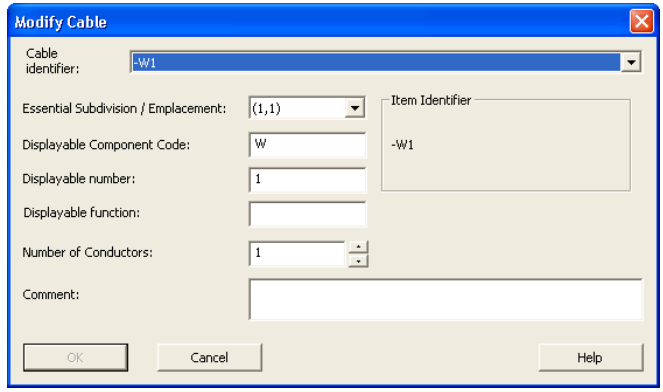


Figure 4-14 : Cable Modification Dialog Box

2. Select the required cable from the drop-down menu “Cable identifier”;

3. Adjust the current information ;

4. Click on “OK”.

4.2.4 Cable Elimination

When eliminating a cable, all wire associations to this cable are also eliminated.

To eliminate a cable:

1. Open the “Modify Cable” dialog box by clicking on the menu “Electrotechnical” → “Cable” → “Delete”;

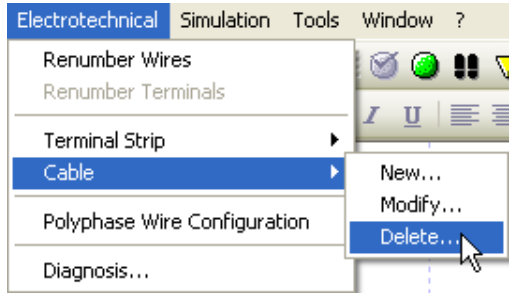


Figure 4-15 : Cable Deletion Menu

2. Select the cable to be eliminated from the “cable identifier” drop-down menu;

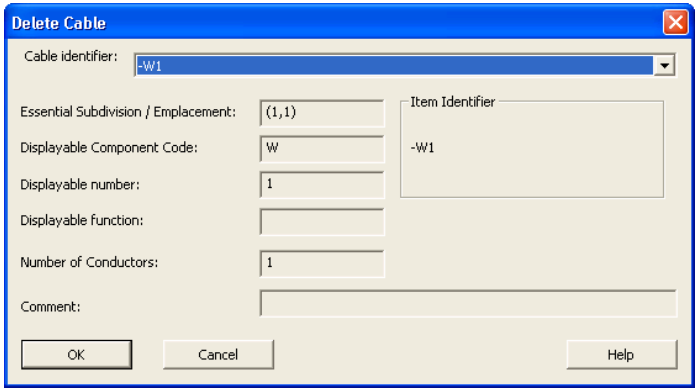


Figure 4-16 : Cable Deletion Dialog Box

3. Click on “OK”.

4.2.5 Wiring Diagram Creation

A wiring diagram can only be created from an existing cable.

To create a cable:

1. Create a new wiring diagram from the menu “File” → “New” → “Electrotechnical” → “Cable Diagram”;

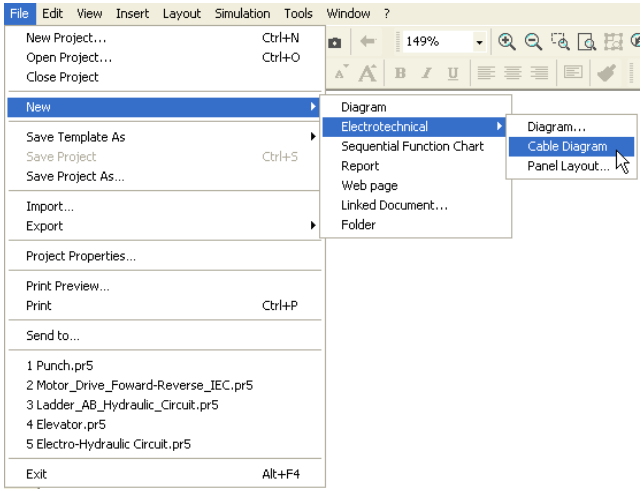


Figure 4-17 : Wiring Diagram Creation Menu

A new "Document Properties" dialog box is created and displayed. It contains a new wiring diagram.

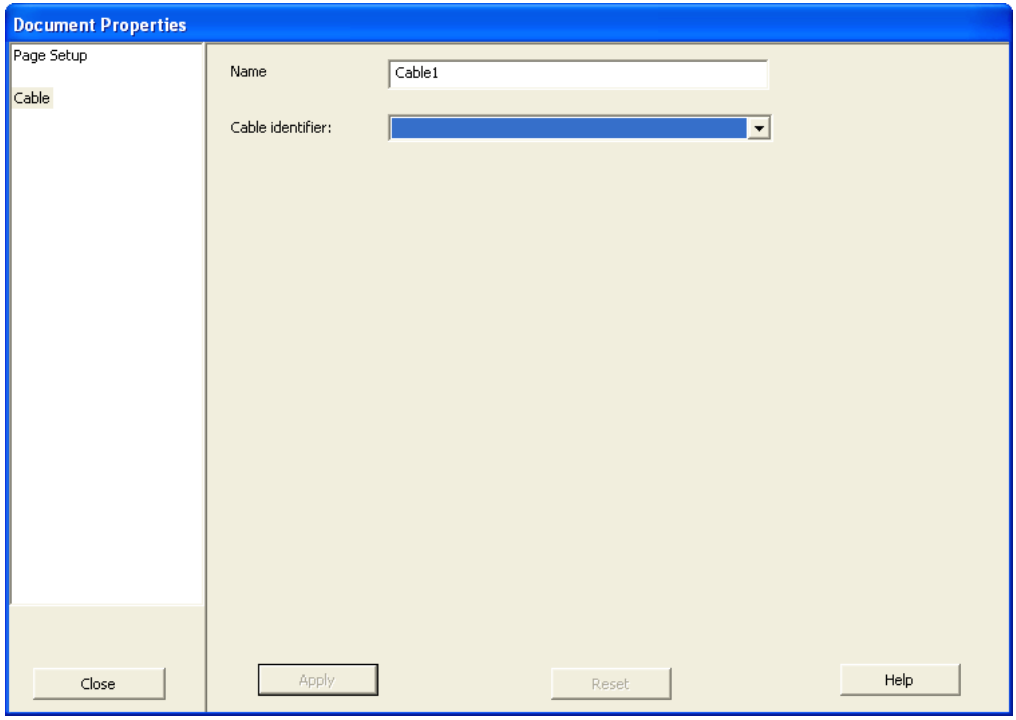


Figure 4-18 : New Cable Diagram Creation Dialog Box

2. Select the cable from the "Cable Identifier" drop-down menu;

3. Click on the "Apply" button to confirm your selection;

4. Click on the "Closed" to close the "Document Properties" dialog box and create the wiring diagram.

The wiring diagram is created and displayed. The wiring diagram information is automatically updated when the cable or its properties are modified.

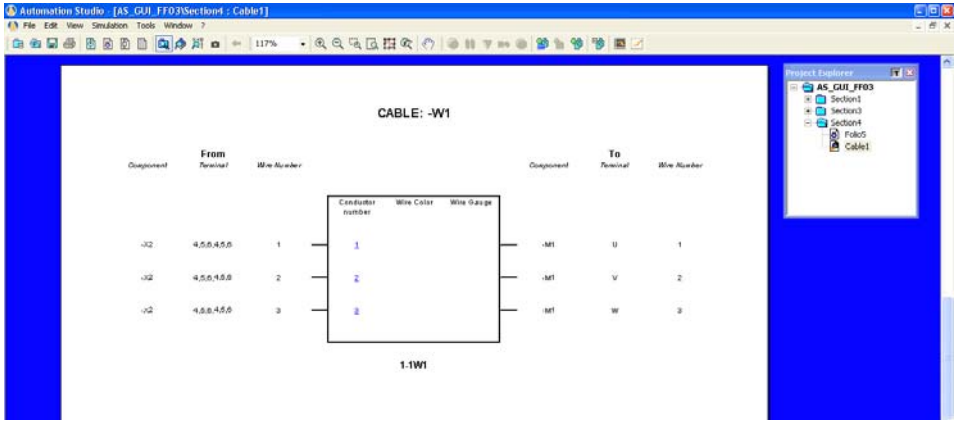


Figure 4-19 : Wiring Diagram

5 Common Tabs: Displayed Information & Catalog

This chapter explains how to identify the electrotechnical components and to display their parameters on the diagram.

5.1 Catalog Information

The catalog information dialog boxes describe the "Catalog" properties attached to every component. Then, these information can be used to automatically generate several types of reports such as bills of materials, purchase lists, etc..

The catalog information is accessible through the "Component Properties" → "Catalog Information" dialog box.

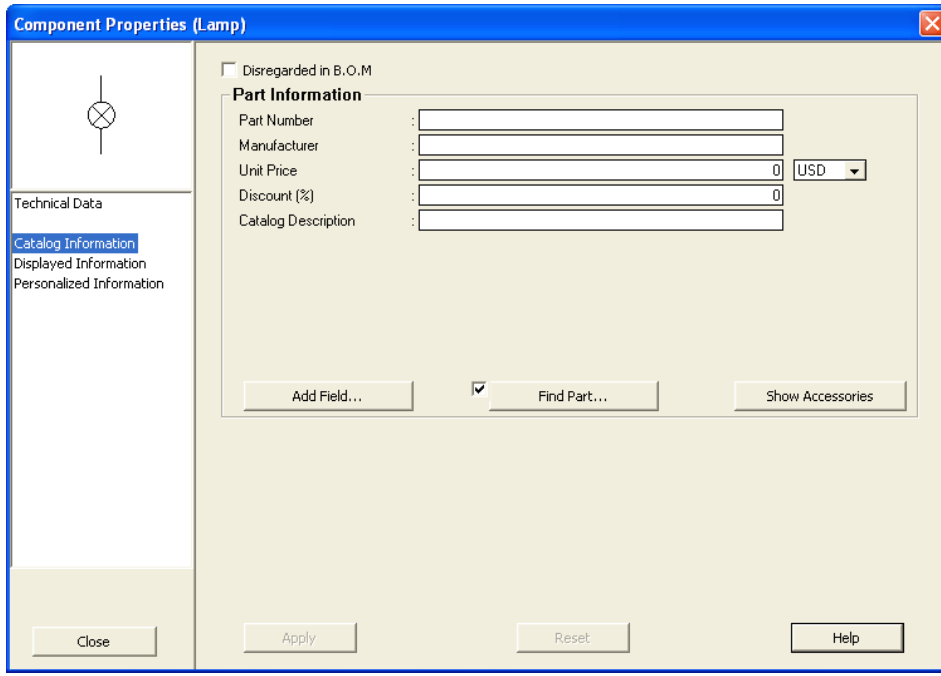


Figure 5-1: Catalog Information Dialog Box for Configured Components

Refer to the *Automation Studio™ User's Guide* for more information.

5.2 Displayed Information

The displayed information dialog box permits the enabling or disabling of the component parameters display on the diagram. A check box, located near the parameter lines, permits this selection.

The displayable information of the electrotechnical components that are not using a builder is viewed using the "Component Properties" → "Displayed Information" dialog box.

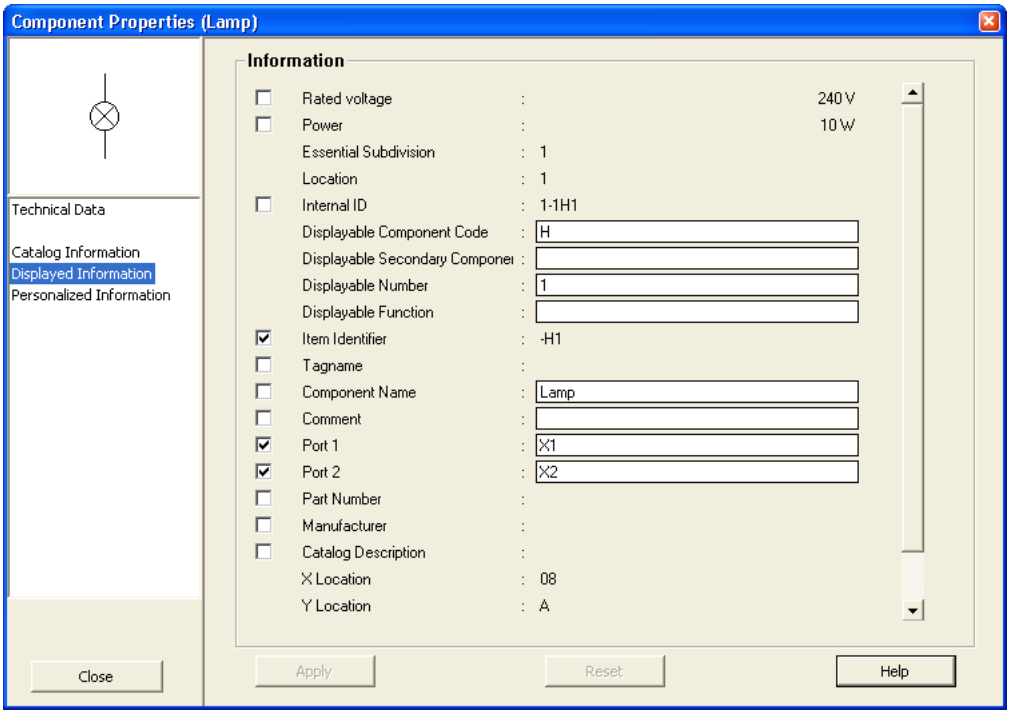


Figure 5-2: Component Properties – Displayed Information

Refer to the *Automation Studio™ User's Guide* for more information.

6 Black Box

A black box is a configurable component to be used on an electrotechnical diagram. This component allows the creation of complex components that can be used in Automation Studio™. The user can configure the component's appearance, connection ports and behavior.

The black box internal circuitry is built using specialized components available in the electrotechnical library, in both IEC and NEMA formats. The programming required to assign a specific behavior to a black box must be done in SFC. HMI components can also be included in the black box as well as basic display elements, which can be used to design a front representation.

The examples in this chapter are built using the IEC standard. When using the NEMA standard, the design techniques and tools are the same, only the symbols vary.

Here is an example of a simple black box representing a "Cigar Lighter".

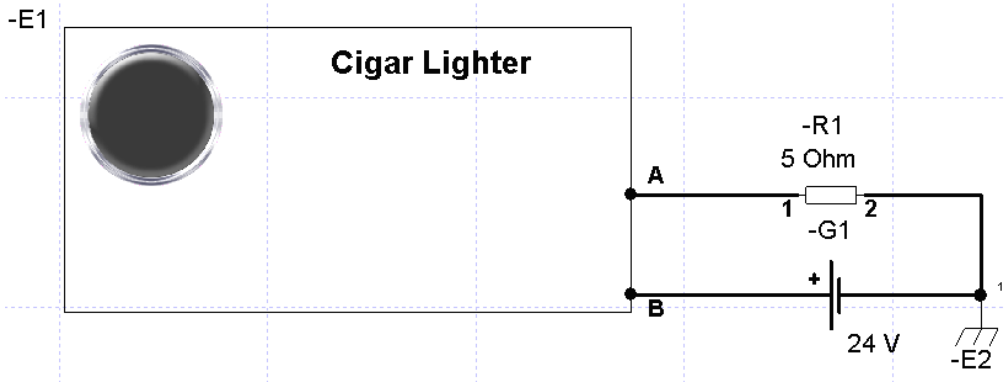


Figure 6-1: Black Box Example – Cigar Lighter

6.1 Specialized Components

Black box specialized components must be used to build the circuitry inside a black box. All other electrotechnical components which are not part of the black box branch are not compatible.

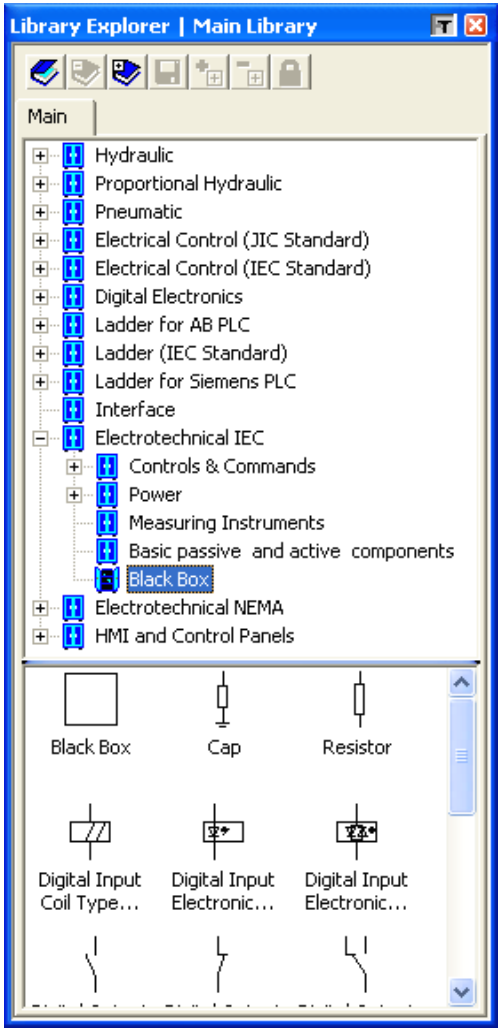


Figure 6-2: Black Box Library

The black box specialized components are made especially for extensive variable assignments, which provide the flexibility needed when creating the SFC programming. The specialized components can be categorized as follows: black box, input component, output component, and the grounding electrode ("Cap").

6.1.1 Black Box Frame

The black box frame is the component containing all general data and components which determine what the black box behavior will be.

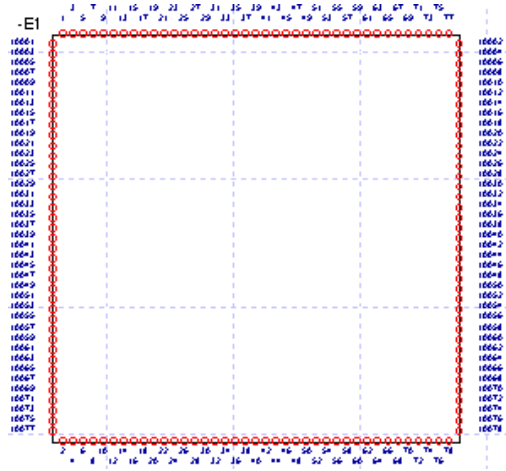


Figure 6-3: Black box - Initial Format

From the black box component properties dialog box, you can:

- Edit the default component characteristics;
- Edit the default port name;
- Edit the default catalog information.

The black box frame can be resized by first selecting it and then using the sizing handle to resize it to the desired size. The number of ports will be adjusted to the container size. It is recommended to first size the black box properly before adding any component. The connection ports can be hidden by un-checking the option “Connection ports” from the “View” menu in the menu bar.

All components, display elements and lines required to generate the black box must be inside the container. The specialized component ports must be directly connected to one of the black box ports.

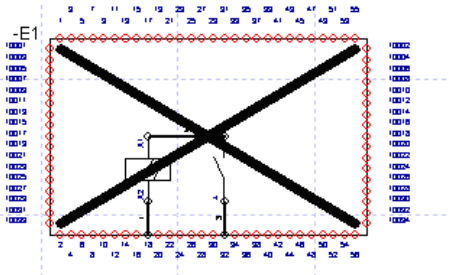


Figure 6-4: Bad Design

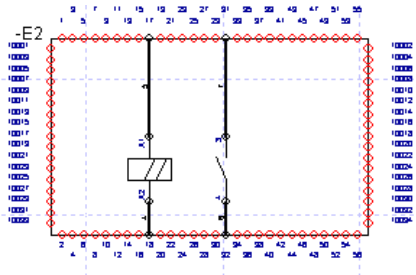


Figure 6-5: Good Design

6.1.2 Input Specialized Components

All input components must be linked to global variables used in a SFC so as to assign a behavior to the black box. A component is defined as an input according to this global variable. If a component needs to receive an input from the SFC to act, it is an input component. The input components are found in the black box library:

- Digital Output Make Contact Type AC-DC;
- Digital Output Break Contact Type AC-DC;
- Digital Output Change-over Contact Type AC-DC;
- Digital Output Electronic Type DC NPN;
- Digital Output Electronic Type DC PNP;
- Digital Output Electronic Type AC;
- Analog Voltage Output DC;
- Analog Current Output DC.

For example, the “Analog Voltage Output DC” component outputs a voltage only if it receives (an input) a control commands from the SFC.

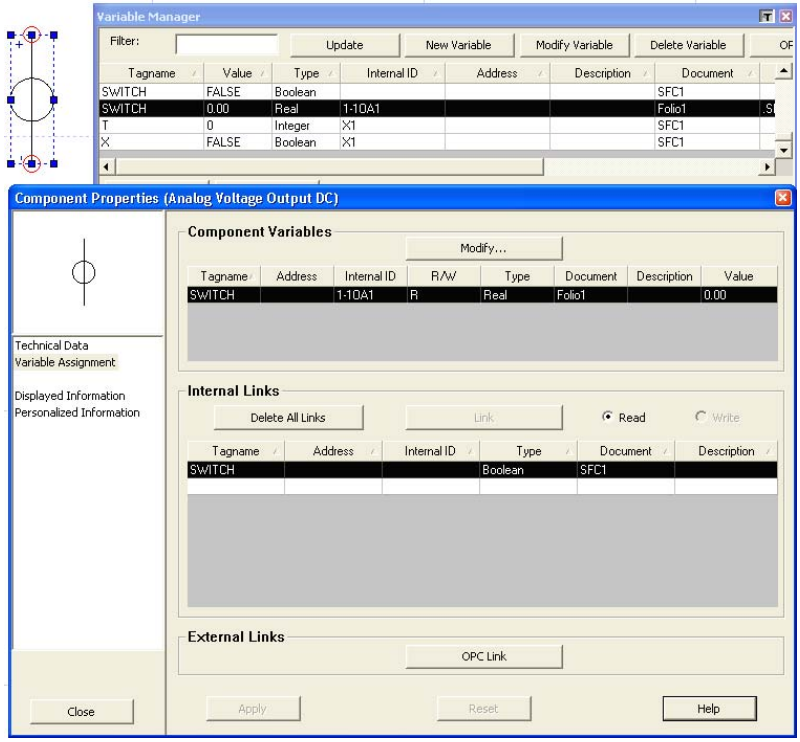


Figure 6-6: Input Component Linked to an SFC Variable

6.1.3 Output Specialized Component

When placing an output component onto the diagram, a variable name is required. This variable is output to the SFC, which uses it to know the circuit status. The components are:

- Digital Input Coil Type AC-DC;
- Digital Input Electronic Type DC;
- Digital Input Electronic Type AC;
- Analog Voltage Input Analog DC;
- Analog Current Input DC;
- Analog Voltage Input AC;
- Analog Current Input AC.

For example, an "Analog Voltage Input AC" can simulate an AC voltmeter which outputs a variable to the SFC, providing the circuit voltage.

6.1.4 Grounding Electrode and Resistor

The “Cap” and the resistor are neither input nor output components; they can not be linked to a variable. They are used to connect ports that are not used by the SFC.

6.2 Black Box Generation Dialog Box

To generate a black box component, you first have to select all the black box components, and then right-click on the mouse. From the pop-up menu that opens, select the option “Generate Black Box”.

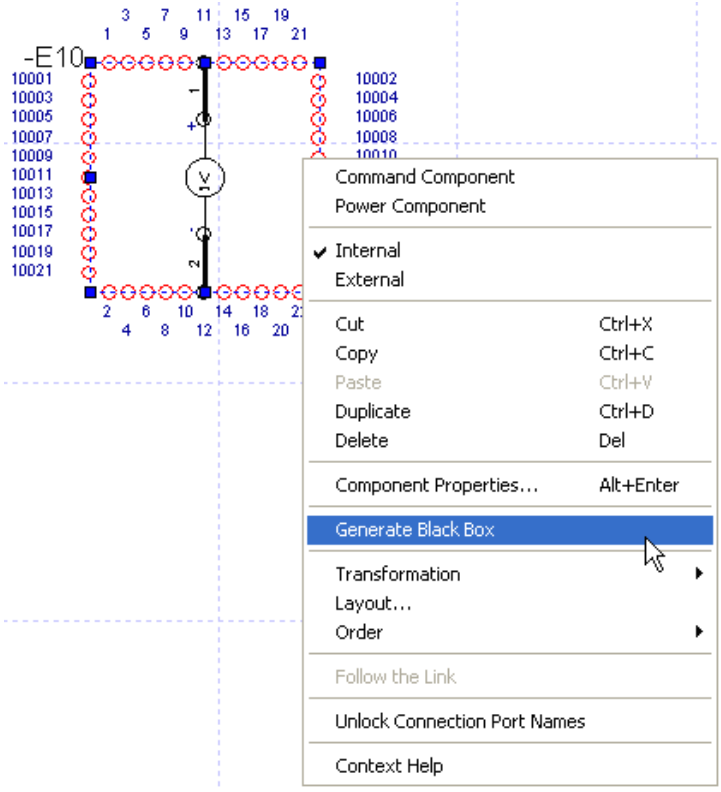


Figure 6-7: “Generate Black Box” Option

A dialog box which offers a selection of the logic source documents required for the black box opens.

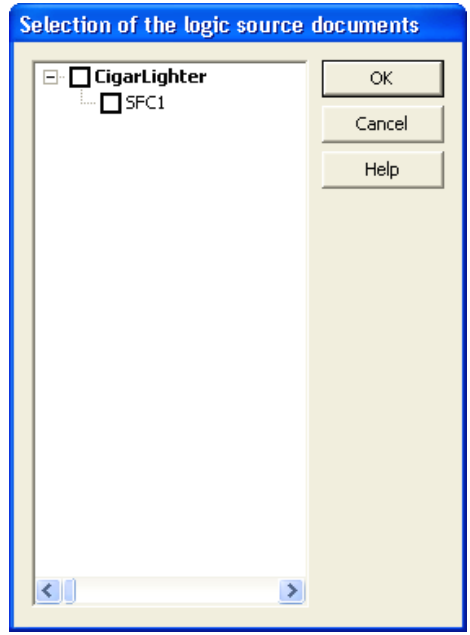


Figure 6-8: Selection of the Logic Source Documents

After selecting the logic source documents and confirming your selection, a new dialog box opens to offer the options for black box generation.

☞ The SFC diagrams used as the logic sources for the black box must be localized in the root of the "Project Explorer". It should not be located in a folder.

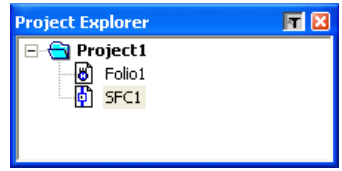


Figure 6-9: SFC diagram in the "Project Explorer" root location

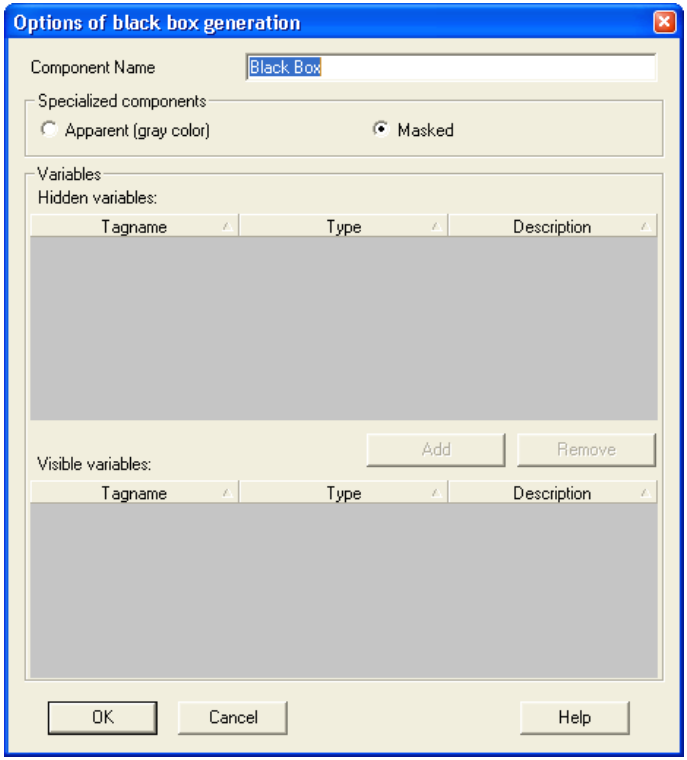


Figure 6-10: Option of Black Box Generation

The information found in this dialog box is:

Field	Description
Component Name	Field in which you may enter the component name. This name appears with the simplified component diagram in the personalized library.
Specialized Components	Area in which you may select whether or not the component and connection lines are visible. Masked: the specialized components and their connection lines are not visible. Apparent; (gray color) The specialized components and their

Field	Description
	connection lines visible are either gray or red. The component is red during simulation and when the component is activated.
Hidden Variables	Lists all black box variables available to the SFC, but which will not be available in the project's Variable Manager where the black box is used.
Visible Variables	Lists all black box variables available to the SFC, and which are available in the project's "Variable Manager" where the black box is used.
"Add" Button	Moves a variable from the hidden variables list to the visible variables list.
"Remove" Button	Moves a variable from the visible variables list to the hidden variables.

6.3 Creating a Black Box Component

In this section, we will create and generate the "Cigar Lighter", which includes the creation of the associated SFC.

In our example, a cigar lighter is a circuit made of a push button which activates a switch for a 10s-period. The switch enables power to an external heating element.

6.3.1 Creating a Black Box Container

It is recommended, but not compulsory, to use layers in a black box drawing. This will simplify the editing operation. The three recommended layers are layer 1 for the container, layer 2 for the specialized components and layer 3 for the graphic elements. Only the layer being edited needs to be available.

To create the "Cigar Lighter" black box:

1. Create an electrotechnical diagram;

2. Insert a black box container which can be found in the library under "Electrotechnical IEC or NEMA" → "Black Box";

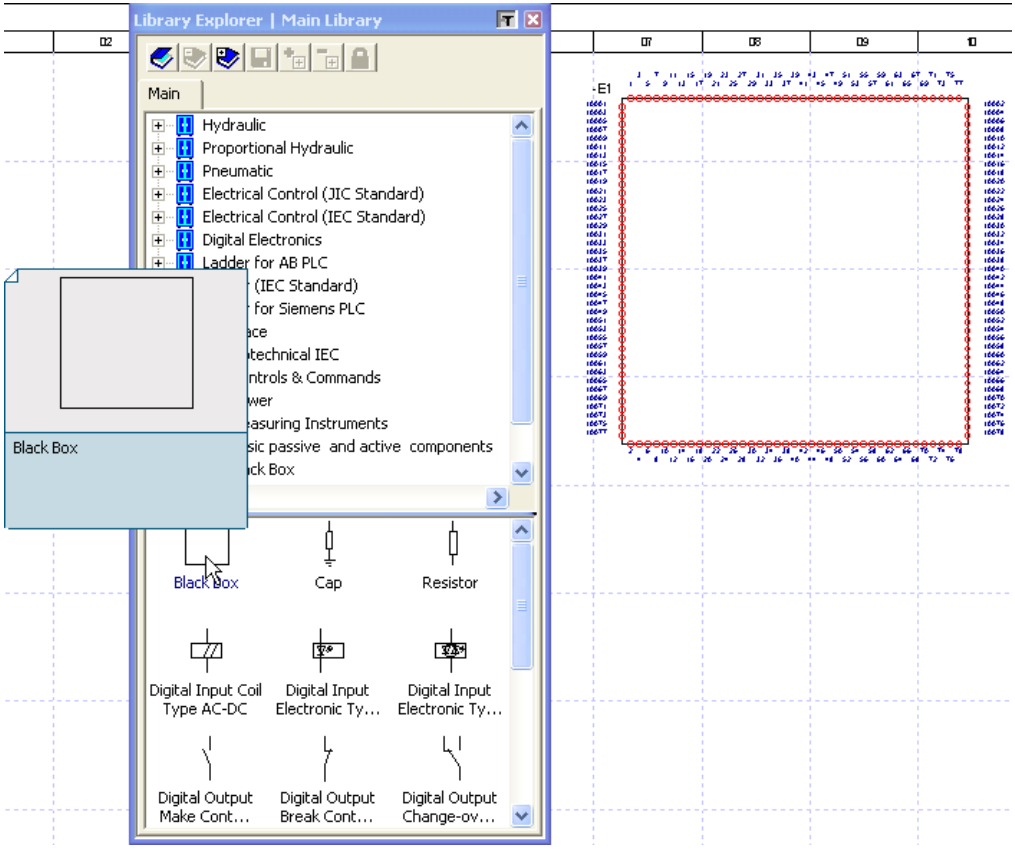


Figure 6-11: Black Box Container Insertion

3. Insert a digital output make contact AC-DC switch in the black box container;

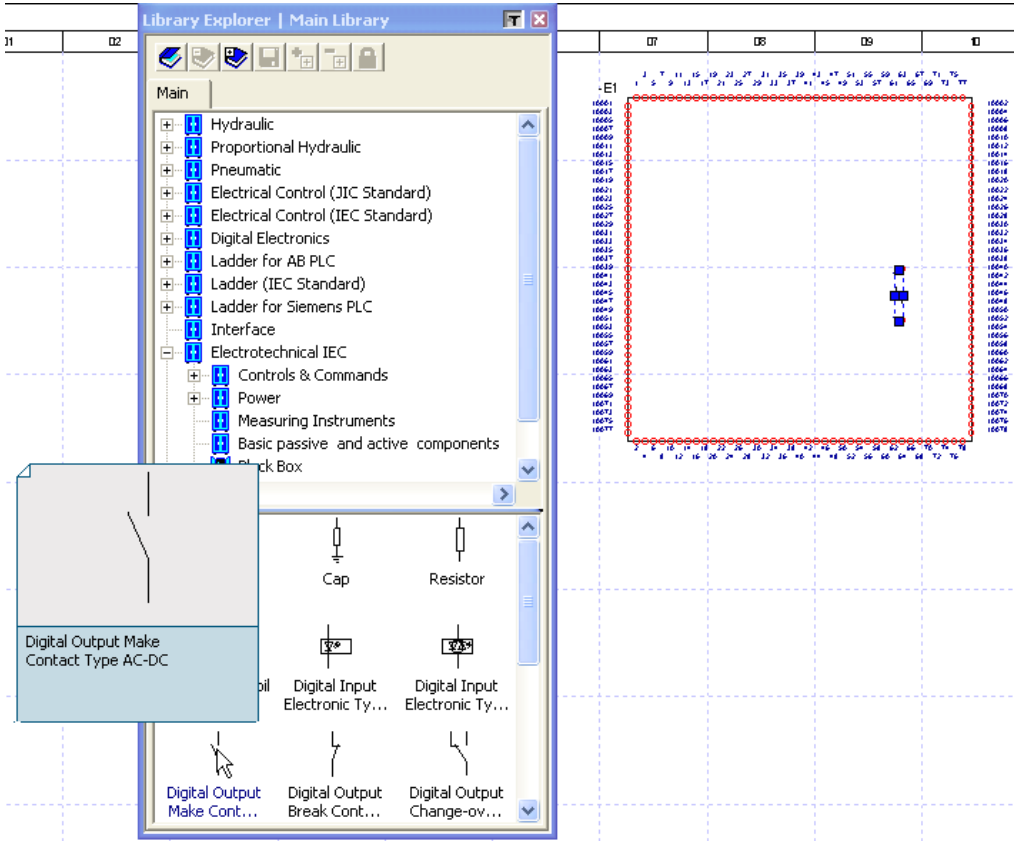


Figure 6-12: DO Make Contact Insertion

4. Insert a push-button HMI component from the main library "HMI and Control Panels" → "Controls & Commands" → "Buttons" branch and type **Pushbutton** as the "Tag name" in the pop-up dialog box.

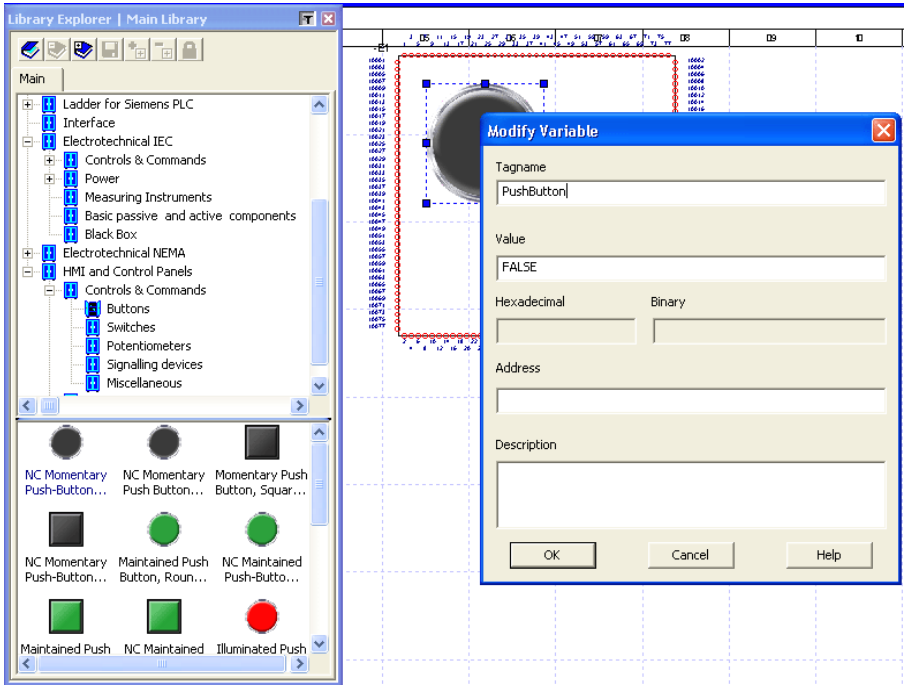


Figure 6-13: Push-Button Insertion

5. Change the push button characteristics and size to match the one found on the next drawing;

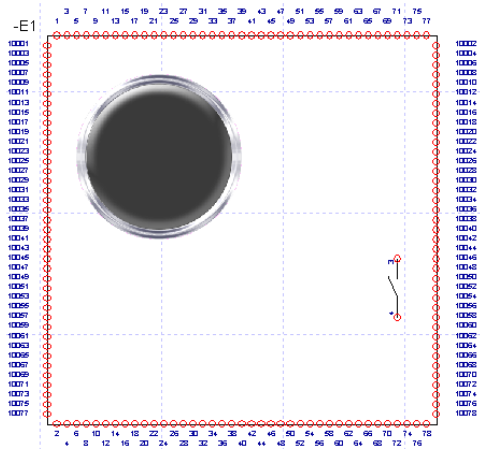


Figure 6-14: Push-Button Configuration

6. Using the graphic element tools, draw a simple identification block, as shown in the following figure;

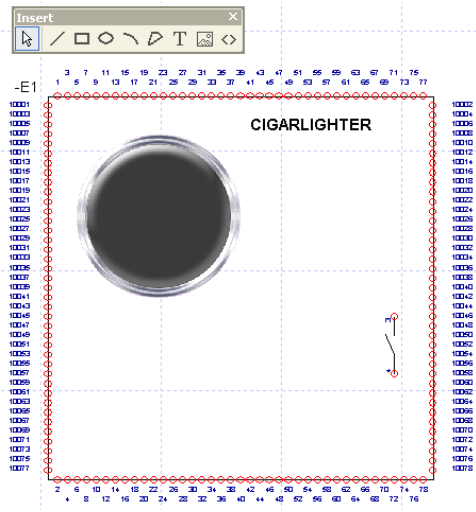


Figure 6-15: Adding Basic Graphic Elements

7. Resize the black box container so as to make it just large enough to contain all the specialized and HMI components;

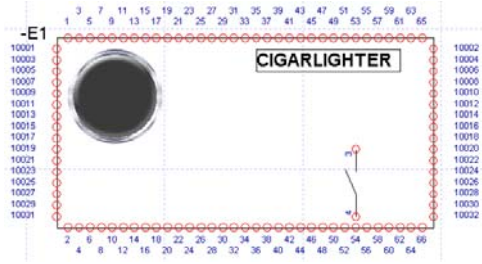


Figure 6-16: Resized Black Box

8. Link the contact connection ports to the black box ports;

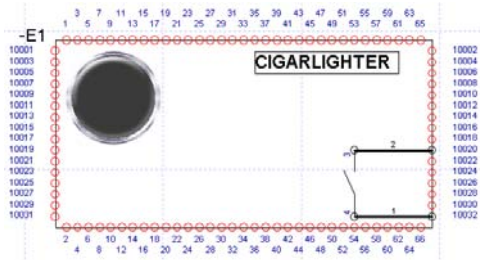


Figure 6-17: Complete Black Box Circuitry

9. Rename the black box ports connected to the switch using the "Component Properties (Black Box) " → "Displayed Information" dialog box, to Port A and Port B;

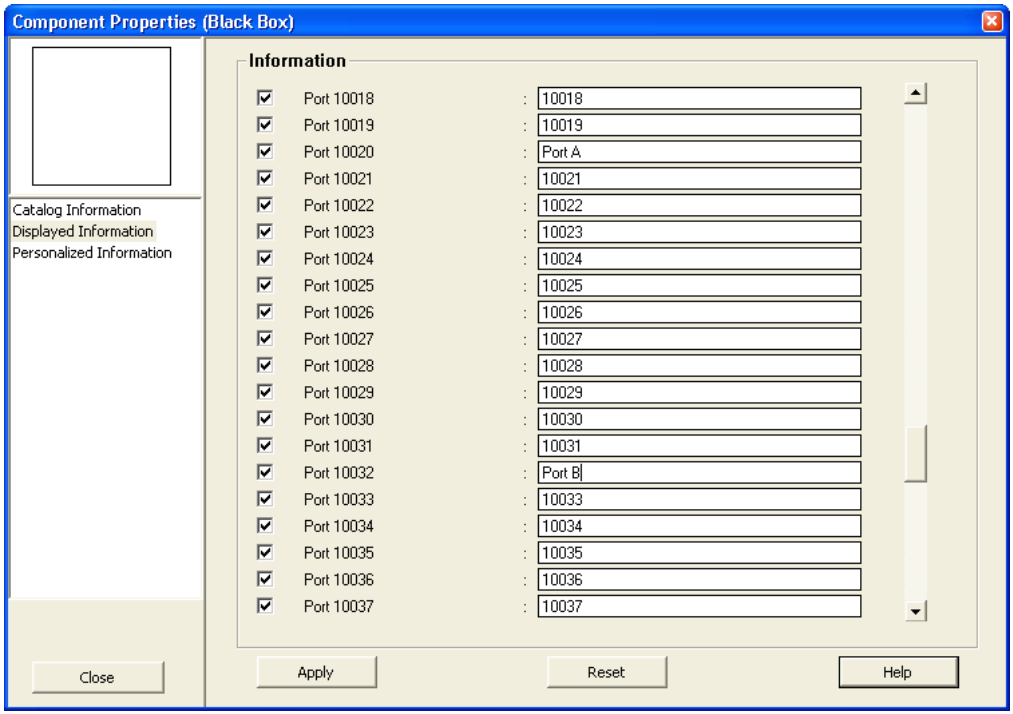


Figure 6-18: Renaming Connection Ports

10. Create a new SFC diagram in the "Project Explorer" root location;

11. Insert an initial step followed by a transition, a normal step and a transition, then link the end to the beginning to permit looping;

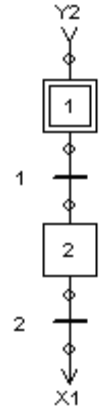


Figure 6-19: SFC Initial Drawing

12. Create a Boolean variable named **Contact** that is associated to the SFC1 diagram;

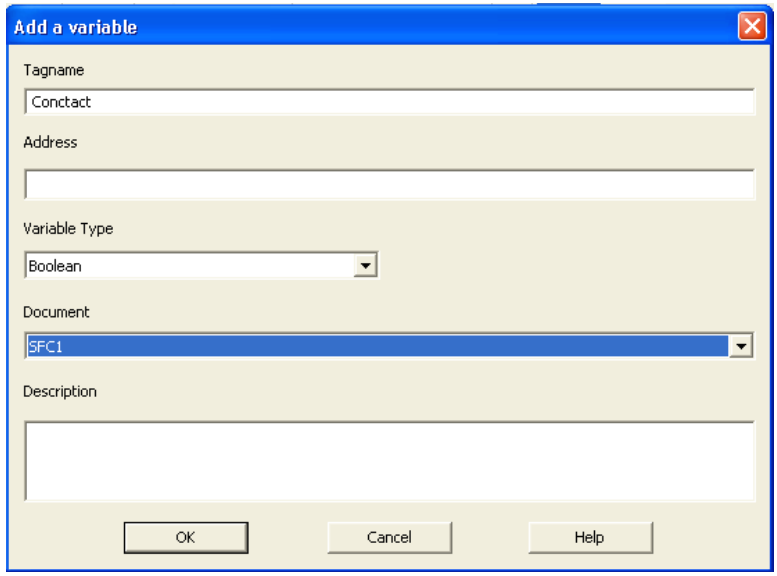


Figure 6-20: Variable Contact Creation

13. Edit the SFC diagram so that it corresponds to the following SFC logic;

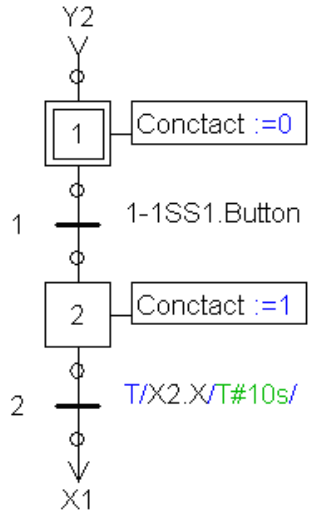


Figure 6-21: SFC Logic

14. Link the "Contact" variable to the contact in the black box;

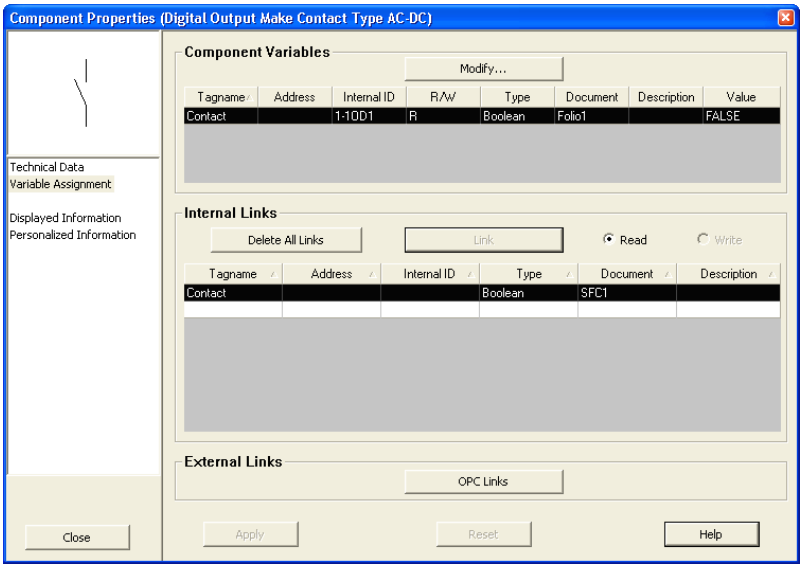


Figure 6-22: "Contact" Variable Creation

The "Cigar Lighter" component is now created. The next section will describe the final component generation.

6.3.2 Generating a Black Box

In this section, we will validate and generate the "Cigar Lighter" created in the previous section. Automation Studio™ performs validations at all steps of the black box simulation and generation. If errors or warnings are generated, they will be displayed in the "Messages" window.

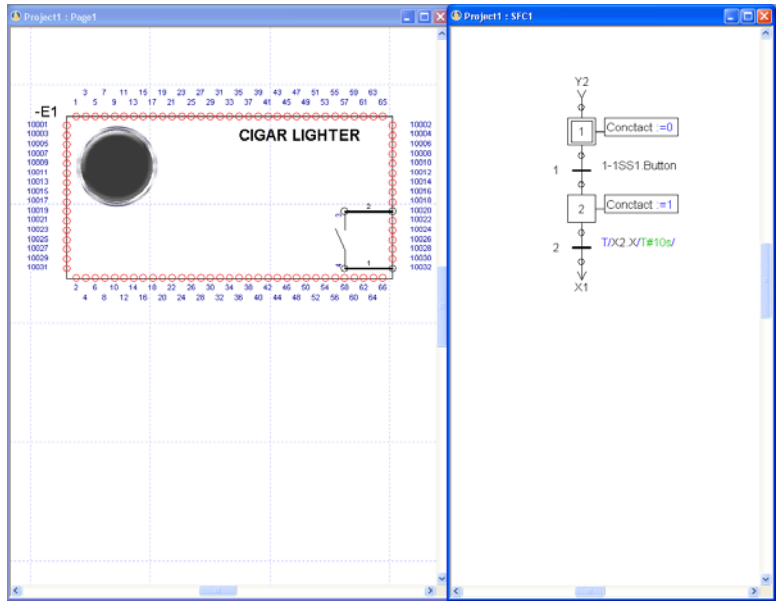


Figure 6-23: Black Box Circuit and Corresponding SFC

1. Validate the cigar lighter circuitry operation by activating the simulation mode;

2. Put diagrams, black box circuitry and SFC side by side;

3. Select the "Simulate the project" option from the toolbar;

4. Start the simulation. Press on the green push button and observe that the circuitry and SFC are acting as programmed. The switch should close, thus appearing in red for 10 seconds;

5. Stop the simulation;

6. On the black box circuitry drawing, select the black box container and all its internal components;

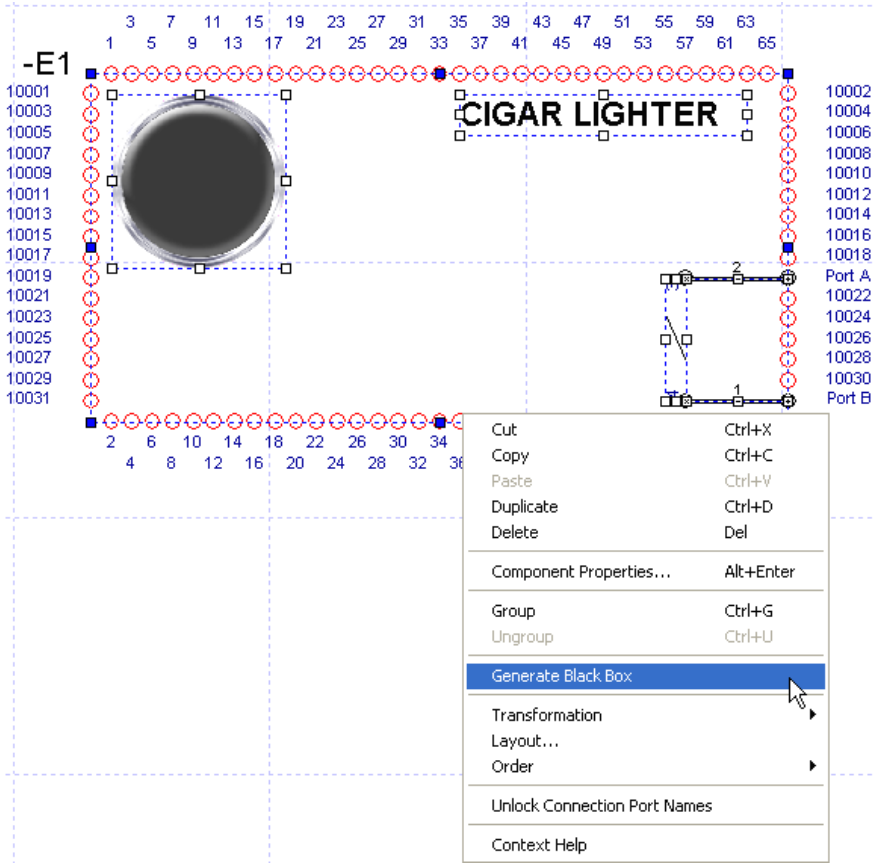


Figure 6-24: Black Box Validation

7. Click on “Generate Black Box” from the component pop-up menu;

   A dialog box offering to assign a logic source document (SFC diagram) opens.

8. Select the SFC diagram which defines the black box logic then click on “OK”;

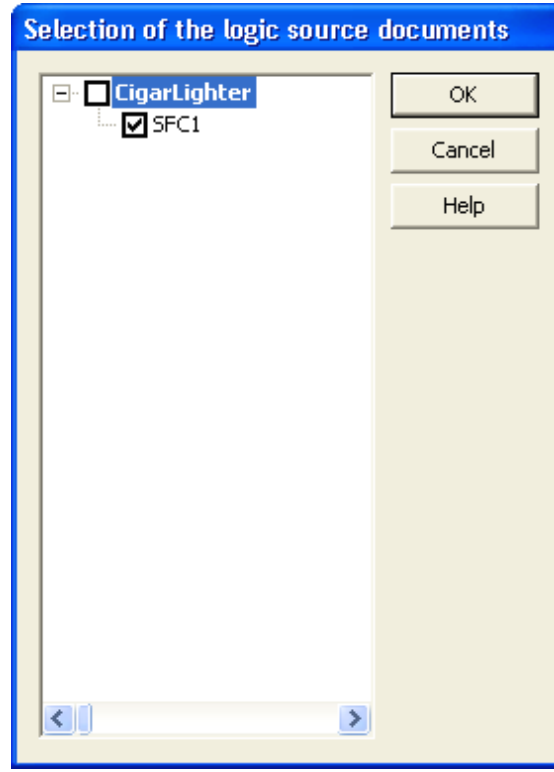


Figure 6-25: "Selection of the Logic Source Documents" Dialog Box

A dialog box offering various optional behaviors for the black box opens.

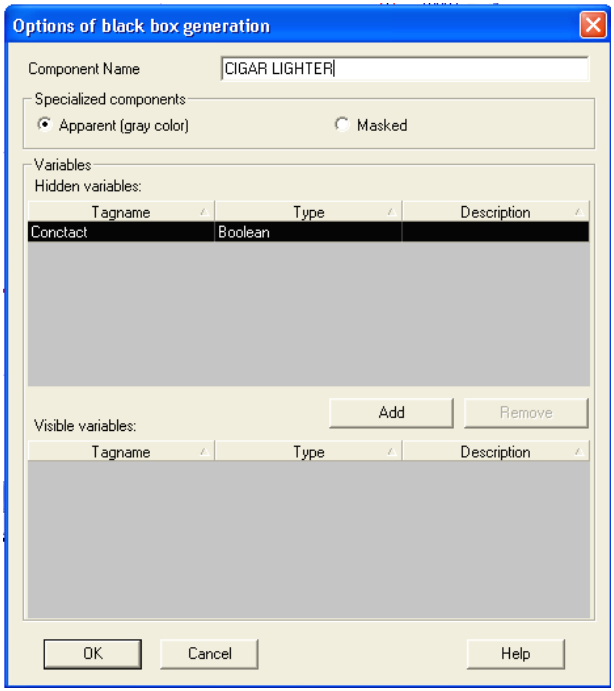


Figure 6-26: "Option of Black Box Generation" Dialog Box

9. Adjust the various fields in the "Options of black box generation" dialog box as follows:

- In the "Component Name" field, enter **Cigar Lighter**;
- Select the "Apparent" (gray color) radio button.

10. Click on OK, then a confirmation message is displayed;

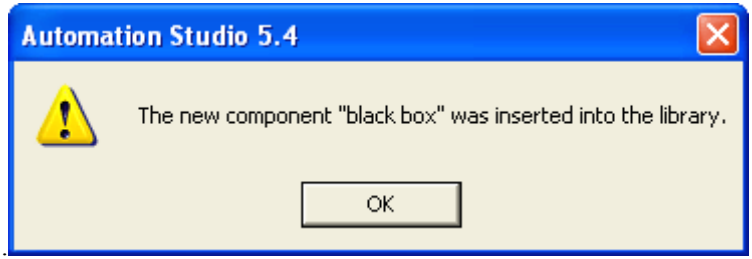


Figure 6-27: Confirmation Message of Black Box Creation

6.3.3 Inserting a Black Box onto a Diagram

When created, the black box is automatically inserted into a personalized library entitled "Black Box".

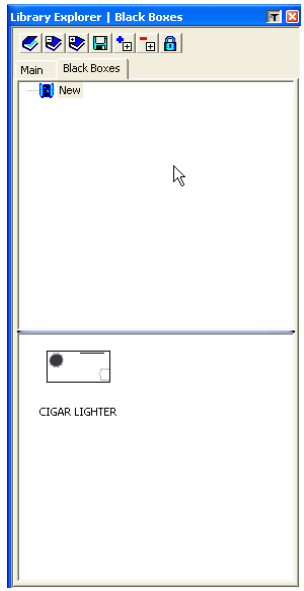


Figure 6-28: Black Box Personalized Library

The library should contain the black box component that has just been created. The component can now be inserted onto a diagram just like any other component.

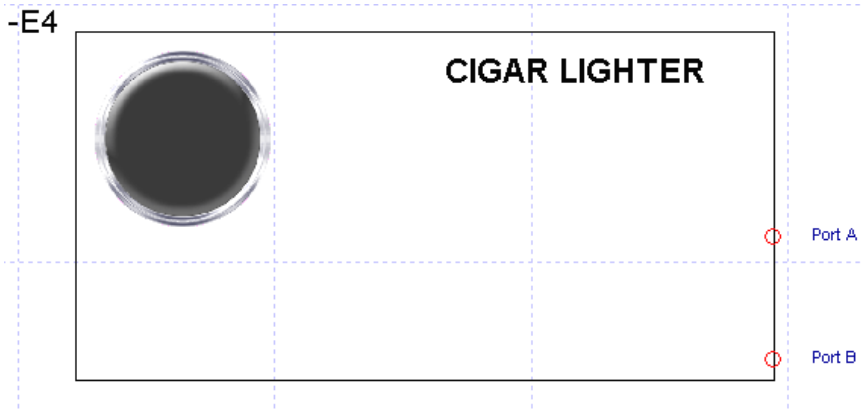


Figure 6-29: New Component "Cigar Lighter"

The black box component is autonomous; the logic programming is now part of it. The black box's unused connection ports are no longer visible. The black box can be moved into a personalized library, and can be used as many times as required. The black box characteristics can be edited just like those of any other Automation Studio™ component.



Please note that performing a rotation or a flip may not be possible if it is not authorized for one of the components forming the black box. In our example, it is the case for the switch. Therefore it will not be possible to rotate the black box.



Important: Save and keep the diagram and logic documents forming the black box. It is not possible to go back to the original documents from the black box.

7 Creating Your First Electrotechnical Diagram

The goal of this example is to familiarize you with the software commands while presenting a step-by-step method to build your first electrotechnical diagram.

7.1 Introduction to the Electrotechnical Workshop in Automation Studio™

An electrotechnical diagram is a graphical representation that uses the basic elements of the workshop to represent a system or part of a system.

The basic elements of the Electrotechnical workshop are:

- Components;
- Wires linking components;
- Terminals, which allow the connection of components to internal wires and to cables or external wires;
- Terminal strips, which regroup the terminals used in diagrams;
- Cables grouping the normally external wires to cabinets.

Generally, electrotechnical diagrams can be divided into two interdependent sections:

- The power supply section, generally built with components from the “Power equipment” library;
- The command section, generally built with components from the “Controls & Commands” library.

7.2 Creating a Project and a new Diagram

To create a new project, which will contain the diagram, do the following:



1. Select File → New Project.

The diagram editor window displays the diagram management menus and commands. The new window created is a folder that will contain the list of all the diagrams of the project.

To create a new electrotechnical diagram:



2. Select File → New → Electrotechnical → Diagram

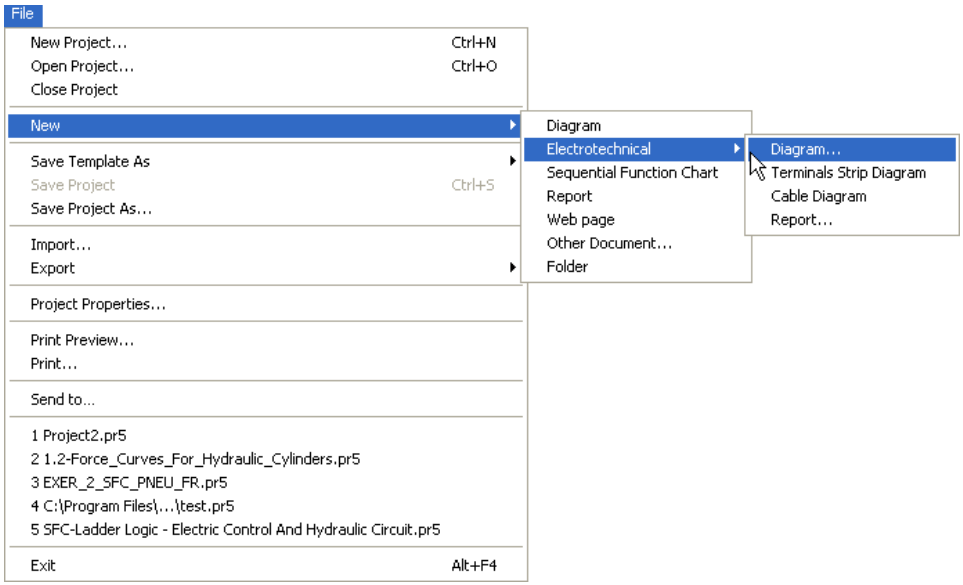


Figure 7-1: Creation of a new Electrotechnical diagram

The dialog box “Electrotechnical Diagram Templates” opens.

3. Select the desired template.

In this case, templates have not yet been created, so the only choice is “None”.

4. Click on OK.

The dialog box "Document Properties" containing the properties of the electrotechnical diagram opens.

5. In the "Component", "1st Line Identifier Format" field, select – *Component code Number: Terminal / . Conductor*. In the "2nd Line Identifier Format field", select *Empty*.(*"Electrotechnical Diagram part 3"*)

6. Click on Apply

7. In the "Link Reference" "Format" field, select *Page number coordinate location"* (*"Electrotechnical Diagram part 1"*)

8. Click on Apply

9. Click on the branch *"Electrotechnical Diagram part 2"* then, in the "Terminal Number" "Method" field, select *Sequential – same number for terminals connected by a jumper*.

10. Click on the "Apply" button to validate the changes.

11. Click on the "Close" button to close the dialog box and return to the diagram.

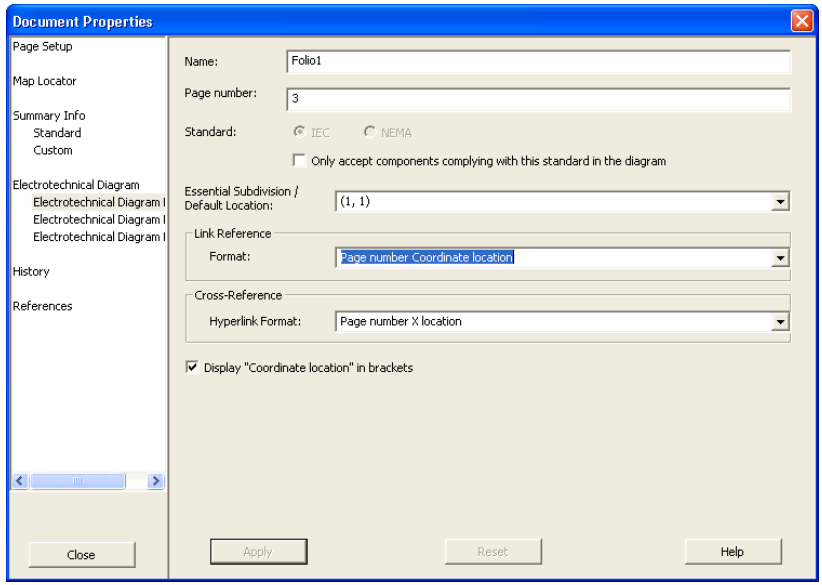


Figure 7-2: "Document Properties" Dialog box. Electrotechnical Diagram part 1

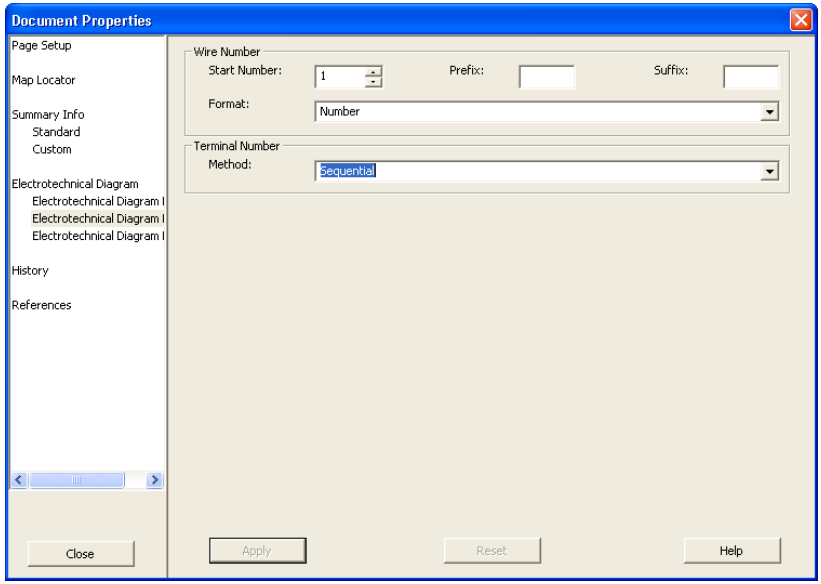


Figure 7-3: "Document Properties" Dialog box. Electrotechnical Diagram part 2

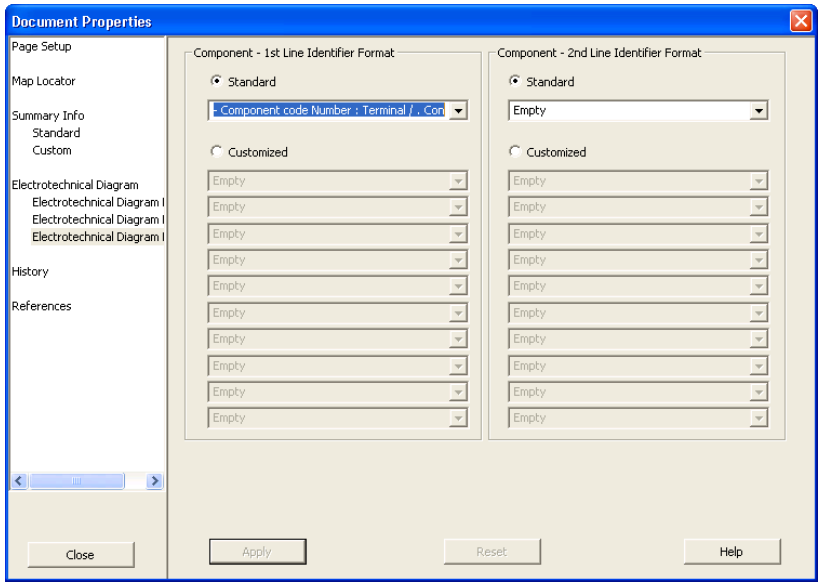


Figure 7-4: "Document Properties" Dialog box. Electrotechnical Diagram part 3

7.3 Description of the Diagram to be Created

The file MOT_FWD_IEC.PRX that contains the circuit to be created is included with the software. This file is located in the Exercises folder. You will also find the same diagram using the NEMA standard in the same location (MOT_FWD_NEMA.PRX). You can use the examples found in this folder as a basis to build your own electrotechnical diagrams.

The circuit to be created allows to control a motor in forward motion. Even if the diagram is simple, it still uses all the functions provided with the Electrotechnical workshop.

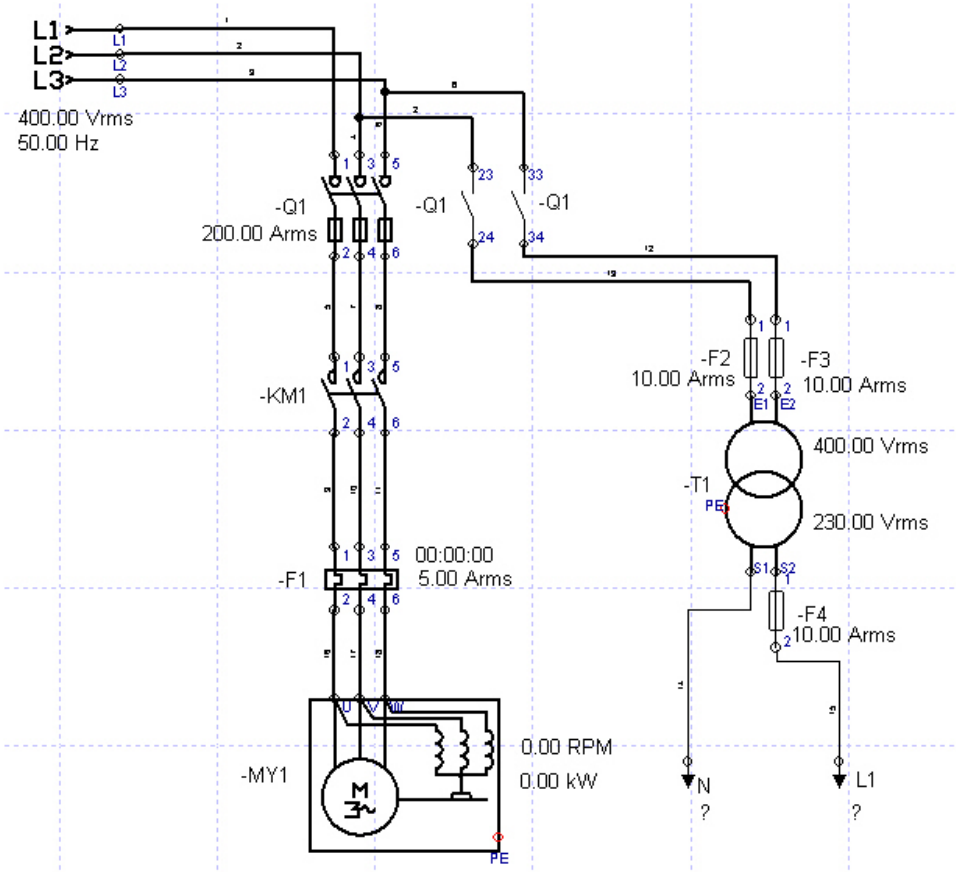


Figure 7-5: Motor command, power supply section

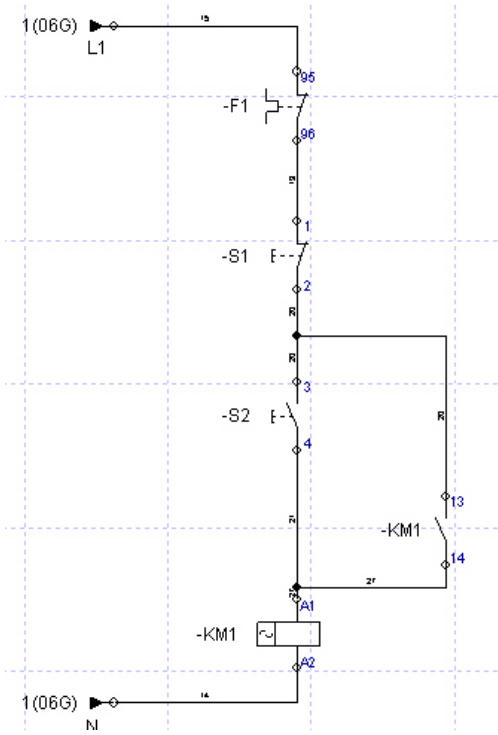


Figure 7-6: Motor command, command section

7.4 Presentation of the Electrotechnical Components

The following figure shows the IEC Electrotechnical workshop component families. These components will be used in the circuit to be created.

The same components can be found in the NEMA Electrotechnical workshop.

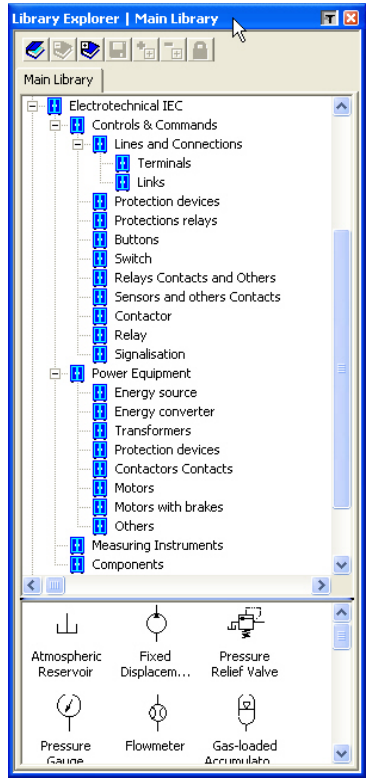


Figure 7-7: Electrotechnical Components, IEC Standard

☞ By default, it is possible to insert NEMA components on an IEC electrotechnical diagram and vice versa. You make the document accept only one standard by modifying the "Only accept components complying with this standard in the diagram" in the "Electrotechnical Diagram Part 1" branch of the "Document Properties" dialog.

7.5 Layout of Electrotechnical Diagram

7.5.1 Layout of Power Components

1. On the new electrotechnical diagram that you have created, lay out the power components as per the MOT_FWD_IEC.PRX file. Give the same names to the components.

You can copy the components from the furnished diagram and paste them on your new diagram

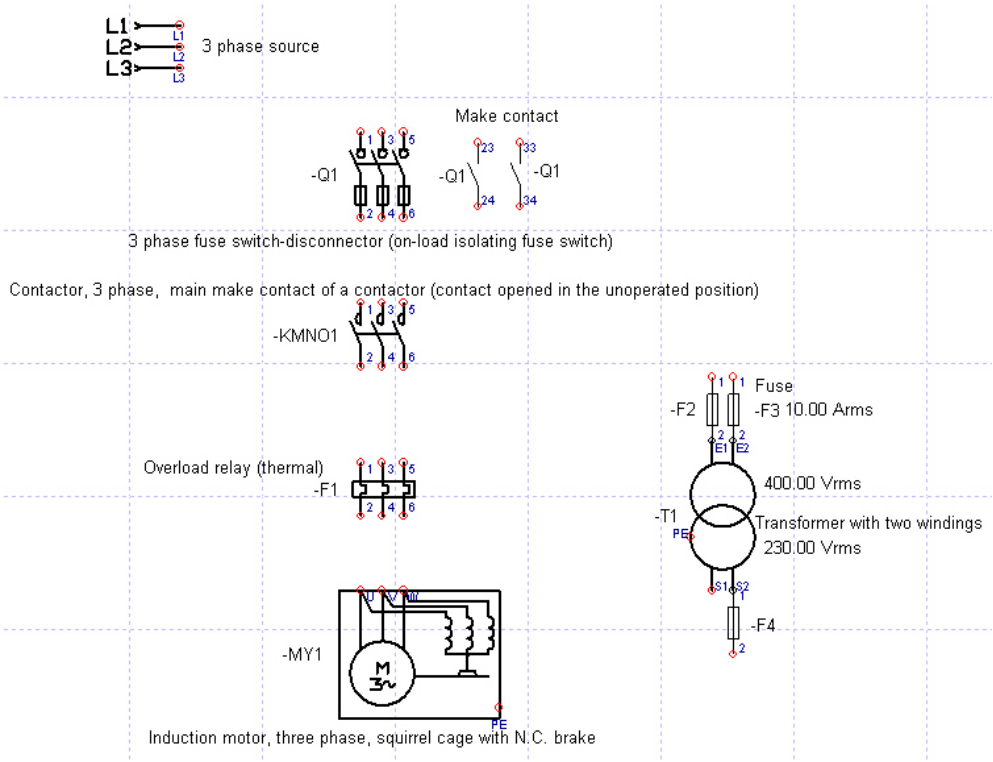


Figure 7-8: Layout of power supply components

2. Insert a "3 phase source" from the *Energy Source* branch of the *Power Equipment* library. Open the Component Properties dialog box from the *Edit* menu or from the contextual menu (right-click on the mouse). In the *Technical Data* branch, adjust the voltage to 400 V at 50Hz.

3. Insert a "3 phase fuse switch-disconnector" from the *protection devices* branch of the *power equipment* library. Enter Q1 as its Tag Name and then click on OK. In the *Technical Data* branch of its properties, adjust the nominal current to 200 A.

4. To the right of the "3 phase fuse switch-disconnector", insert two "make contact" from the *Relay Contacts and Others* branch of the *Controls & Commands* library. These two contacts must be linked to the "3 phase fuse switch-disconnector". In the *Variable Assignment* branch of the properties, click on the line of the internal links

table that contains the name of the "3 phase fuse switch-disconnector", Q1, click on the link button and close the dialog box.

5. Below the "3 phase fuse switch-disconnector", insert a "Contactor", 3 phases, main make contact of a contactor from the *Contactors Contacts* branch of the *Power Equipment* library. These contacts are linked after layout of the command components.

6. Below the contactor, insert an "Overload relay (thermal)" from the *Protection Devices* branch of the *Power Equipment* library.

   Enter F1 for the Tag Name and click on OK. In the Technical Data branch of the properties, adjust the nominal current to 5 A.

☞ The time curve (in seconds) of the thermal relay is calculated as follows:

$$t = e^{\frac{1}{0.0527+0.0694\bullet\frac{I}{I_N}}}$$

☞ If desired, you can display the countdown value directly on the diagram simply by making this property available for viewing. You can do so by accessing the "Displayed information" tab of the thermal relay.

7. Bellow the overload relay, insert an "Induction motor, three phases, squirrel cage with N.C. brake" from the *Motors with Brake* branch of the *Power Equipment* library.
   In the Technical Data branch of the properties, you can see the default parameters of the motor.

8. To the right of the overload relay, insert a "Transformer with two windings" from the *Transformers* branch of the *Power Equipment* library.
   In the Technical Data branch of the properties, adjust the primary voltage to 400 V, the secondary voltage to 230 V, the frequency to 50 Hz and the yield to 1.

9. To the primary of the Transformer with two windings, insert two Fuses, general symbol from the *Protection Devices* branch of the *Control & Commands* library. In the Technical Data branch of the properties, adjust the nominal current to 10 A.

10. To the right secondary of the Transformer with two windings, insert a Fuse, general symbol from the *Protection Devices* branch of the *Control & Commands* library. In the Technical Data branch of the properties, adjust the nominal current to 10 A.

You have finished inserting and using power components. The next section describes the layout of Command components.

7.5.2 Layout of Command Components

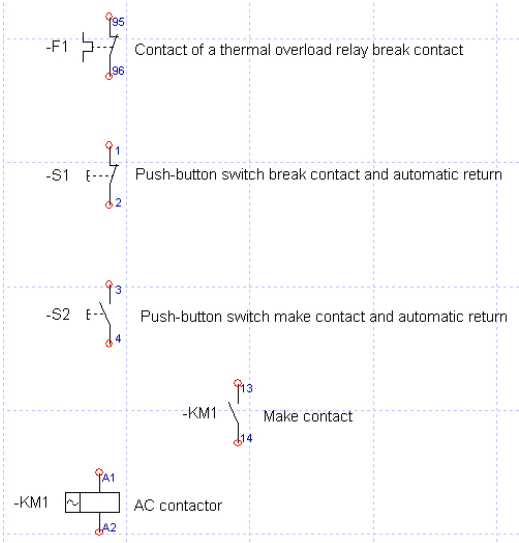


Figure 7-9: Layout of Command Components

1. On the same diagram, lay out the command components as per the MOT_FWD_IEC.PRX file. Give the same names to the components.

2. To the bottom right of the motor, insert a ”Contact of a thermal overload relay break contact” from the *Relay Contacts and Others* branch of the *Controls & Commands* library.
   This contact must be linked to a Thermal Relay. To do so, select the *Variable Assignment* branch of the properties, click on the line of the internal links table that contains the name of the Thermal Relay F1, then click on the “Link” button and close the dialog box.

3. Below the “Contact of a thermal overload relay break contact”, insert a “Push-button switch break contact and automatic return” from the *Buttons* branch of the *Controls & Commands* library. Enter PB_STOP as a Tag Name and click on OK. In the *Technical Data* branch of the properties, click on the “Plus” button to access the advanced data and adjust the maximum voltage to 230 V and the nominal current to 10 A. Click on “Apply” and close the dialog box.

4. Below the push-button, insert a “Push-button switch make contact and automatic return”

from the *Buttons* branch of the *Controls & Commands* family.
Enter PB_START as a Tag Name and click on OK. In the *Technical Data* branch of the properties, click on the "Plus" button to access the advanced data and adjust the maximum voltage to 230 V and the nominal current to 10 A. Click on "Apply" and close the dialog box.

5. Below the last push-button, insert an "AC Contactor" from the *Contactor* branch of the *Controls & Commands* library.
   Enter START as a Tag Name and click on OK. In the *Technical Data* branch of the properties, click on the "Plus" button to access the advanced data and adjust the nominal voltage to 230 V, the nominal current to 0.10 A, the operate voltage to 180 V and the release voltage to 100 V. Click on "Apply" and close the dialog box.

6. To the right of the BP_START push button, between the button and the relay, insert a "Make Contact" from the *Relays Contacts and Others* branch of the *Controls & Commands* library.
   This contact must be associated with the contactor. To do so, select the *Variable Assignment* branch of the properties, click on the line of the internal links table that contains the name of the Contactor START, then click on the "Link" button and close the dialog box.

7. In the power section, the "Contactor, 3 phases, main make contact of a contactor" must be linked to the "Contactor". To do so, select the *Variable Assignment* branch of the properties, click on the line of the internal links table that contains the name of the "Contactor" START, then click on the "Link" button and close the dialog box.

You have finished inserting components. You may now proceed with the layout of wires.

7.5.3 Layout of Wires

7.5.3.1 Layout of Power Wires

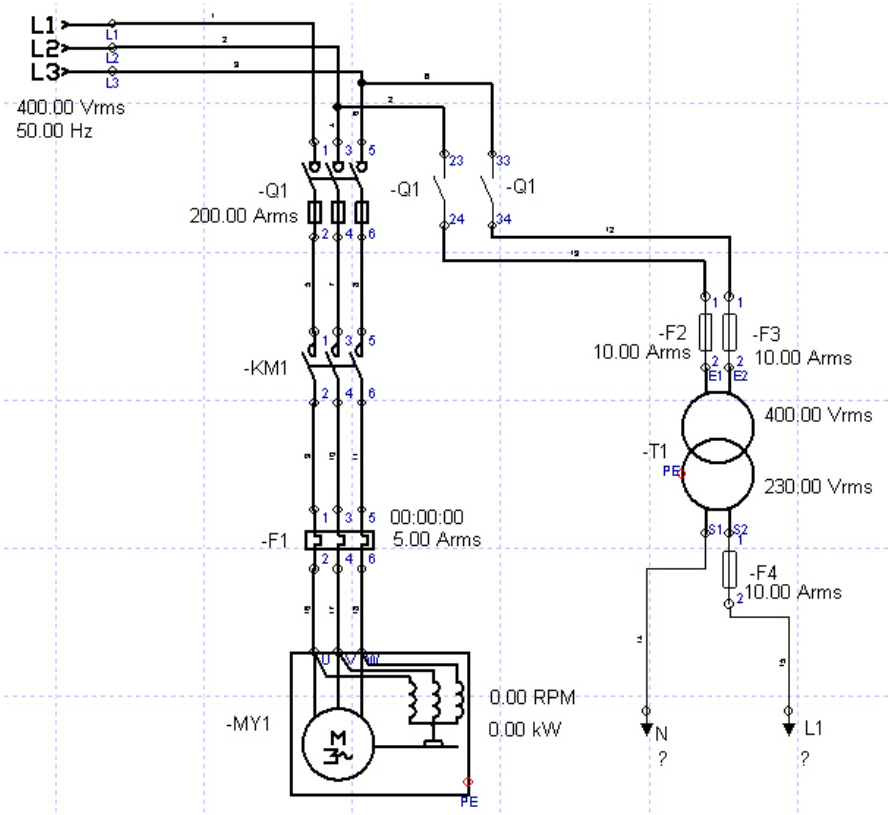


Figure 7-10: Layout of power wires

1. In the "Insert" menu, select the "Power Wire" option. All wires placed after selecting this command are power wires.

2. Position the pointer on terminal 1 of the voltage source, left-click and move the pointer to terminal 1 of the 3-phase fuse switch-disconnect, left-click again. A power wire is placed between the two terminals. This wire is automatically numbered. Insert another wire between terminals 2 and 3, then between terminals 3 and 5.



You can connect these 3-phase components in this exercise by using the polyphase wire tool (See section 3.3.6 Polyphase Links); you will obviously avoid repetitive steps

and save time.

3. Insert a wire between terminal 3 of the 3-phase fuse switch-disconnect and terminal 23 of the first auxiliary contact, then between terminal 5 and terminal 33 of the second auxiliary contact.

4. Insert the wires between the 3-phase fuse switch-disconnect and the contacts of the contactor (as shown in Figure 7-8: Layout of power supply components)

5. Insert wires between the contacts of the contactor and the overload relay (thermal).

6. Insert the wires between the overload relay (thermal) and the motor.

7. Insert the wires between the auxiliary contacts and the primary of the transformer.

8. Below and to the left of the transformer secondary, insert a link #2 – output from the *Link* sub branch of the *Lines and Connectors* branch of the *Controls & Commands* library.
   Using the rotation tool (), rotate the component to the right. Open the component property dialog box to the Link branch. Enter an N and click the "Apply" button.
   This allows you to identify the link in order to associate it with an input link. Open the Displayed Information branch and tick the Tag Name display option. Click on the "Apply" button and close the dialog box.

 You can move and rotate any displayed information around the components.

9. Below and to the right of the transformer secondary, insert a second link #2 – output and identify it as L1. Open the Displayed Information branch and tick the Tag Name display option.

10. In the Insert menu, select the Command Wire option. Once selected, all the following wires are command wires.

11. Insert a wire between the terminal S1 of the transformer and the link N, then another wire between terminal 2 of the fuse and the link L1.

You have finished inserting power wires. You may now proceed with the layout of command wires.

7.5.3.2 Layout of Command Wires

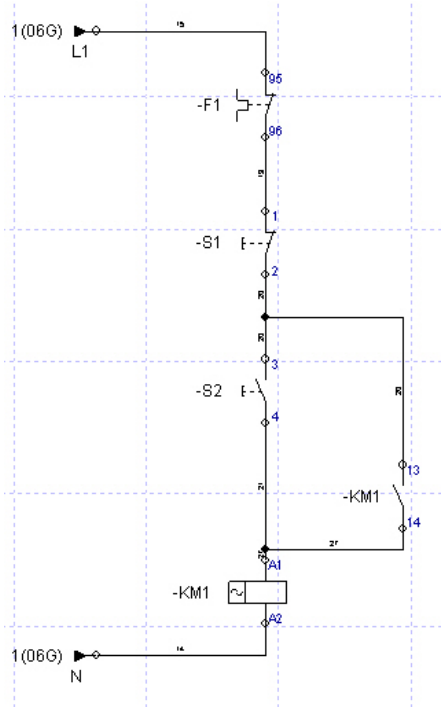


Figure 7-11: Layout of Command Wires

1. In the insert menu, select the Command Wire option. Wires inserted afterwards are command wires.

2. Above and to the left of the overload relay break contact, insert a link #2 – input from the *Link* sub branch of the *Lines and Connectors* branch of the *Controls & Commands* library. Open the component property dialog to the Link branch. Enter L1 and click on "Apply".
   This allows you to identify the link in order to associate it with an output link. Open the Displayed Information branch and tick the Tag Name display option. Click on the "Apply" button and close the dialog box.

☞ Once an input link is associated with an output link, it is no longer possible to modify the association; the link must be delete one has to delete the link and insert a new one.

3. Below and to the left of the contactor, insert a second link #2 – input and identify it as N.

Open the Displayed Information branch and tick the Tag Name display option.

4. Insert a wire between link L1 and the terminal 95 of the Contact of a thermal overload relay break contact.

5. Insert a wire between terminal 96 of the Contact of a thermal overload relay break contact and terminal 1 of the Push-button switch break contact and automatic return.

6. Insert a wire between terminal 2 of the Push-button switch break contact and automatic return and terminal 3 of the Push-button switch make contact and automatic return. Insert a second wire between terminal 2 of the Push-button switch break contact and automatic return and terminal 13 of the Make Contact (see Figure 7-9: Layout of Command Components).

7. Insert a wire between terminal 4 of the Push-button switch make contact and automatic return to terminal A1 of the relay. Insert a second wire between terminal 14 of the Make Contact and terminal A1 of the relay.

8. Insert a wire between terminal A2 of the relay to the link N.

You have finished inserting command wires. You can now proceed with the layout of terminals.

According to the IEC 60757 standard, the standard colors for wires are:

Black:	BK
Brown:	BN
Red:	RD
Orange:	OG
Yellow:	YE
Green:	GN
Bleu:	BU
Violet:	VT

Black:	BK
Grey:	GY
White:	WH
Pink:	PK
Turquoise:	TQ

7.5.4 Layout of Terminals

In order to link terminals to terminal strips, you have first to create a terminal strip. To do so:

1. Open the New Terminal dialog box from the Electrotechnical menu → Terminal Strip → New. The New Terminal Strip dialog box appears on the screen.

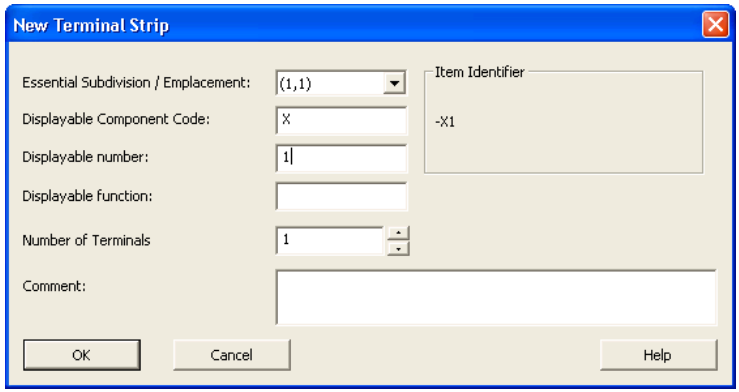


Figure 7-12: New terminal strip

2. Enter the number of terminals of the terminal strip (16).

3. Edit the comment regarding the terminal strip if desired.

4. Click on OK button to create the terminal strip.

You can now select this terminal strip and associate it with terminals.

A terminal strip is a virtual component. Only terminals of the terminal strip appear on electrotechnical diagrams. To see a terminal strip, you have to create a terminal strip diagram.

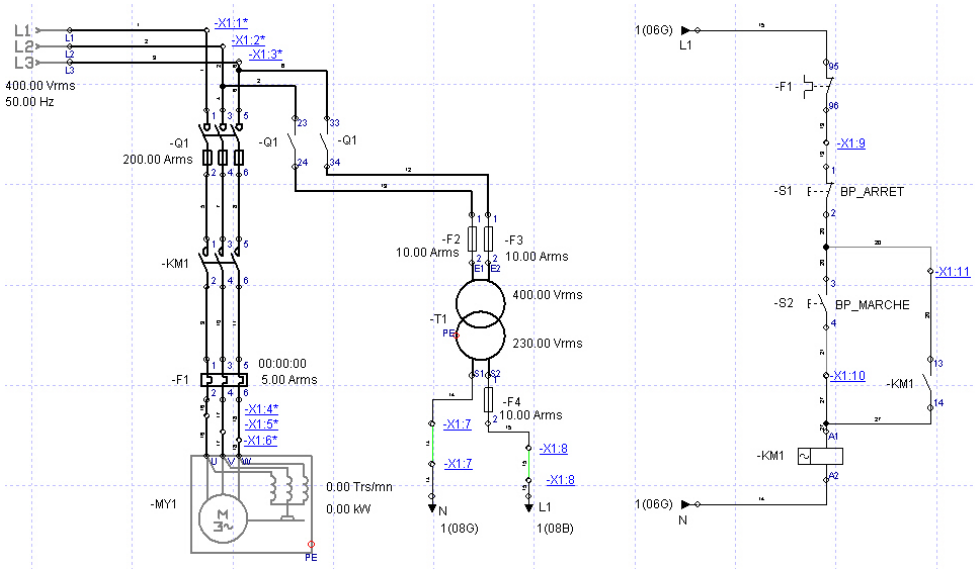


Figure 7-13: Layout of terminals

1. On the first power supply wire, insert a terminal from the *Terminal* sub branch of the *Lines and Connectors* branch of the *Controls & Commands* library (see Figure 7-13: Layout of terminals).The terminals property dialog box opens.

2. Select the terminal strip from the drop-down menu of the Terminal Strip Identifier.

3. Select the first position of the terminal strip from the drop-down menu of the Terminal Position. Click on the Apply button and close the dialog box. Lock the terminal number using the terminal shortcut menu (right mouse button). Locked terminal numbers are not modified when automatic renumbering is performed. The positions used are automatically removed from the list.

4. The wire is automatically cut at the point of insertion and the two wires have the same number. You can move the terminal identifier that is displayed on the diagram.

5. Insert the next two power supply terminals. Select the positions of the terminals sequentially. Lock the number of the terminal from the terminal shortcut menu.

6. From the wire shortcut menu, define the section of the wires between the power supply and the terminals as external. This will allow the proper positioning of the wire on the terminal strip once the terminal strip diagram is created.

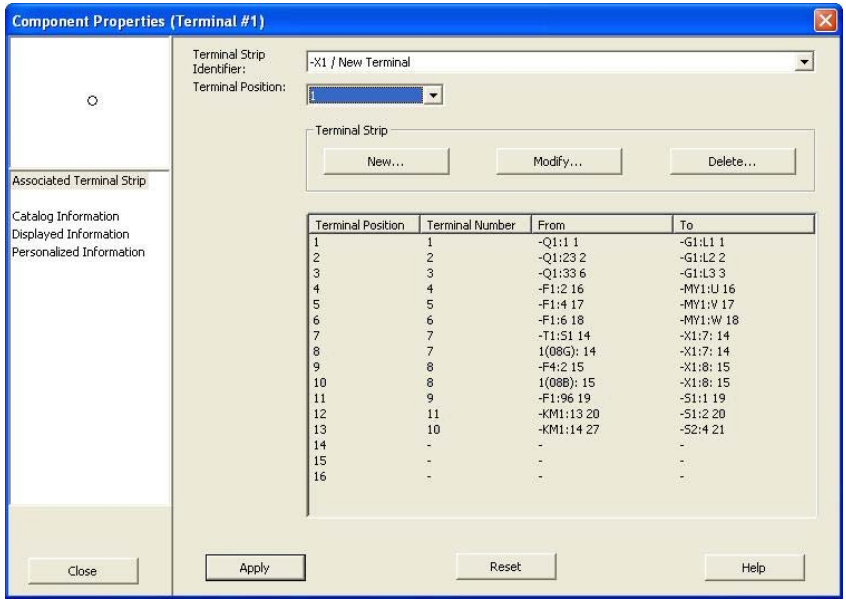


Figure 7-14: Terminal dialog box

7. From left to right insert the terminals on the power supply wires of the motor. Select the positions of the terminals sequentially. Lock the terminal number from the terminal shortcut menu. From the wire shortcut menu, define the section of the wires between the motor and the terminals as external.

8. Insert two terminals on the left wire on the transformer's secondary (from top to bottom). Select the positions of the terminals sequentially. Do not lock the number of the terminals. From the wire shortcut menu, define the wire between the two terminals as a clip jumper.

9. Insert two terminals on the right wire on the transformer's secondary. From the wire shortcut menu, define the wire between the two terminals as a clip jumper.

10. Insert a terminal on the wire between the Contacts of a thermal overload relay break contact and the Push-button switch break contact and automatic return.

11. To the right of the second push button, insert a terminal on the wire on top of the Make Contact.

12. Insert a terminal on the wire between the Push-button switch break contact and automatic return and the relay.

13. Choose the option Renumber Terminals from the Electrotechnical menu. The terminals linked by jumpers now have the same number.

You have finished inserting terminals. You may now create the terminal strip diagram.

☞ You can insert a first terminal, link it to a terminal and copy it. The terminal is linked to the first free space on the terminal strip and the number of the terminal is incremented automatically.

7.5.5 Creating a terminal Strip Diagram

To create a terminal strip diagram:

1. Choose File → New → Electrotechnical → Terminal Strip Diagram. Or from the Project Explorer shortcut menu, choose New → Electrotechnical → Terminal Strip.

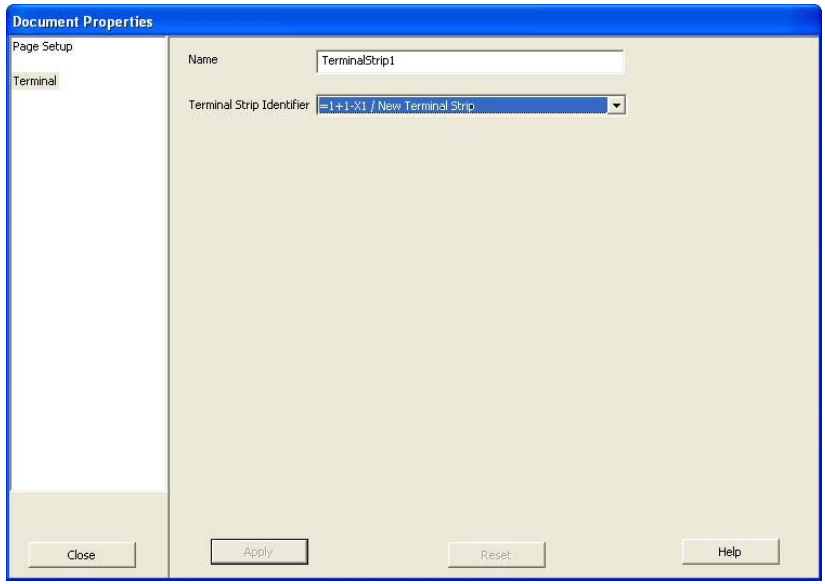


Figure 7-15: Creating a terminal strip diagram

2. You can modify the name of the diagram in the "Name" field.

3. In the Terminal Strip Identifier field, select the identifier of the desired terminal from the drop-down menu, click the Apply button and close the dialog box.

The terminal strip diagram is now displayed on the screen.

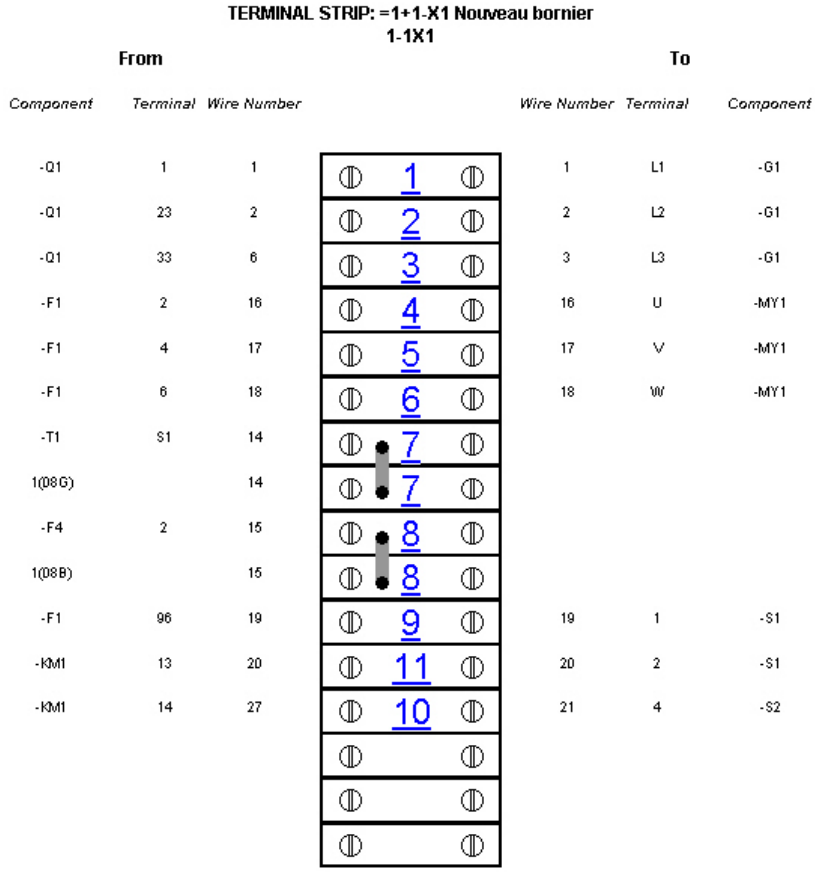


Figure 7-16: Terminal strip diagram

A clip jumper is automatically inserted onto the diagram when a section of the wire between two terminals is defined as a clip jumper. The linked terminals are usually contiguous, but this is not an obligation. The black points indicate the connection points of the clip jumper.

You can manually modify the numbers of the terminals in the Displayed Information

branch of the properties dialog box of the terminal (see Figure 3-24: Terminal properties, Displayed Information branch).

We will now define the cables.

7.5.6 Defining Cables

In order to associate wires with a cable, you have to create a cable. To create a cable:

1. Choose Electrotechnical → Cable → New.

The New Cable dialog box is displayed on the screen.

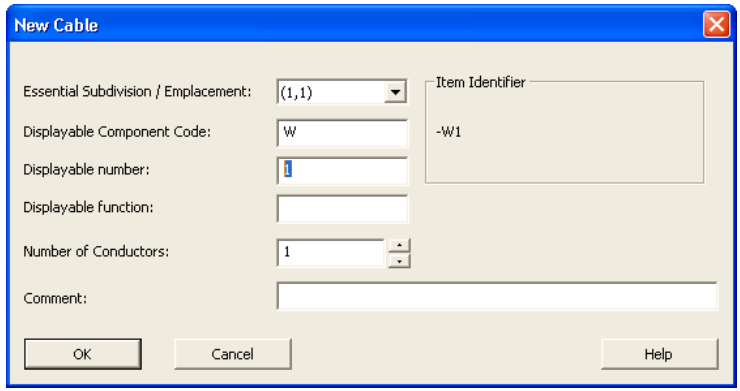


Figure 7-17: Creating a cable

2. In the Number of Conductors field enter 3;

3. In the Comment field enter Power. This cable serves as the power supply to the cabinet.

4. Create a second cable with three conductors. In the Comment field enter Motor Power. This cable serves as the power supply to the motor.

Now that the cables have been created, you can associate the wire with them. To associate a wire with a cable:

1. On the diagram, select the power source. From the shortcut menu, define the component as being external to the cabinet. This allows to define what the conductors of the power cable are connected *From* what they are connected *To*.

2. On the diagram, select the wire attached to terminal "1" of the power source. Open the

wire properties dialog box to the Associated Cable branch. In the Cable Identifier field select the power cable from the drop-down list.

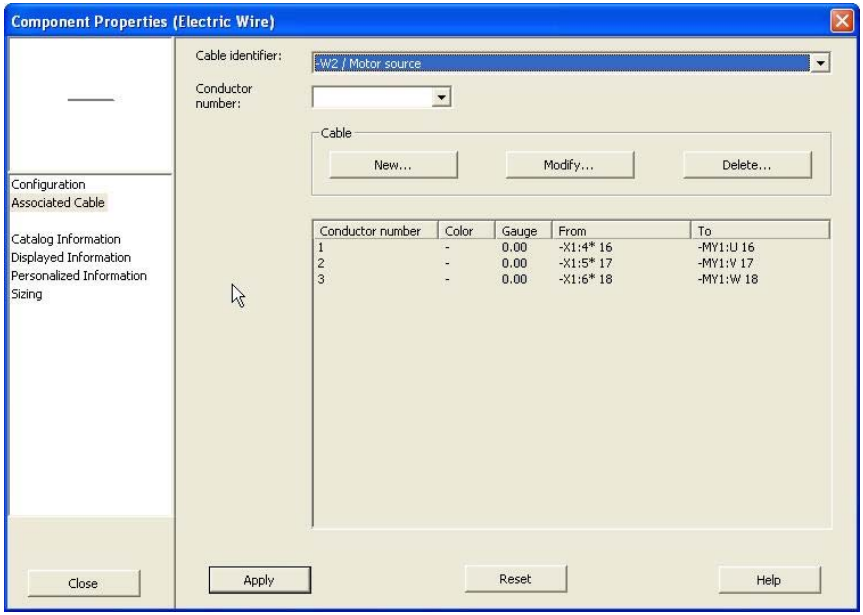


Figure 7-18: Association of wires to cables

3. In the Conductor Number field select the conductor 1 from the drop-down menu. Click on the Apply button. Information relating to the conductor is updated in the cable association table. Close the dialog box.



You can modify the color of each wire in the Configuration field of the properties dialog box. (See section 3.3.4 Configuration Branch).

4. Repeat the procedure for wires linked to terminals L2 and L3 of the power supply. Select the numbers of the cable conductors sequentially.

5. Now you must associate the wires with the motor power supply cable.

6. On the diagram, select the motor. From the shortcut menu, define the component as being external to the cabinet. This allows you to define what the conductors of the motor power supply cable are connected *From* and what they are connected *To*.

7. On the diagram, select the wire connected to terminal U of the motor. Open the wire properties dialog box to the Associated Cable branch. In the Cable Identifier field select

the motor power supply cable from the drop-down list.

8. In the Conductor Number field select conductor 1 from the drop-down menu. Click on the Apply button and close the dialog box.

9. Repeat this operation for the wires linked to terminals V and W of the motor. Select the numbers of the cable conductors sequentially.

You have finished inserting cables. You will now create the cable diagrams.

7.5.7 Creating a cable Diagram

To create a cable connection diagram:

1. Choose File →New → Electrotechnical → Cable Diagram or from the shortcut menu of the project explorer, choose New → Electrotechnical → Cable Diagram.

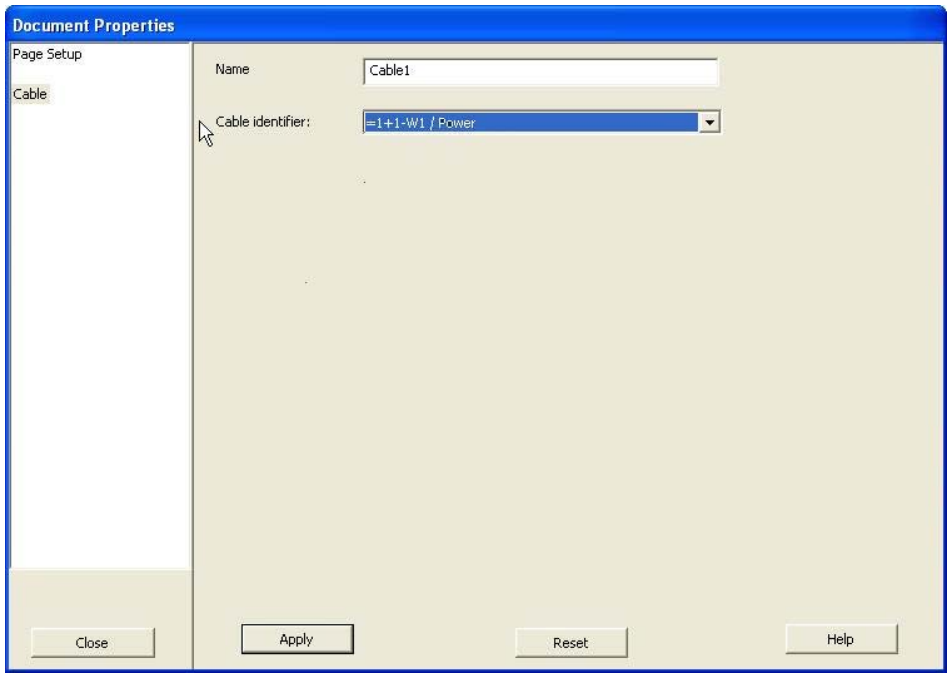


Figure 7-19: Creating a cable diagram

2. The name of the diagram can be modified in the "Name" field.

3. In the Cable Identifier field, select the cable identifier from the drop-down menu. Click on the Apply button and close the dialog box.

The cable connection diagram is displayed on the screen:

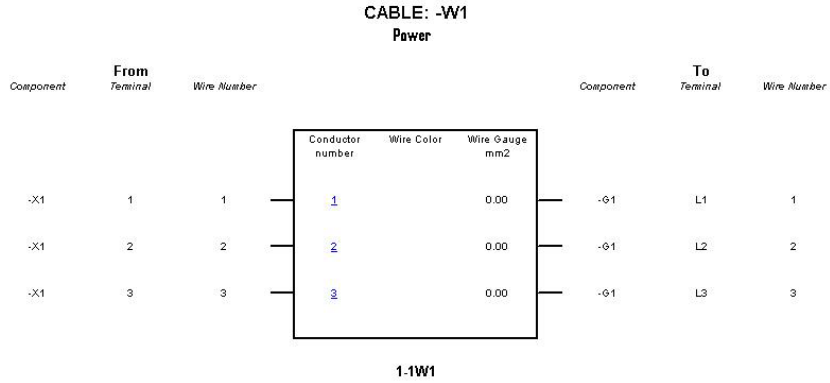


Figure 7-20: Cable diagram

1. Repeat the procedure to create the cable connection diagram of the motor power supply.

You have finished editing the circuit. You will now verify the diagram in order to detect any potential faults.

7.6 Verifying Errors

You can perform a diagnosis of your diagrams at any time in order to verify if there are any errors such as missing connections between components or receivers not linked to emitters. To verify the errors:

1. Choose Electrotechnical → Diagnosis. The diagnosis dialog box is displayed on the screen.

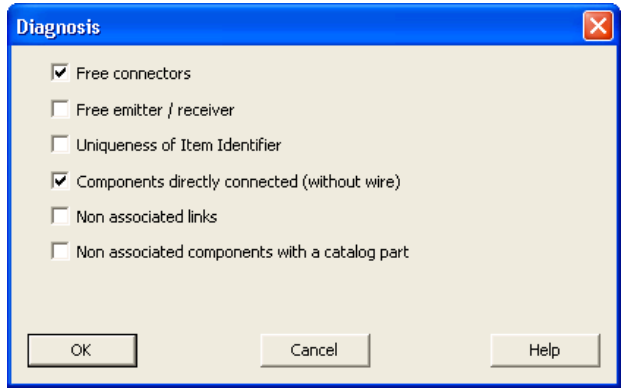


Figure 7-21: Diagnosis dialog box

2. Check the Free connectors and the Components directly connected (without wire) options and then click on OK.

The message window appears and displays the selected diagnostic results. For more information, see the *Message Window* in *Automation Studio™ User's Guide*. On the diagram, components that are directly connected to each other are displayed in a color used to display faulty components.

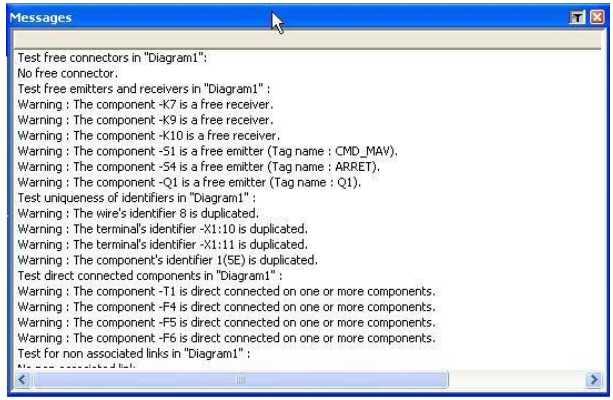


Figure 7-22: Message window displaying diagnosis results

Normally, the components of a cabinet are not directly connected to each other, they are connected by wires. Nevertheless, this does not affect the simulation of the circuit.

8 Simulation Guide

This chapter describes the simulation characteristics that are directly related to the electrotechnical module. The purpose is to introduce the simulation of an electrotechnical circuit.

8.1 Simulation Characteristics of an Electrotechnical Circuit

The electrotechnical diagram simulation is numeric. The voltage and current value are calculated according to the power source voltage, the components present in the circuit, and their inter-relation. The electrotechnical module offers the simulation of RLC circuit in AC or DC current. The simulation specifics for the electrotechnical diagrams are:

- In simulation mode, the user can change selected component characteristics;
- Measuring instruments are available to efficiently provide direct or average values;
- The measured values can be traced using the electrotechnical plotter, which include sinusoidal curves for current, voltage, power, etc;
- The simulator ensures that all limits are respected.

Also, note that:

- Some components, such as power sources and the measuring instruments, have internal resistance that may affect your measurements;
- Transient state calculations are approximated;
- Signals' frequency is limited between 50 and 60 Hz.

8.2 Operation

8.2.1 Changing Component Characteristics during Simulation

During simulation, you can dynamically change any selected component characteristics, which include resistor value, motor resistive charge, etc.

As an example, we will use the circuit created in the previous chapter. Start the simulation. To modify the motor resistive torque:

1. Click on the motor. The "Setting" dialog box opens;

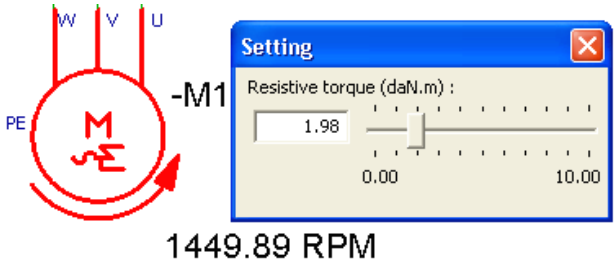


Figure 8-1: Setting Dialog Box during Simulation

2. Change the resistive torque by entering a value in the field box; or by sliding the cursor to the left or to the right to modify the field value.

8.2.2 Changing Component State during Simulation

As in other Automation Studio™ modules, the component state can be changed from "Activated" to "Deactivated".

As an example, we will use the circuit created in the previous chapter and simulate it. To modify the "Start" switch state:

1. Move the cursor on the STOP switch;

The cursor changes to a pointing hand. The hand indicates that the state of the pointed component can be modified during simulation.

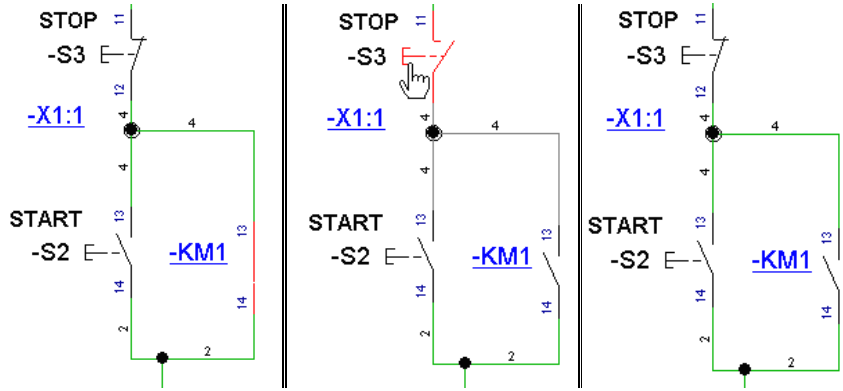


Figure 8-2: Activating a Switch

2. Click on the push button, and release. The push button's state changes;

3. Release the mouse button. The push button's state returns to its initial state.

8.3 Plotter

Using a plotter in the electrotechnical module is done in the same manner as for any other module, but the electrotechnical plotter has the following specific characteristics:

- The time scale is in "ms" and can be set from 10 to 500 ms;
- The plotter only accepts components from the electrotechnical module;
- Due to the amount of calculation required to plot a signal, the plotting is not done according to the simulation time, but 20 times slower. For example, 500ms period on the plotter is equivalent to a 10s simulation period;
- The amplitude scale must be set for each signal;

Refer to *Automation Studio™ User's Guide* for details on the plotter.

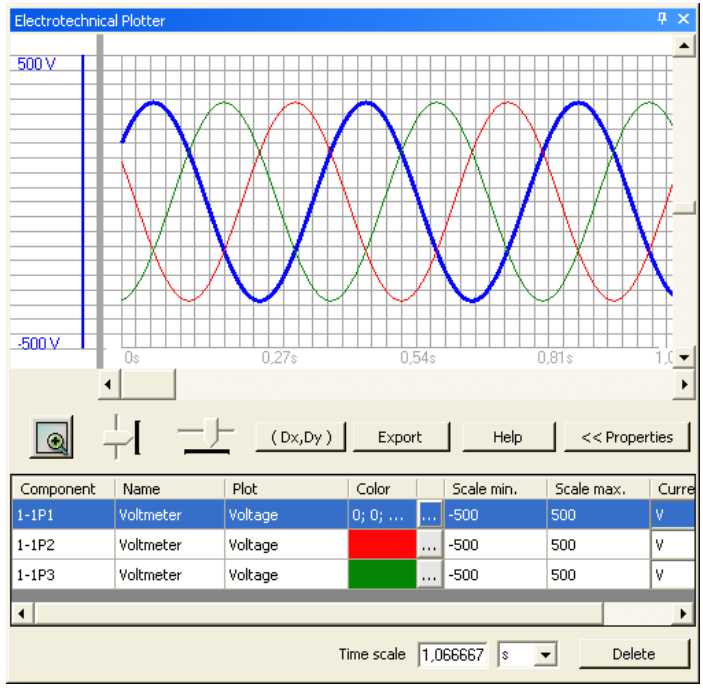


Figure 8-3: Electrotechnical Plotter

8.4 Advanced Analysis Tools

Advanced analysis tools allow you to perform thorough systems analyses, which are divided into two parallel and distinct parts.

- The first part is composed of the “Steady State” and “Snapshot” tools. You can trigger a data simulation snapshot according to many criteria. These values can be used to restart a simulation with saved initial conditions. This data can also be exported to be included in a complete simulation analysis report.

- The second part is the “Measuring Instruments” tool. You can insert dynamic measuring instruments on desired components and connection ports. Each instrument record variable evolution in simulation mode. After simulation, the measured values can be saved. These values can be plot or export for analysis.



For details on creation and management of the advanced analysis tools, refer to the *User’s Guide*. This section contains only information on measuring instruments own to *Electrotechnical Workshop*.

8.4.1 Dynamic Measuring Instruments

Dynamic measuring instruments are specialized components satellites that allow to obtain useful information on the variables used in a simulation. In order to avoid inserting useless components in real schematics, these instruments, accessible by a dedicated tool bar, can be inserted in a circuit during simulation.

The main purpose of these dynamic measuring instruments is to display and record a given property. It is then possible for you to plot evolution curves of variables and to build a complete analysis report of the simulation. Unit and refreshment period can be specified.

The following figure shows an example of utilization of measuring instruments.

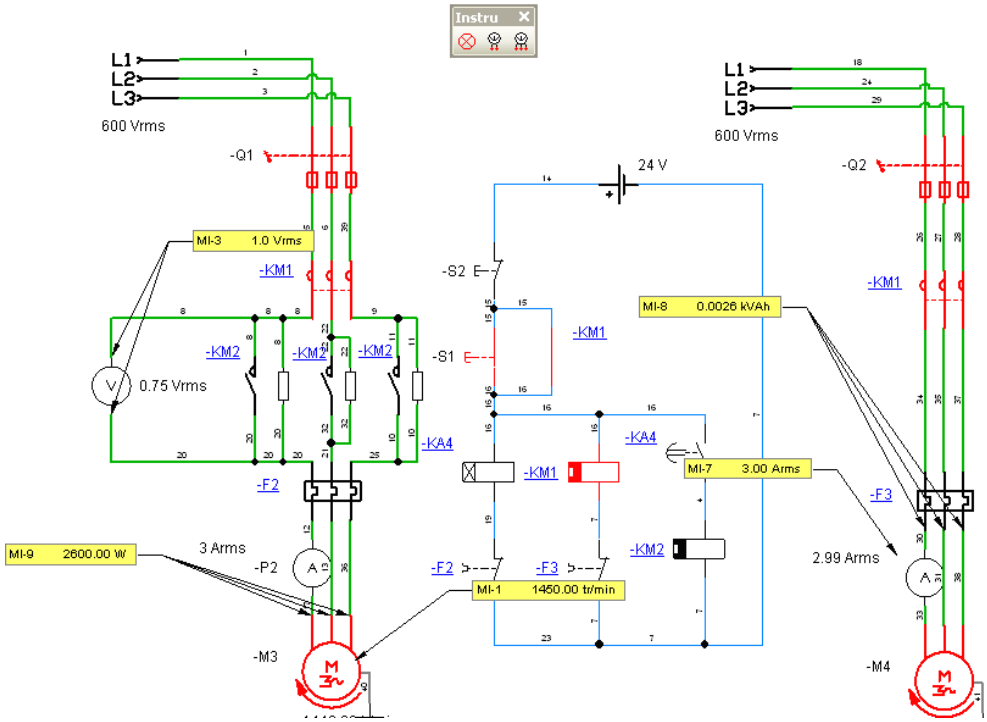


Figure 8-4: Example of measuring instruments in simulation mode

8.4.1.1 Measuring Instruments Toolbar

The toolbar contains options which allow you to insert dynamic measuring instruments on a diagram.



Figure 8-5: "Dynamic Measuring Instruments" Toolbar

The "Dynamic Measuring Instruments" Toolbar contains the following buttons.

	Command	Description
	Component Dynamic Measuring Instrument	Allows you to insert a measuring instrument directly on a component.
	Two Connectors	Allows you to insert a measuring instrument between

	Command	Description
	Dynamic Measuring Instrument	two connectors.
	Three Connectors Dynamic Measuring Instrument	Allows you to insert a measuring instrument between three connectors.

8.4.1.2 Creating Dynamic Measurement Instruments

Component Measuring Instrument

To insert measuring instrument on a component:

1. Choose "Insert" → "Measuring Instruments" → "Component Dynamic Measuring Instrument" or click on the  button in the "Dynamic Measuring Instruments" toolbar;

   The cursor takes this shape  .

2. Click on a component for which you want to measure a property;

   The selected element turns blue.

3. Move the cursor to the location where you want the measuring instrument arrow tip to be;

4. Click and, by holding mouse button, drag and drop to determine its location on the diagram;

   The dialog box for measuring instrument configuration appears.

5. Modify the property to measure as needed (if necessary);

 For this instrument, the available properties are the same as the ones that can be plotted.

6. Click on "OK".

   The measuring instrument takes the following shape.

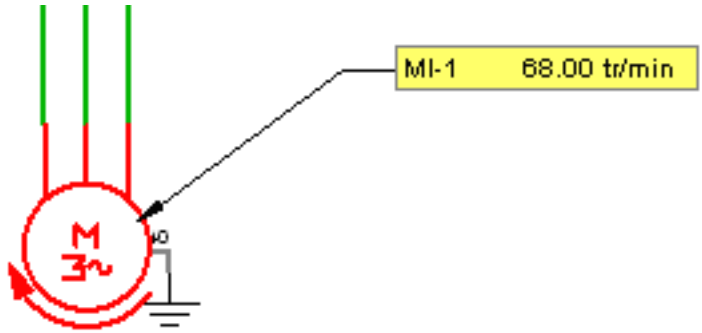


Figure 8-6: Component Dynamic Measurement Instrument Creation

Dynamic Measuring Instrument on Two Connection Ports

To insert measuring instrument between two connection ports:

1. Choose "Insert" → "Measuring Instruments" → "Differential Dynamic Measuring Instrument on Two Connectors" or click on the button in the "Dynamic Measuring Instruments" toolbar;

   The cursor takes this shape .

2. Choose a connector for which you want to measure a property, by moving the cursor on the desired connector;

   The cursor takes this shape .

3. Click and, by holding the mouse button, drag and drop to determine its location on the diagram;

   The instrument location is fixed.

4. Click on a second connector for which you want to measure differential property;

   The dialog box for measuring instrument configuration appears.

5. Modify the property to measure as needed;

☞ For this instrument, the available properties are shown in the following figure.

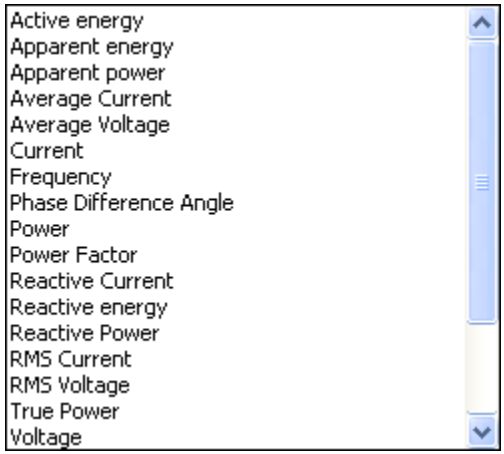


Figure 8-7 : Properties Between Two Connectors

6. Click on “OK”.

   The measuring instrument takes the following shape.

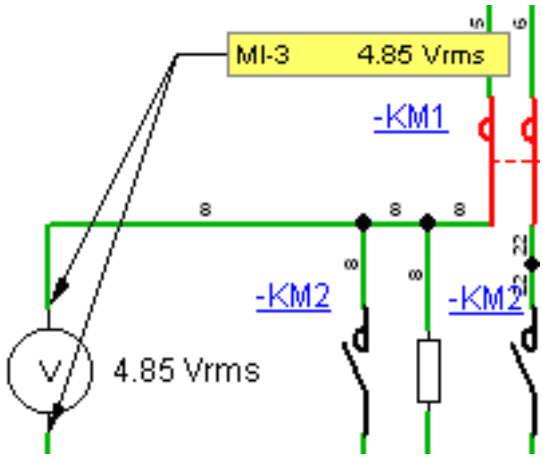


Figure 8-8: Dynamic Measuring Instrument Creation on Two Connectors

Dynamic Measuring Instrument on Three Connection Ports

To insert measuring instrument between three connection ports:

7. Choose "Insert" → "Measuring Instruments" → "Dynamic Measuring Instrument on Three Connectors" or click on the  button in the "Dynamic Measuring Instruments" toolbar;

   The cursor takes this shape  .

8. Choose a connector for which you want to measure a property, by moving the cursor on the desired connector;

   The cursor takes this shape  .

9. Click and, by holding the mouse button, drag and drop to determine its location on the diagram;

   The instrument location is fixed.

10. Click on a second and a third connector for which you want to measure a property;

    The dialog box for measuring instrument configuration appears.

11. Modify the property to measure as needed;

☞ For this instrument, the available properties are shown in the following figure.

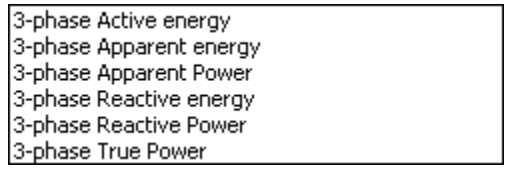
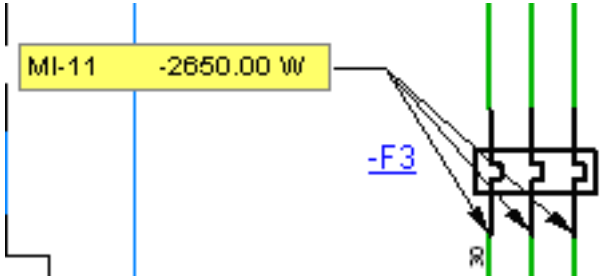


Figure 8-9 : Properties Between Three Connectors

12. Click on "OK".

    The measuring instrument takes the following shape.



Dynamic Measuring Instrument Creation on Three Connectors

8.5 Component Limit Monitoring

This function detects when a component is used beyond its limits.

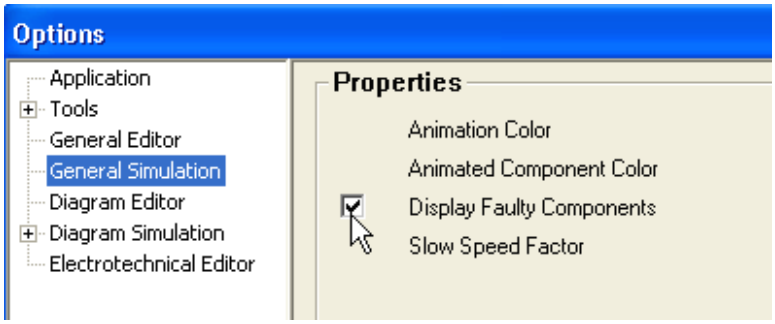


Figure 8-10: Options – Display Faulty Components

The limit monitoring is performed on all components except wires, measuring instruments, and virtual components (terminal strips and cables).

During simulation, the current value is compared to those of the component's characteristics; if the current value is outside any of the limits, an error is detected. If the value returns within the limits, the error disappears.

The component operating beyond its limits is listed in the messages dialog box and a flashing symbol is displayed on top of the faulty component.

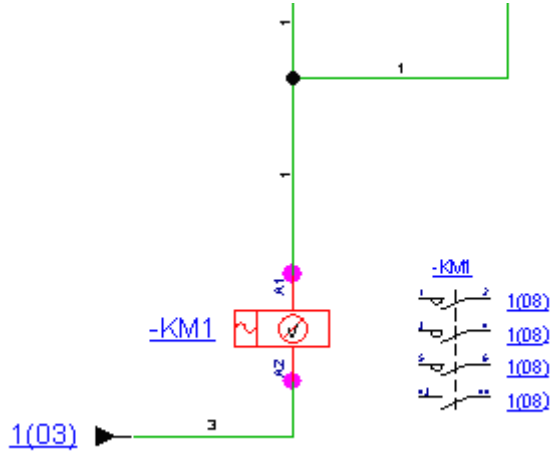


Figure 8-11: Faulty Component

8.6 Common Faults during Simulation

The possible simulation errors are:

- Free connectors;
- Links that are not associated;
- Input components that are not associated with output components;
- Activation value is not set properly;
- Limits are too restrictive;
- Component characteristics that were not properly set.

All faults are listed in the messages dialog box.

9 Panel Layout

This chapter contains the information and procedure to create a panel layout using electrotechnical components found on electrotechnical drawing from a same project.

9.1 Introduction Automation Studio Panel Layout

9.1.1 Automation Studio Panel Layout

A panel layout is a drawing of section of a project that is used to design and display a cabinet containing the physical model of the electrotechnical component found on a circuit drawing

With Automation Studio, you can define and view:

- The cabinet casing;
- A Physical model of the electrotechnical component found on the project's drawings;
- Terminal strips with their terminals;
- PLCs; micro-PLC's, PLC modules, cards and racks;
- Cabinet accessories; mounting rails, cable-trays, ducting, cabinet labels, …;
- Mechanical accessories; cable glands, conduits, bolts, …;
- Cable accessories; cable end sleeves, cable labels.

Automation Studio™ tools are used to:

- Draw the component to scale;
- Draw in a 3D view; (cross-section views)
- Manage specific components: chassis, mounting panels, rails, ducting; …
- Manage functional perimeters;
- Manage dimension markings;
- Help to select the cabinet and panel mounting.

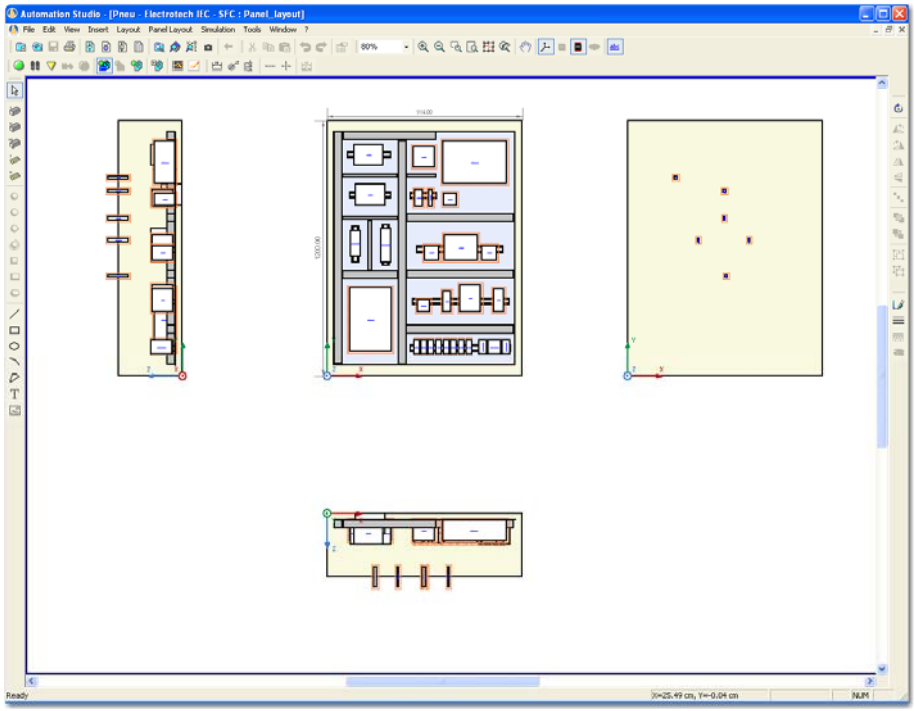


Figure 9-1: Panel Layout - Example

9.1.2 Panel Layout Drawing

The panel layout drawing is a specialize document. Panel layouts are drawn to scale.

9.1.3 Standards

A panel layout drawing is not dependant upon the IEC or NEMA standard. The ISO or North-American standard are applied to the view positioning and dimensions.

9.1.4 Panel Layout Pre-Requisites

The panel layout function has its own library. This library is populated with component layouts of the components included in the electrotechnical drawings of the same project. A component must be present in a drawing to have this component layout in the panel layout library.

A panel layout can be created without an electrotechnical drawing available in the project, but

the library will be empty, therefore components cannot not be included in the panel.

9.1.5 Available component family

Electrotechnical components with a physical representation will also have a layout representation in a panel layout drawing. Components without a physical representation can not have a layout representation. These components are mostly wires and cables.

All of the components found on an electrotechnical drawing have a physical representation as well as the other standard hardware used in panel manufacturing, such as:

- The chassis;
- Mounting panels;
- Mounting rails;
- Wire and cable ducting.

9.1.6 Coordinates

Components are positioned according to the 3D Cartesian coordinate (x, y, and z).

A component reference point, when looking at the front side of the component, is at the left bottom in the back.

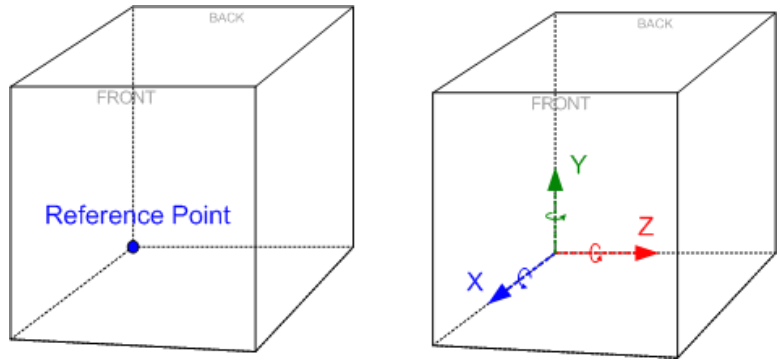


Figure 9-2: Reference point and coordinate

Each component is positioned according to another component. Its coordinates are provided according to its parent component. When the parent component is moved, the referred components follow the move, their coordinates do not change.

The prime positioning component is the chassis.

For example: The chassis in placed on the drawing, the mounting panel is automatically inserted in the chassis, the mounting rail is positioned on the mounting panel, and the component is mounted on the rail.

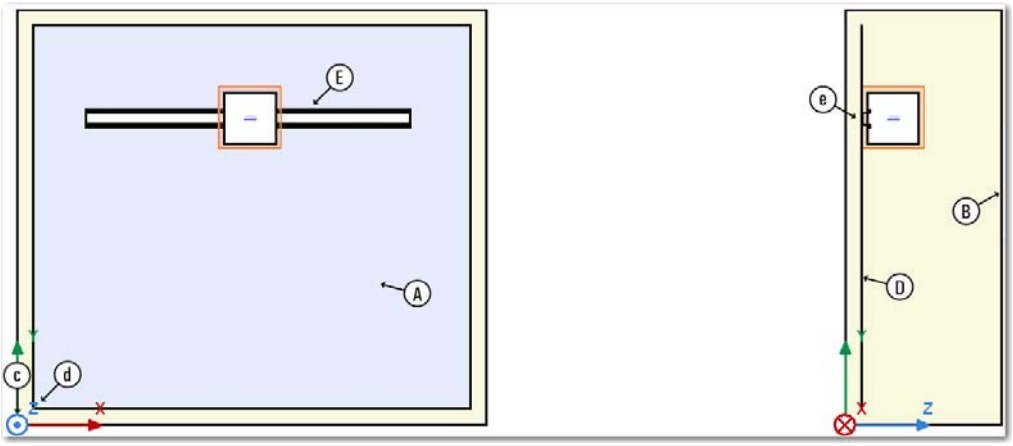


Figure 9-3: Chassis cross-section view

Zone	Name	Description/Coordinates		
A	Cross-section view	Front		
B	Cross-section view	Left side		
c	Axes	Chassis reference point		
D	Mounting panel	X = 30mm	Y = 30mm	Z = 30mm
d	Axes	Mounting panel reference point		
E	rail	X = 40mm	Y = 616mm	Z = -32mm
e	Component	Rail reference point		

9.2 Interface

Automation Studio's interface changes according to the type of drawing that is presently being edited. The following sections cover the panel layout interface and its unique menu

options.

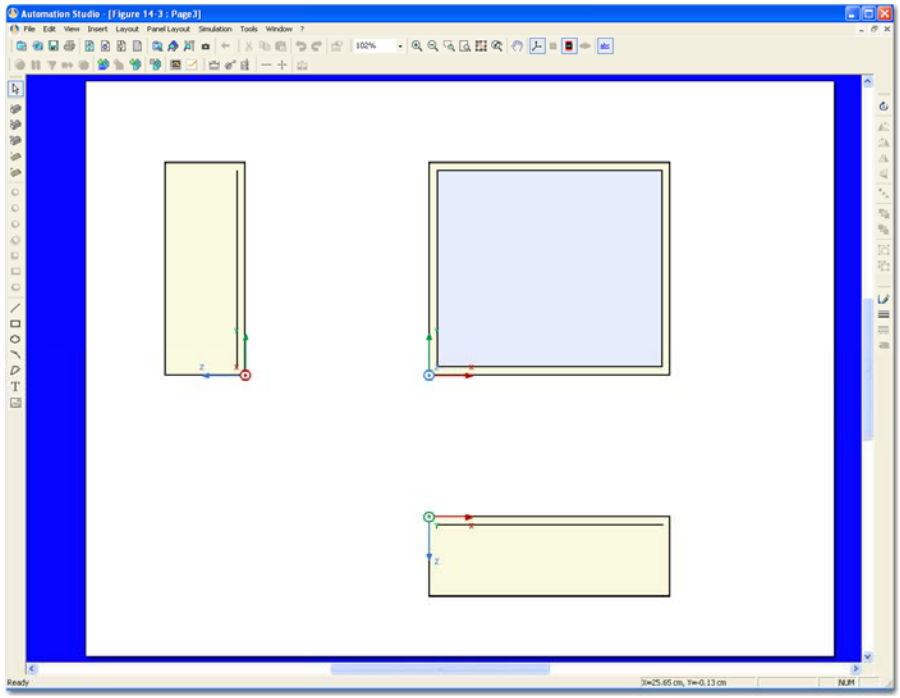


Figure 9-4: Basic chassis

9.2.1 Panel Layout Specific Environment

Panel Layout Specific Environment - Display

The display edition option, unique to the panel layout drawing, is available through:

- the “View” menu in the menu bar;
- from the drawing pop-up menu using the “View” menu;
- from the “View” tool bar.

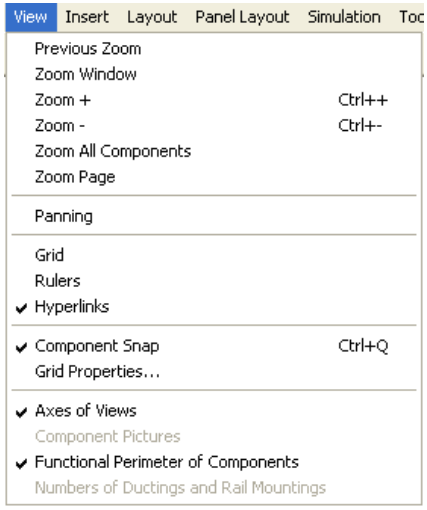


Figure 9-5: Bar Menu - View

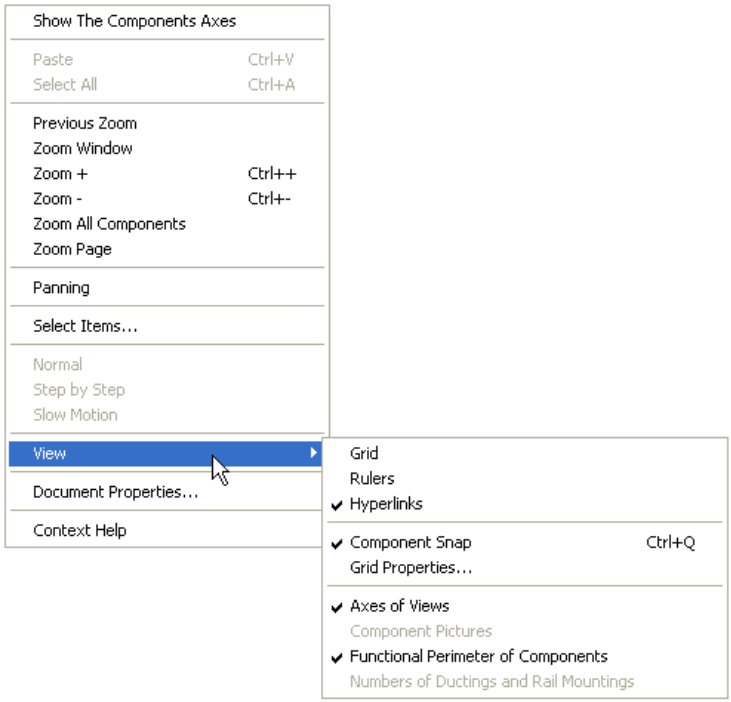


Figure 9-6: Pop-up menu - View



Figure 9-7: Tool bar - View

The display editing tools that are specific to panel layout construction are:

Menu option	**Tool bar button**	**Description**
View – Coordinate axes		Display the coordinate axes for the chassis views.
View – Component Picture		Option not available.
View – Component functional perimeter		Display the component's functional perimeter.
View – Rail number and ducting		Option not available.

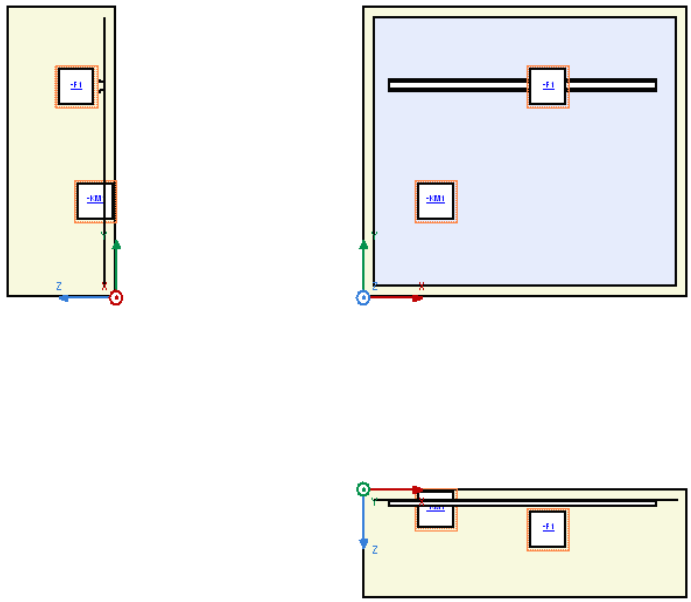


Figure 9-8: View – Axes of view & component perimeter

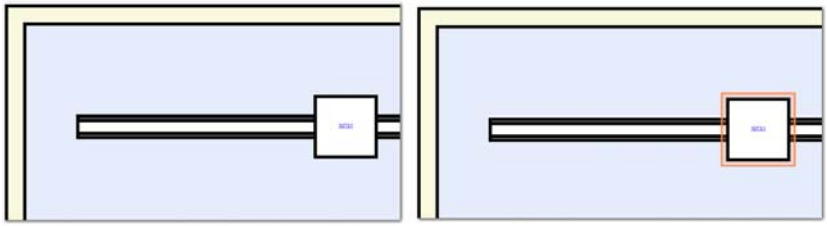


Figure 9-9: View – Component without and with a perimeter

Panel Layout Specific Environment - Insert

The insertion option, unique to the panel layout drawing, is available through:

- the “Insert” menu in the menu bar;
- from the “Citation” tool bar;
- from the “Insertion” tool bar.

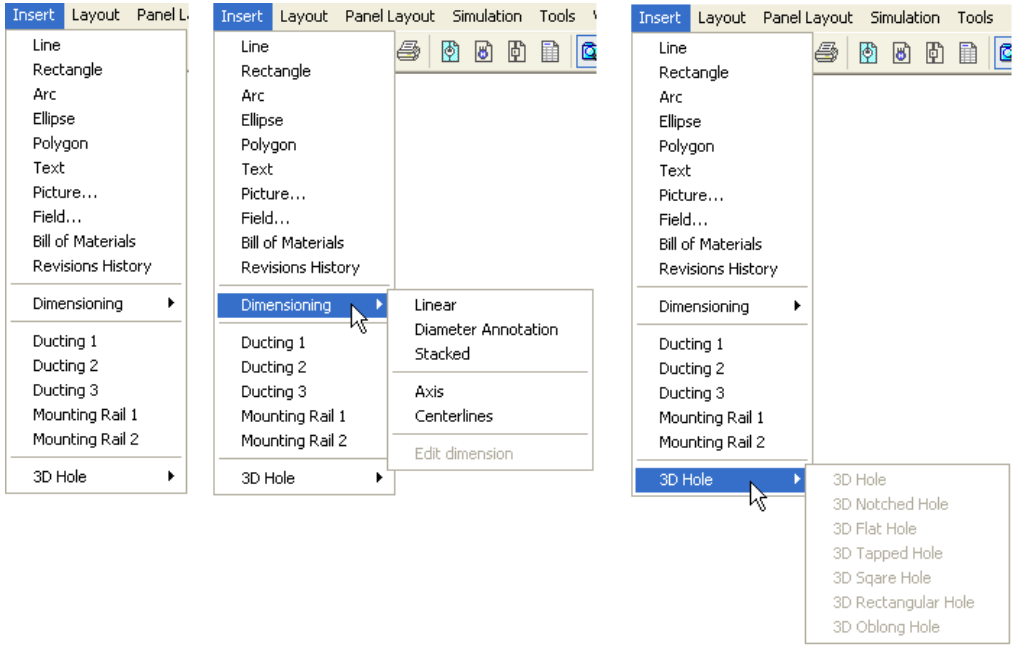


Figure 9-10: Insertion Menu



Figure 9-11: Dimensioning menu bar



Figure 9-12: Insertion menu bar

The control panel specific insertion options available are:

Name	Tool bar	Description
Insert – Dimensioning – Linear		Insert a linear dimension.
Insert – Dimensioning – Diameter		Insert a diameter annotation.

Name	Tool bar	Description
annotation		
Insert – Dimensioning – Stacked dimension		Insert stacked dimensions.
Insert – Dimensioning – Axis		Insert an axis dimension.
Insert – Dimensioning – Centerline		Insert a centerline.
Insert – Dimensioning – Edit dimension		Option not available.
Insert – Ducting 1, 2, 3		Insert ducting.
Insert – Rail 1,2		Insert a mounting rail.
Insert – 3D Hole – 3D Hole		Option not available.
Insert – 3D Hole – Notched Hole		Option not available.
Insert – 3D Hole – 3D Flat Hole		Option not available.
Insert – 3D Hole – 3D Tapped Hole		Option not available.
Insert – 3D Hole - 3D Square Hole		Option not available.
Insert – 3D Hole – 3D Rectangular Hole		Option not available.

Name	Tool bar	Description
Insert – 3D Hole – 3D Oblong Hole		Option not available.

Panel Layout Specific Environment – Panel Layout

The panel layout option, unique to the panel layout drawing, is available through the "Panel Layout" menu in the menu bar.

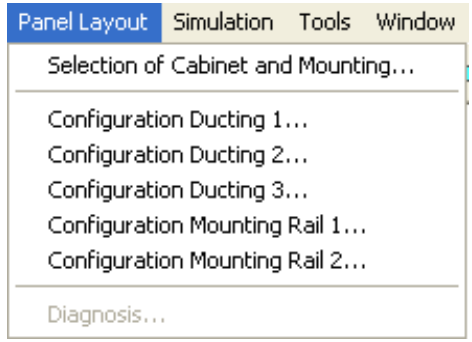


Figure 9-13: Panel Layout menu

The panel layout options are:

Name	Description
Selection of Cabinet and Mounting Panel	Open the "Cabinet and Mounting Panel Selection" dialog box.
Configuration Ducting 1, 2, 3	Open the "Wire Duct Configuration" dialog box.
Mounting Rail 1, 2	Open the "Rail Configuration" dialog box.
Diagnosis	Option not available.

Panel Layout Specific Environment – Localization

The localization option, unique to the panel layout drawing, is available through the menu bar

under "Window" → "Display" → "Localization".

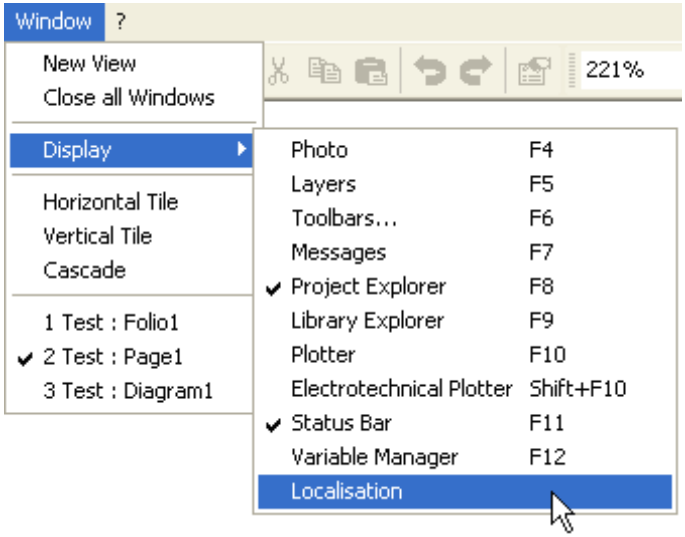


Figure 9-14: Menu to display the Localization tool

9.2.2 "Document Properties" → "Panel Layout"

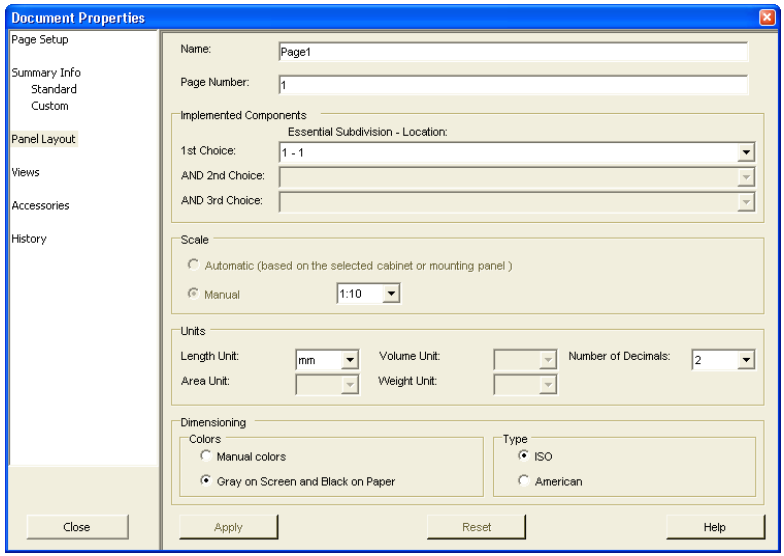


Figure 9-15: Document Properties – Panel Layout

Name	Description
Name	Field to display and edit the document name.
Page Number	Field to display and edit the page number
Implemented Components	Defines the location of the components that will be included in the panel layout library.
Scale	Defines the scale definition standard; manual or automatic. In the manual mode, the user chooses using pre-defined selections. When the scale is not compatible with the present chassis dimension, the various views may be only partly visible. In the automatic mode, Automation Studio defines the scale. It is assigned based on the chassis dimension.
Scale	Drop-down list that is used to assign a scale when in manual mode.
Length Unit	Drop-down list that is used to select the length unit.
Surface Unit	Drop-down list that is used to select the surface unit.
Volume Unit	Drop-down list that is used to select the volume unit.
Weight Unit	Drop-down list that is used to select the weight unit.
Number of Decimals	Drop-down list that is used to select the number of decimals displayed with the dimension.
Dimensioning Colors	Radio button that is used to select the color standard for the dimensioning values: color or gray.
Dimensioning Type	Radio button that is used to select the dimensioning type: ISO.

9.2.3 "Document Properties" → "Views"

Provide the means to change the drawing configuration with option on the displayed views, interstices dimension, and display standard.

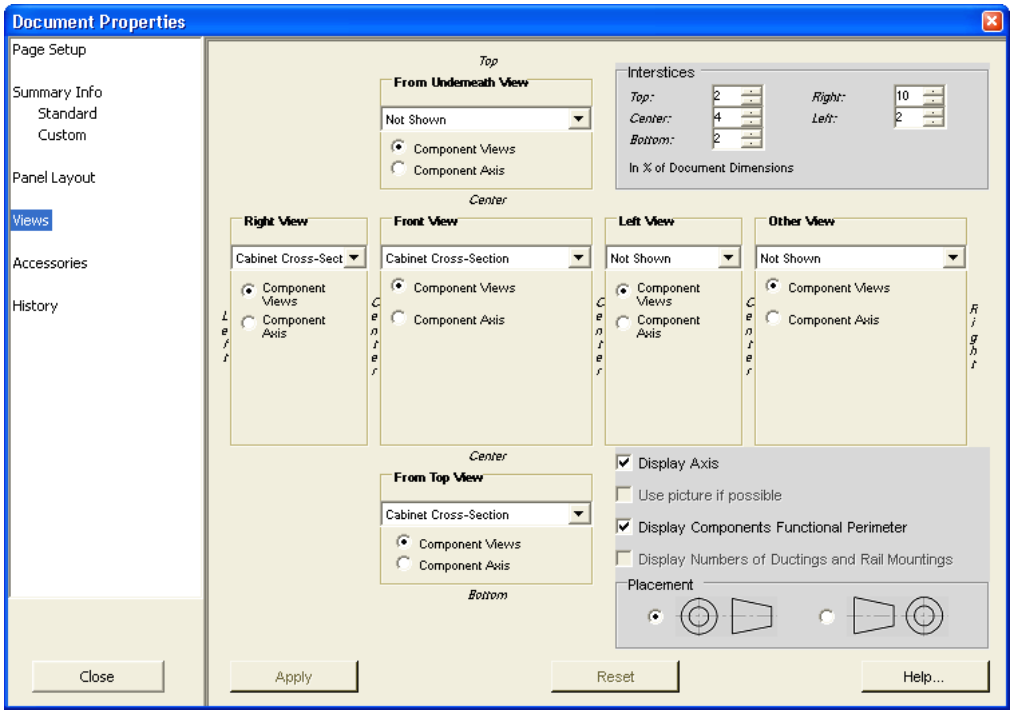


Figure 9-16: Document Properties - Views

Field	Description
From Underneath View Right View Front View Left View From Top View	Drop-down menu that is used to select the type of diagram that will be presented for this view, including "none".

Field	Description
Other View	
Component Views Component Axis	Option to either display the component or only the axis of the component.
Interstices	Fields that are used to define the space in between the components (center) and for the drawing margins (top, bottom, left, right). The interstice is a percentage of the space that is not used by the cabinet cross-section views.
Display Axis	Option to display, or not, the cross-section view axis.
Display Components Functional Perimeter	Option to display or hide the functional perimeter.
Display Numbers of Ducting and Rail Mountings	Option to display or hide the numbers of ducting and rail mounting.
Use picture if possible	Option to display or hide the component picture instead of the component layout, if a picture is available.
Placement	This option allows the selection of either the ISO or North-American representation.  ISO North-American

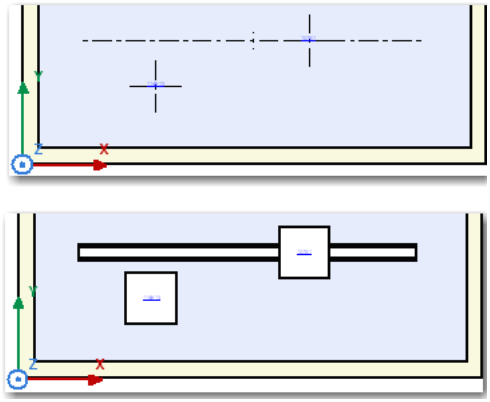
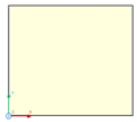
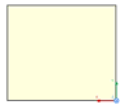
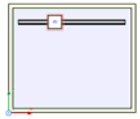
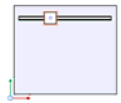
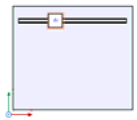
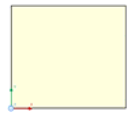


Figure 9-17: Axis OR View Drawing

The different views available are:

View	Representation	View	Representation
Closed Cabinet		Cabinet Back View	
Cabinet Cross-Section		Mounting Panel Front View	
Mounting Panel		Mounting Panel Back View	
Cabinet Back View		Inside Front Door Picture	

View	Representation	View	Representation
Closed Cabinet Front View		Not Shown	

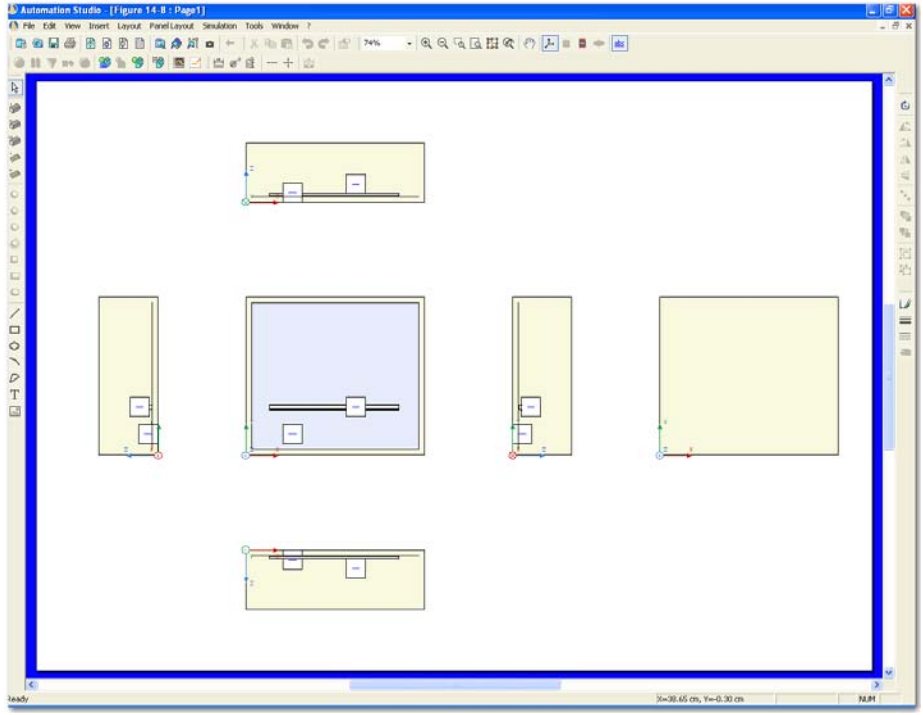


Figure 9-18: All possible cross-section views

9.2.4 "Localization"

This dialog box provides the means to edit the association, dimension, position, and orientation of the selected component.

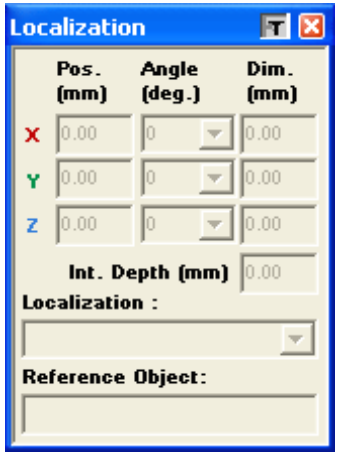


Figure 9-19: "Localization" Dialog Box

Name	Description
X Position	Field that is used to edit the component's X position according to the Reference Object.
Y Position	Field that is used to edit the component's Y position according to the Reference Object.
Z Position	Field that is used to edit the component's Z position according to the Reference Object.
X Angle	Field that is used to edit the component's X angular position according to the Reference Object.
Y Angle	Field that is used to edit the component's Y angular position according to the Reference Object.
Z Angle	Field that is used to edit the component's Z angular position according to the Reference Object.
X Dimension	Field that is used to edit the X component's dimension.
Y Dimension Y	Field that is used to edit the Y component's dimension.

Name	Description
Z Exterior Dimension	Field that is used to edit the Z exterior component's dimension.
Z Interior Dimension	Field that is used to edit the Z interior component's dimension.
Localization	Field that is used to edit the component's localization.
Reference component	Identifies the reference component to the selected component.

9.2.5 "Cabinet and Mounting Panel Selection"

This dialog box provides the means to edit the chassis and mounting panel configuration.

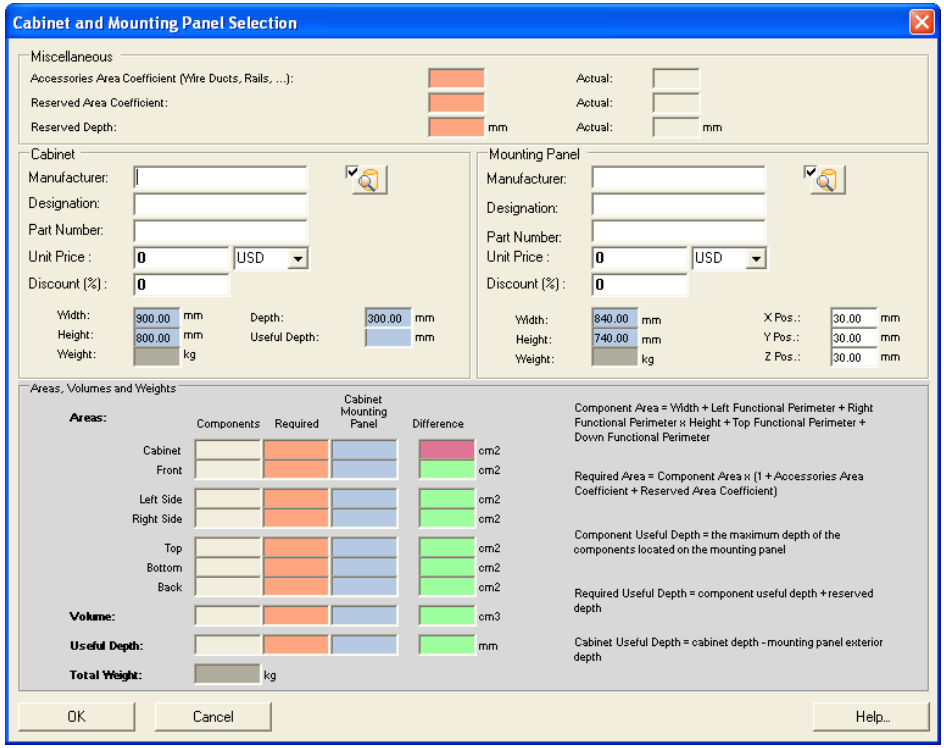


Figure 9-20: Cabinet and Mounting Panel Selection

Name	Description
Miscellaneous	This section is not functional. It defines the space proportion that is not used for components.
Cabinet	This section defines the cabinet characteristics.
Manufacturer	Field that is used to identify the manufacturer.
Designation	Field that is used to identify the cabinet name/designation.
Part Number	Field that is used to identify the cabinet part number.
Unit Price and Currency	Fields that are used to identify the cabinet unit price and currency used.
Discount (%)	Field that is used to identify the discount given to this unit.
Height	Field that is used to identify the cabinet height.
Width	Field that is used to identify the cabinet width.
Weight	Option not activated. Field that is used to identify the cabinet weight.
Depth	Field that is used to identify the cabinet depth.
Useful Depth	Option not activated. Field that is used to identify the cabinet useful depth; the cabinet depth less the mounting panel installation depth.
	Option not activated. Button that is used to associate a catalog component to the cabinet.
Mounting panel	Section that is used to set the mounting panel characteristics.
Manufacturer	Field that is used to identify the manufacturer.

Name	Description
Designation	Field that is used to identify the mounting panel name/designation.
Part Number	Field that is used to identify the mounting panel part number.
Unit Price and Currency	Fields that is used to identify the mounting panel unit price and currency used.
Discount (%)	Field that is used to identify the discount given to this unit.
Height	Field that is used to identify the mounting panel height.
Width	Section that is used to set the mounting panel characteristics.
Weight	Option not activated. Field that is used to identify the cabinet weight.
X Pos.	Field that is used to identify the mounting panel X position in reference to the cabinet reference point.
Y Pos.	Field that is used to identify the mounting panel Y position in reference to the cabinet reference point.
Z Pos.	Field that is used to identify the mounting panel Z position in reference to the cabinet reference point.
	Option not activated. Button that is used to associate a catalog component to the cabinet.
Areas, Volumes and Weights	Area not activated. Section that is used to identify the cabinet calculated values for areas, volumes, and weights.

9.2.6 Component Properties – Component Layout

This dialog box is used to define a component's physical dimensions.

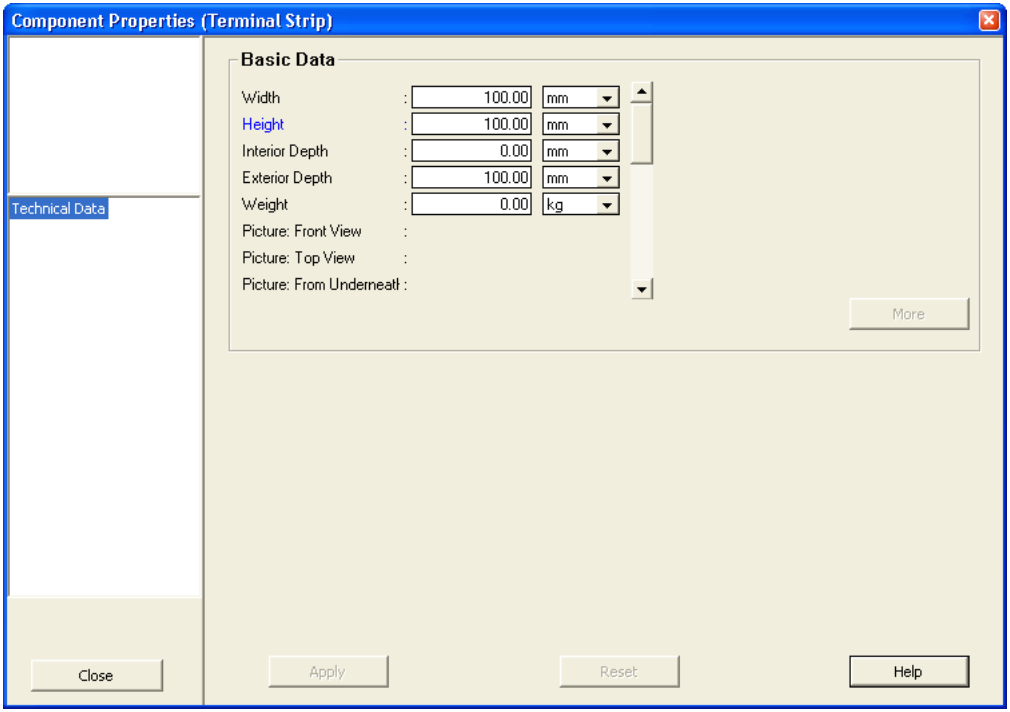


Figure 9-21: Component Properties – Component Layout

Name	Description
Width	Refer to figure “Component Dimension – Example 1 & 2”.
Height	Refer to figure “Component Dimension – Example 1 & 2”.
Interior Depth	Refer to figure “Component Dimension – Example 1 & 2”.
Exterior Depth	Refer to figure “Component Dimension – Example 1 & 2”.
Right functional perimeter	Refer to figure “Component Dimension – Example 1 & 2”.
Left functional perimeter	Refer to figure “Component Dimension – Example 1 & 2”.
Top functional perimeter	Refer to figure “Component Dimension – Example 1 & 2”.

Name	Description
Bottom functional perimeter	Refer to figure “Component Dimension – Example 1 & 2”.
Front functional perimeter	Refer to figure “Component Dimension – Example 1 & 2”.
Back functional perimeter	Refer to figure “Component Dimension – Example 1 & 2”.
‘G’ axis position	Refer to figure “Rail Mounting”.
‘C’ axis position	Refer to figure “Rail Mounting”.
Penetration	Refer to figure “Rail Mounting”.
Picture front view	Refer to figure “Picture position”.
Picture top view	Refer to figure “Picture position”.
Picture underneath view	Refer to figure “Picture position”.
Picture left view	Refer to figure “Picture position”.
Picture right view	Refer to figure “Picture position”.

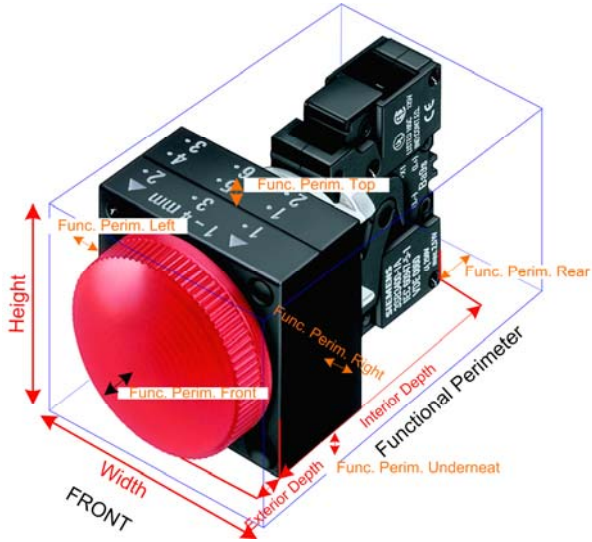


Figure 9-22: Component dimensions – Example 1

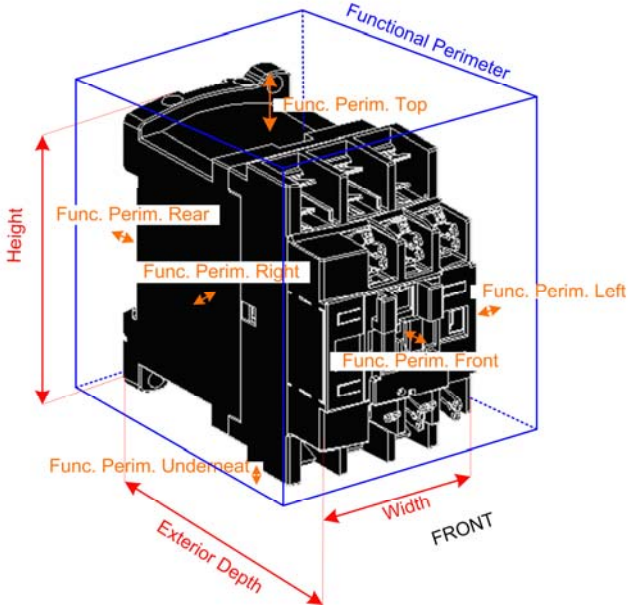


Figure 9-23: Component dimensions – Example 2

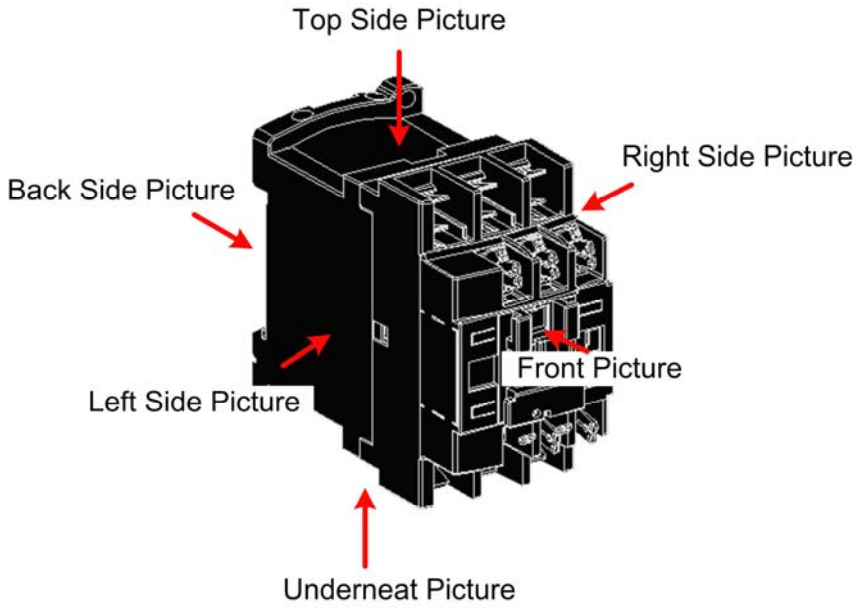


Figure 9-24: Picture positions

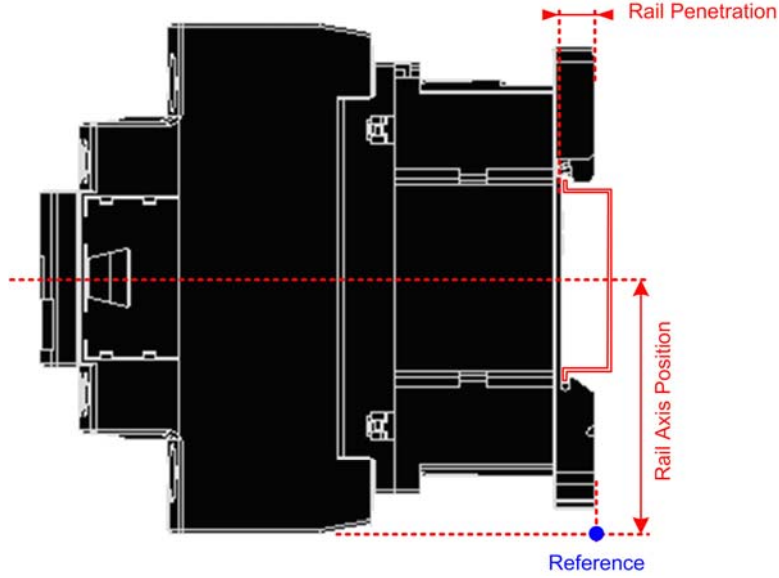


Figure 9-25: Rail mounting

9.2.7 Rail

This dialog box is used to set the rail characteristics. These characteristics are attached to the rail insertion action, and they are available either through the tool bar or the insertion menu.

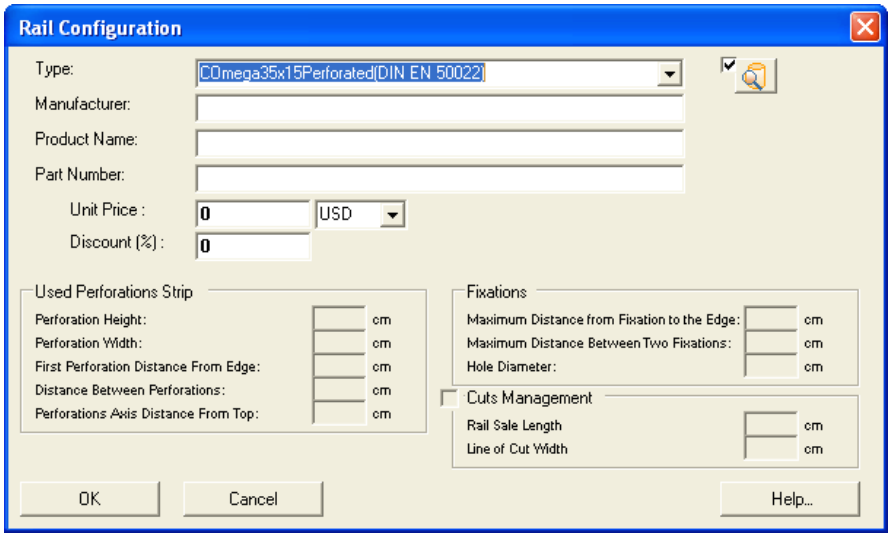


Figure 9-26: Rail Configuration

Name	Description
	Button that is used to assign this rail to a catalog component.
Type	Drop-down menu that is used to select the rail type.
Product Name	Field that is used to enter the rail product name.
Manufacturer	Field that is used to enter the rail manufacturer.
Part Number	Field that is used to enter the fail part number.
Unit Price and Currency	Fields that is used to identify the rail unit price and currency.
Discount (%)	Field that is used to identify the discount given to this unit.

Name	Description
Used Perforations Strip	Option not activated. This section is used to set the rail perforation characteristics.
Fixations	Option not activated. This section is used to the set the fixation characteristics.
Cuts Management	Option not activated. This section is used to the set the cuts characteristics.

9.2.8 Wire Duct

This dialog box is used to set the wire duct characteristics. They are attached to the wire duct insertion action, and they are available either through the tool bar or the insertion menu.

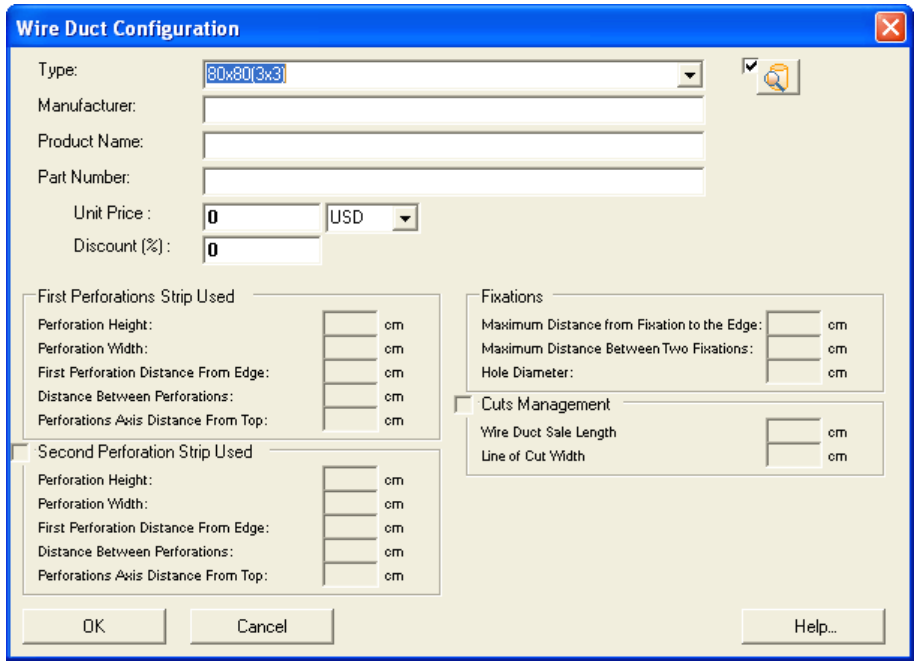


Figure 9-27: Wire Duct Configuration

Name	Description
	Button that is used to assign this wire duct to a catalog component.
Type	Drop-down menu that is used to select the wire duct type.
Product Name	Field that is used to enter the wire duct product name.
Manufacturer	Field that is used to enter the wire duct manufacturer.
Part Number	Field that is used to enter the wire duct part number.
Unit Price and Currency	Fields that are used to identify the wire duct unit price and currency.
Discount (%)	Field that is used to identify the discount given to this unit.
First Perforations Strip Used	Section not activated. This section is used to set the duct strip perforation characteristics.
Second Perforations Strip Used	Section not activated. This section is used to set the duct strip perforation characteristics.
Fixations	Section not activated. This section is used to set the fixations characteristics.
Cut Management	Option not activated. This section is used to the set the cut characteristics.

9.3 Panel Layout Drawing Creation

Creating a panel layout drawing can only be done in the electrotechnical module or drawing.

9.3.1 Guidelines

The basic guidelines are:

- Create the electrotechnical circuit;
- Open a new panel layout drawing;
- Set the panel layout drawing characteristics;
- Set the panel view characteristics;
- Set the cabinet characteristics;
- Insert the components on the drawing and adjust their characteristics;
- Insert the component found in the library;
- Insert the rails and ducts;
- Insert the dimensions.

9.3.2 Opening a new panel layout drawing

To open a new panel layout drawing:

1. In the Menu Bar, select "File" → "New" → "Electrotechnical" → "Panel Layout";

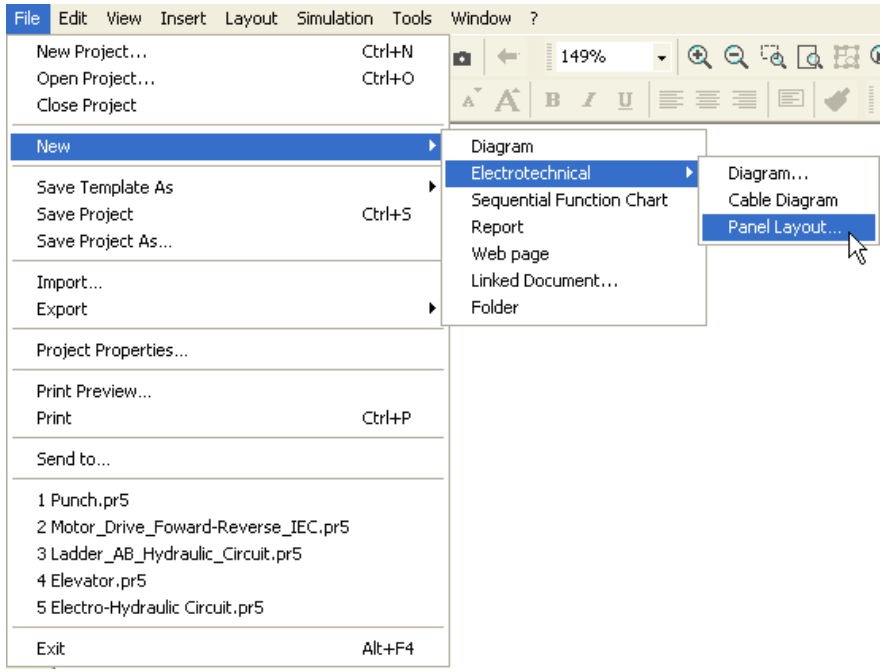


Figure 9-28: Creating a new panel layout drawing

A new panel layout diagram opens with the "Document Properties" dialog box on

top.

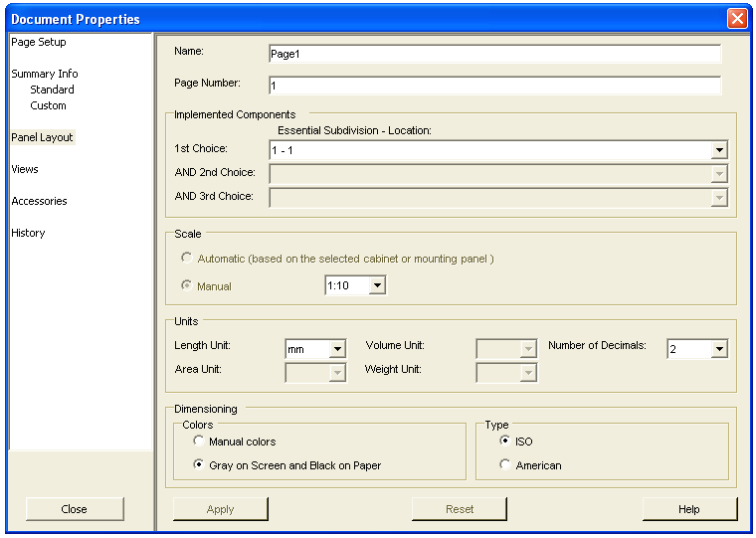


Figure 9-29: Panel Layout Configuration

2. Set the panel layout characteristic to match your needs;

3. Click on “Apply”;

4. In the side menu, select “Views”;

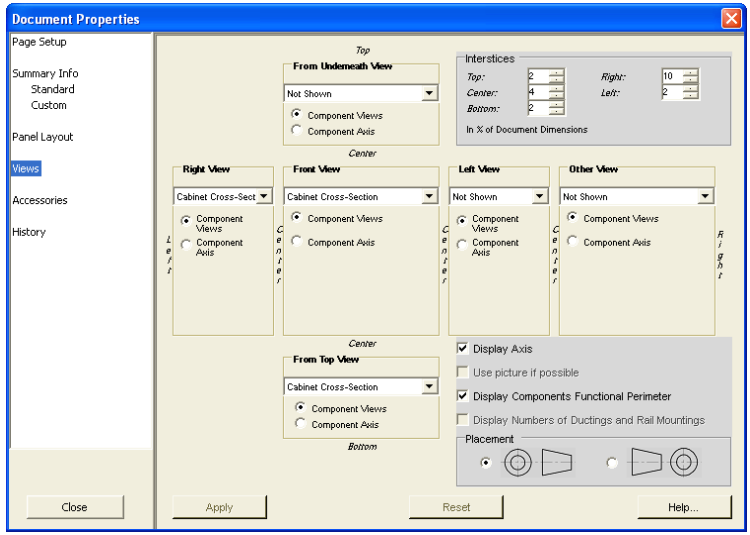


Figure 9-30: "Views" Dialog Box for a New Panel Layout Drawing

5. Set the "Views" characteristics to meet your needs;

6. Click on "Apply", than on "Close".

The dialog box closes and an empty panel layout is displayed.

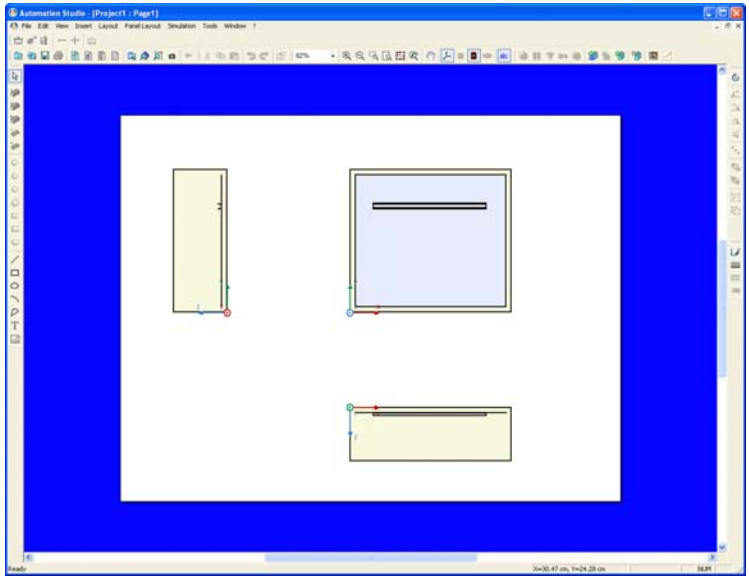


Figure 9-31: Basic panel layout drawing

9.3.3 Set the panel layout drawing characteristics

To set the panel layout drawing characteristics:

1. In the menu bar, select "Panel Layout" → "Selection of Panels and Mounting …";

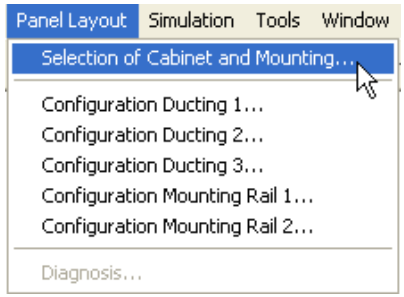


Figure 9-32: Opening the Dialog Box "Selection of Cabinet and Mounting …"

The "Cabinet and Mounting Panel Selection" dialog box opens.



Figure 9-33: Cabinet and Mounting Panel Selection

2. Set the cabinet and mounting panel characteristics to suit your needs;

3. Click on “OK”.

9.3.4 Insert, place and size a wire duct or rail

Before adding a wire duct or rail, you must set the characteristics to meet your needs.



Figure 9-34: Insertion tool bar

To configure the duct or rail:

1. Select “Panel Layout” → “Configuration Mounting Rail 1…”;

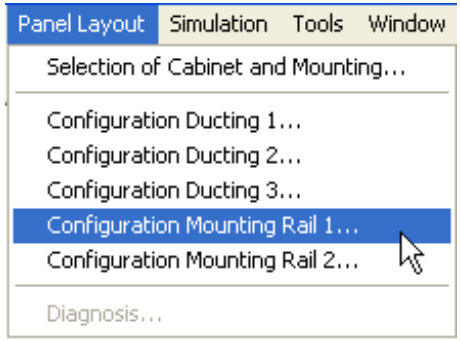


Figure 9-35: Opening the "Configuration Mounting Rail" Dialog Box

The dialog box "Rail Configuration" opens.

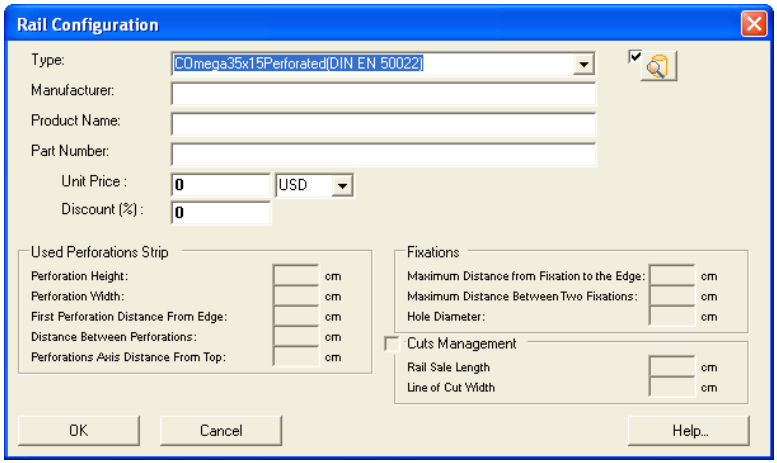


Figure 9-36: Rail configuration

2. Edit the rail configuration to meet your needs;

3. Click on "OK"

To insert a new rail or duct:

1. Select the rail button from the "Insert" tool bar;



Figure 9-37: "Insert" Tool Bar

2. Position the cursor on the mounting panel, at the location where the rail will start;

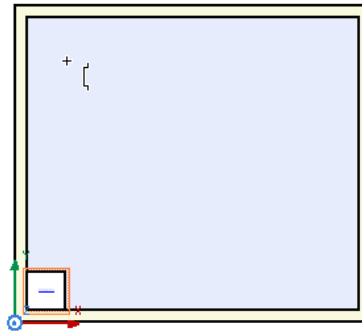


Figure 9-38: Initial Rail Position

3. Click and slide the cursor to the rail end position;

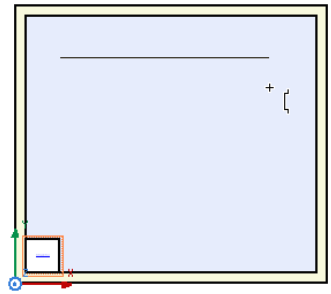


Figure 9-39: End Rail Position

4. Release the mouse button. The rail will appear on the drawing.

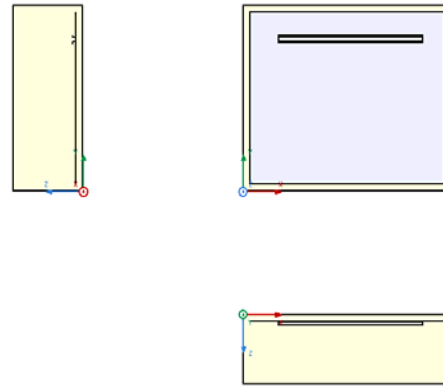


Figure 9-40: Rail Inserted

9.3.5 Insert, place and size components from the library

To insert a component:

1. Open the component library;

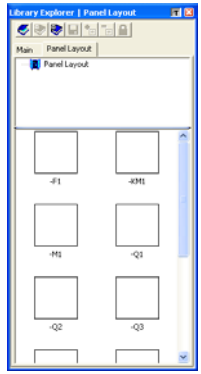


Figure 9-41: Panel Layout Library

2. Select and slide the component onto the mounting panel;

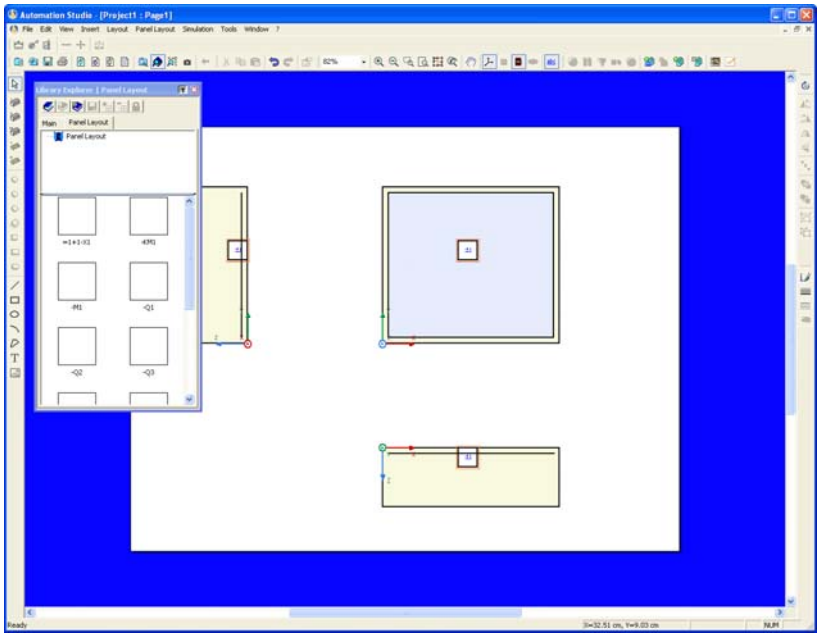


Figure 9-42: First component in a panel layout drawing

3. Move the component in one of the cross-section views to position the component correctly in a 3D environment;

4. Modifying the size of the component;

   To perform a coarse component sizing:

   a. Select the component face to size;

   b. Click on one of the resizing handles, and drag it to resize the component.

   To perform a precise component sizing:

   a. Right-click on the component;

   b. Select “Component Properties”;

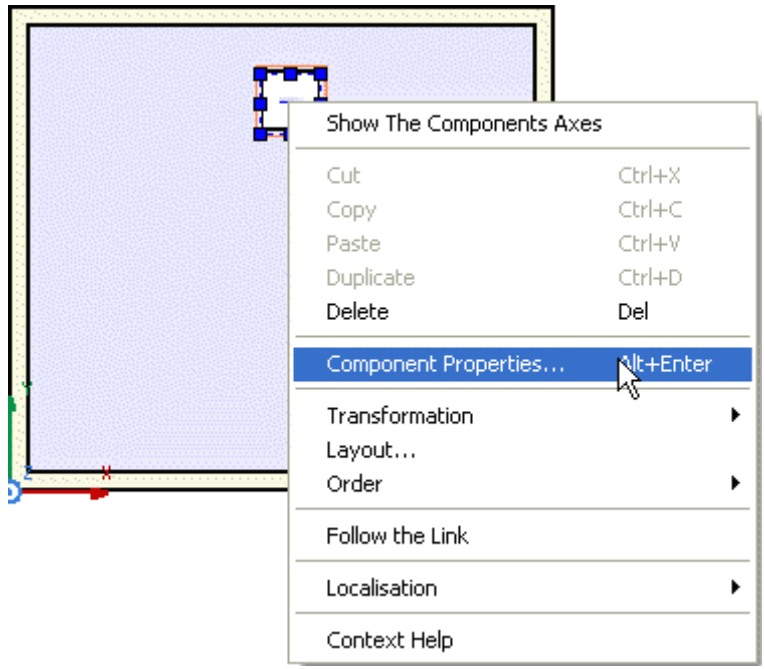


Figure 9-43: “Component Properties” selection

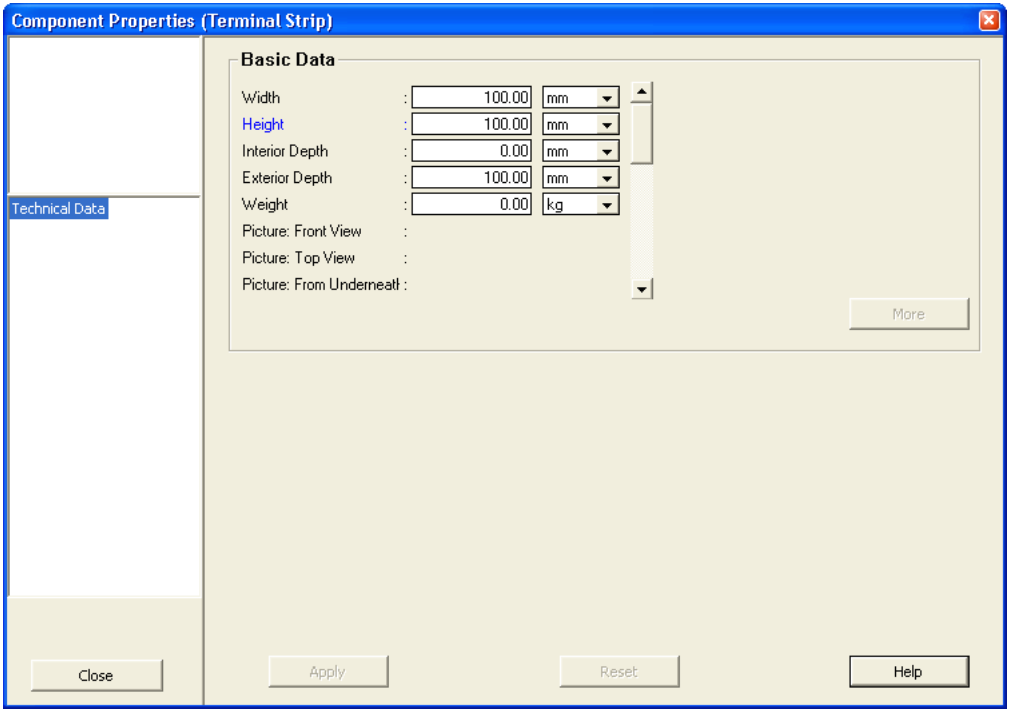


Figure 9-44: Component Properties – Panel Layout Component

c. Edit the sizing to your needs;

d. Click on “Apply”;

e. Click on “Close”.

9.3.6 Dimensioning

Dimensioning can be inserted anywhere on a drawing, but the actual dimension will be displayed only if the dimensioning line is tied to a component end or center. AS will display an “S” (snap) cursor when it is on a snapping point, a corner or a center.

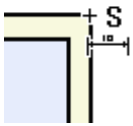


Figure 9-45: Dimensioning – Snap Identification

In a polygon, the dimensioning line can be tied on all sides and centers. On a circle or arc, a line can be tied to the sides. For an arc, it is also possible to tie the dimensioning line to the line connecting the arc to the circle center.

The dimensioning lines are added in three steps:

- Select a dimensioning tool
- Attached the dimensioning line ends
- Size the dimensioning line.



Figure 9-46: Dimensioning Tools

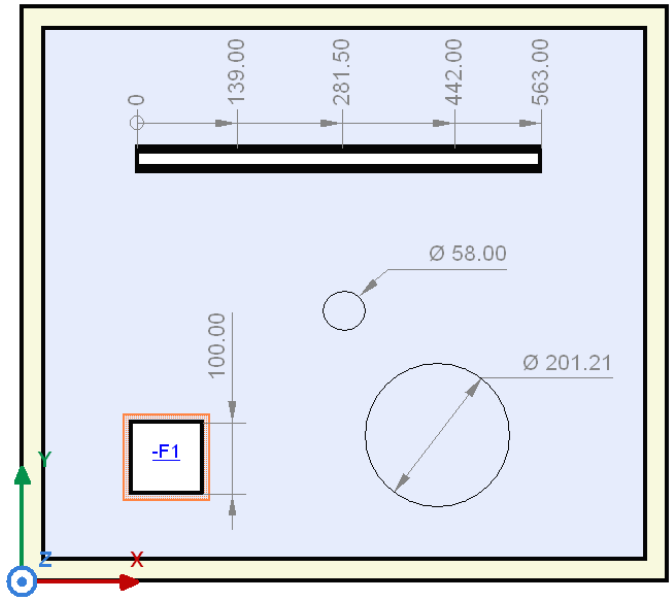


Figure 9-47: Dimensioning Example

9.3.7 Insert a Linear Annotation

Insert linear quote

To display a linear quote:

1. Select the linear dimension tool;



Figure 9-48: Dimensioning Tool Bar

2. Place the cursor at the position where the dimension line is to begin;

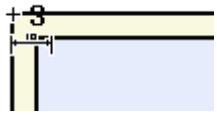


Figure 9-49: Linear Dimension Insertion

3. Click and drag the cursor to the position where the dimension line will end;

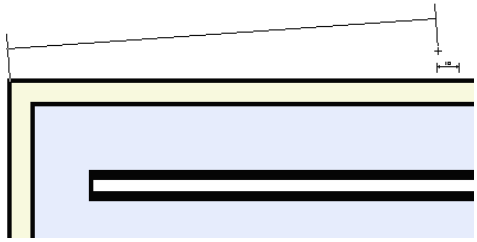


Figure 9-50: Linear Dimension Insertion

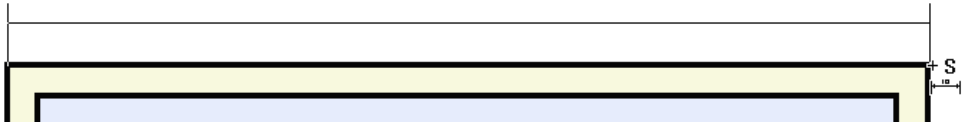


Figure 9-51: Linear Dimension Insertion (2)

4. Release the mouse button. DO NOT CLICK. The dimension line will appear on the drawing.

Size the linear dimension line

1. Move the mouse to size or position the dimension line.

☞ The lines will stay tied to their anchor positions. Everything else will move.

2. Click.

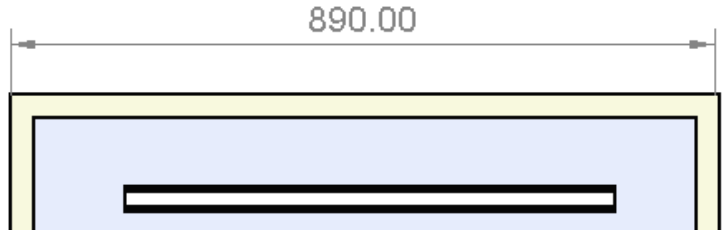


Figure 9-52: Linear Dimension Inserted

9.3.8 Insert Stacked Dimensions

Additional dimensions can be added to dimension lines only if they are vertical or horizontal. To insert Stacked Dimensions:

1. Select the "Stacked Dimension" tool;



Figure 9-53: "Dimension" Tool Bar

2. Place the cursor over the start point;



Figure 9-54: Initial step to a Stacked Dimensions Insertion

3. Click on this point;

4. Slide and release the cursor to set a second tie point.

☞ The setting of the second point does not need to be at a snapping position. This second point will define the axis of the dimension line.

5. Adjust the height where the dimension line will be inserted then click to add more set points;



Figure 9-55: Stacked Dimension Insertion

6. Move the cursor to a second point;

7. Click;

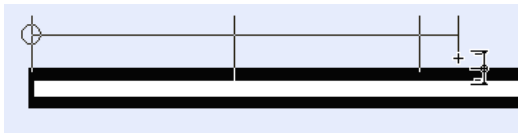


Figure 9-56: Stacked Dimensions Insertion (2)

8. Repeat the previous two steps to add more points;

9. Keep the cursor over the last point, and click to finish the insertion of the Sacked Dimension line.

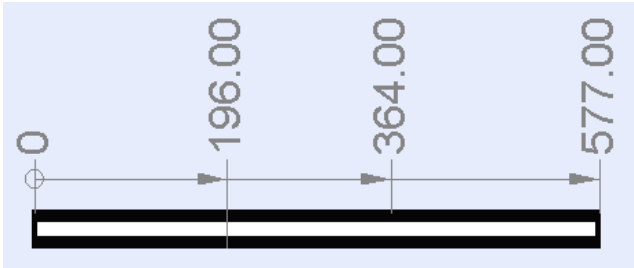


Figure 9-57: Stacked Insertion Inserted

9.3.9 Insert a Diameter Annotation

Two methods may be used to insert a diameter annotation.

Two Points Insertion Method

1. Select the diameter annotation tool;



Figure 9-58: Diameter Annotation Tool

2. Position the cursor at the center of the circle;

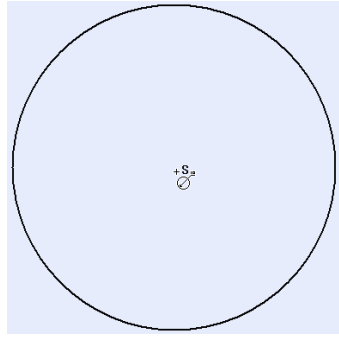


Figure 9-59: Tie in Circle Center Point

3. Click on this point and slide the cursor to the circle outline;

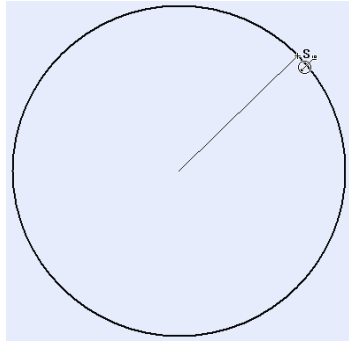


Figure 9-60: Tie on Circle Perimeter

4. Release the mouse button. DO NOT CLICK;

5. Move the mouse to position the dimension line;

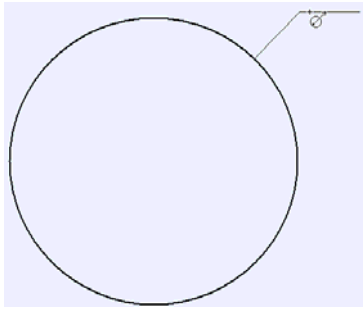


Figure 9-61: Dimension Line Positioning

6. Click.

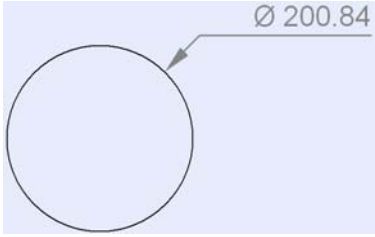


Figure 9-62: Two Point Annotation Result

Three Point Insertion Method

1. Select the diameter annotation tool;



Figure 9-63: Diameter Annotation Tool

2. Position the cursor at the center of the circle;

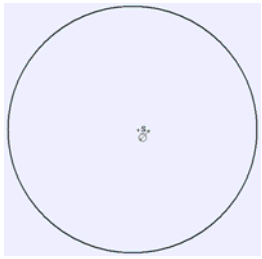


Figure 9-64: Tie in Circle Center Point

3. Click on this point and slide the cursor to the circle outline;

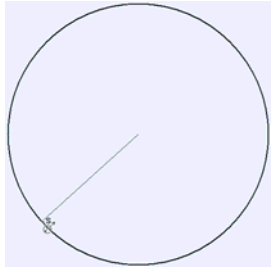


Figure 9-65: First Tie on Circle Perimeter

4. Release the mouse button but do not click, and slide it to the opposite side of the circle;

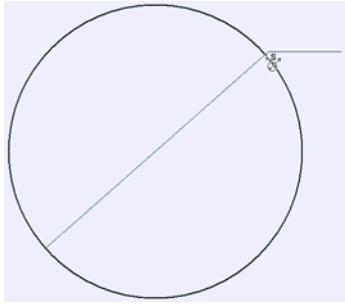


Figure 9-66: Second Tie on Circle Perimeter

5. Click and move the mouse to position the dimension line;

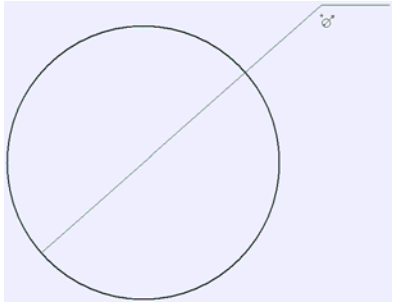


Figure 9-67: Dimension Line Positioning

6. Release the mouse button.

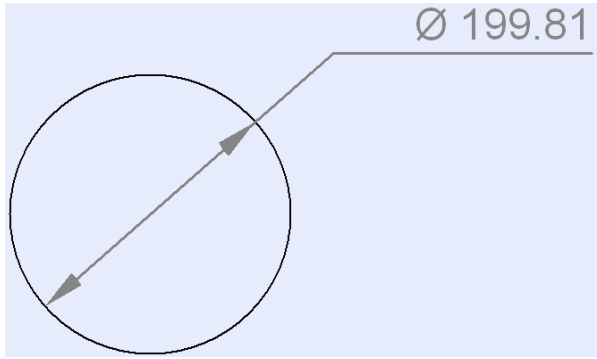


Figure 9-68: Three Point Annotation Result

10 Report

Automation Studio™ offers the possibility to create reports based on the electrotechnical diagrams. Various electrotechnical report types are available. Each of these reports contains precise and relevant data. The report format and the information it contains can be adjusted.

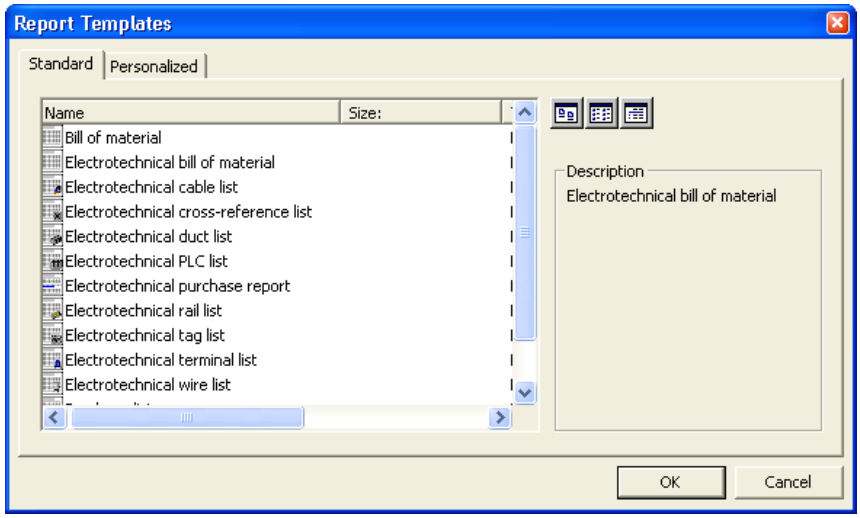


Figure 10-1: Report Templates

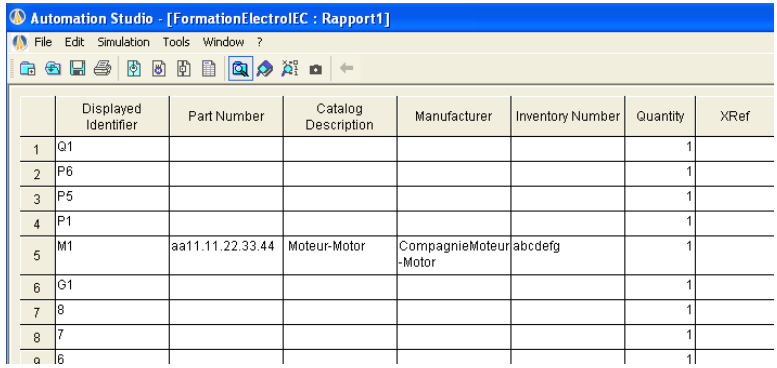
Following is the list of the available electrotechnical reports with a short description for each:

Report Type	Description
Bill of material	The bill of material details all of the current project's components. This list enables the collection of all the required parts prior to the production process.
Purchase Report	The purchase list identifies all of the current project's components.
Terminal List	The terminal list allows the rapid connection of cables and wires to external terminals. It provides a comprehensive view of the terminal strips.
Wire List	The wire list allows the easy identification of the components'

Report Type	Description
	interconnections. The wire list can be classified by color, section and length as required for the manufacturing phase.
Cable List	The cable list allows the identification of the components linked by cables.
Cross-reference list	The cross-reference list identifies all input and output components linked by a same cross-reference function. The cross-reference referring to “terminal/terminal strip” and “cable/wire” are not covered in this list.
Tag List	The tab list contains all the labels needed to identify components.
PLC List	The PLC list allows identifying in details the structure and the functions of all the PLC components inserted in the project diagrams.
Duct List	The duct list allows identifying and describing the ducts used in the panel layout diagrams.
Rail List	The rail list allows identifying and describing the rails used in the panel layout diagrams.



Most of the report requires the catalog information for the component to be present. Therefore you must fill in the catalog data on all components. The catalog form is found in the “Component Properties” dialog box.



	Displayed Identifier	Part Number	Catalog Description	Manufacturer	Inventory Number	Quantity	XRef
1	Q1					1	
2	P6					1	
3	P5					1	
4	P1					1	
5	M1	aa11.11.22.33.44	Moteur-Motor	CompagnieMoteur-Motor	abcdefg	1	
6	G1					1	
7	8					1	
8	7					1	
9	6					1	

Figure 10-2: Bill of Material

For more details on how to manage reports, refer to the *Automation Studio™ User's Guide.*

A Glossary

Alphanumeric

Expression made of various characters recognize by the computer, principally the 10 decimal number and 26 alphabet letters.

AS

Automation Studio™.

Cable

A cable is composed of electric conductors surrounded by a sheath. A cable is used to connect external components such as motors, power supplies, etc. to a terminal strip of the command cabinet.

Clip Jumper

Allows the user to mechanically link many terminals together. It is used when a signal must be transmitted to more than one wire. A wire between two terminals can be defined as a clip jumper. The linked terminals are normally contiguous, but this is not a necessity.

Command wire

Command wires allow you to connect components of low power such as relays, push buttons, indicators, etc.. Command wires are represented by a thin line. A wire can be redefined as a command or power wire.

Component

Basic elements used for creating diagrams. Each component represents a behavior or a function during simulation. Components are part of the library and are provided by the workshops.

Connection

A connection binds two components and/or links them together. Connections are the same color as the element to which they are connected, and they differ if they are not connected. Two connections from non-compatible technologies cannot be connected together. For example, a pneumatic line cannot be connected to an Electrotechnical component and vice-

versa.

Connection Line Shape

It is a line shape identifying the connection path between two points.

Connection Number

It is a number that designates each point where components are connected.

Connection port

Component terminal that permits the connection of a wire. Normally there will be one to two wires connected to a component terminal.

Connector

An element represented by a red circle when it is not linked. It allows the connection of two components. A connector becomes a connection as soon as it is connected to another element.

Diagram

A graphic representation of a circuit or a model by means of elements and components drawn from the Automation Studio™ workshop libraries.

Diagram Editor

The tool used to create, modify and manage project files. The project editor includes the document manager that enumerates all the documents in the project.

Display Tools

Designates the design accessories in the Diagram Editor: the grid, rulers, connections and connection numbers. You can specify how they are displayed from the View menu.

Edit Mode

Operational mode in which the components are inserted and changed.

Electric conductor

Electric wire that is part of a cable. Each cable conductor is numbered.

Emitter

Component that transmits its state or its value to other receiver components that are associated with it.

External component

Components external to the command cabinet (motor, power supply, etc.). External components are normally connected by cables. External components (To) that are connected on the right side of cable diagrams.

External wire

Wire used to connect external components to the back plate of a command cabinet (buttons and indicators on the front of command cabinets, motors, power supply, etc.). External wires are connected on the right side of terminal strip diagrams.

Grid

Horizontal and vertical dotted lines located in the workspace of the editor, used to align the elements of the graph.

Internal component

Components inside the command cabinet such as terminal strips, etc.. Internal components (FROM) are connected on the left side of cable diagrams.

Internal wire

Wires used to connect internal components to the back plate of a command cabinet. Internal wires are connected on the left side of terminal strip diagrams.

Library

Window containing the basic elements used for designing circuits or simulation models. There are three types of elements: components, links and graphic objects. They are included in the workshops.

Link

A link connects components. During simulation, it is used to transfer information from one component to the other (see **wires**).

Menu Bar

A horizontal bar located above the application title bar. The menu bar displays the various menus and commands that are available for the active window.

PLC

Programmable Logic Controller.

Power wire

Power wires allow the user to connect high power components such as motors, power supplies, etc.. Power wires are represented by a thick line. A wire can be redefined as a command or power wire.

Project

A group of documents forming a cohesive whole. A Project is managed by the Diagram Editor.

Project Explorer

The contents of one of the project editor's windows. The document manager lists all of the documents in the project.

Properties

Characteristics or parameters of a component. You can change or view the properties by opening the component properties dialog box.

Receiver

Component that receives its state or its value from an emitter component with which it is associated. It takes its tag name from the emitter component with which it is associated.

Simulation Cycle

One step in the calculation that determines the state of each component.

Simulation Mode

Mode of operation in Automation Studio™ during which a project or a diagram is simulated. Unlike the Edit mode, the Simulation mode, when active, does not allow any changes to the

project.

Status Bar

A horizontal bar located below the utility windows. It contains various info such as: comments, zoom factors, pointer coordinates, etc.

Terminal

Part of a terminal strip allowing the mechanical connection of two wire sections. The terminal is composed of two connectors, one for internal connections and one for external connections. Normally a single wire is connected to each terminal. Terminals can be mechanically linked by clip jumpers when a signal must be transmitted to more than one wire.

Terminal Strip

Assembly of many terminals.

Title Bar

A horizontal bar located above the window which contains the title.

Toolbar

A bar located under the menu bar. It contains buttons for the most frequently used commands.

User Interface

Environment made of windows, dialog boxes, menus, commands, mouse and keys etc. which enable user to communicate with the computer.

Utility

A general designation for different tools, workshops and other functions that operate in different windows in AS.

Wires

Wires allow the linking of components. Wires can be associated with a cable. A wire between two terminals can be defined as a clip jumper.

Worksheet

Entire surface available in Automation Studio for diagram creation.

Workshop

A module that plugs into the Core System. Each workshop contains elements, diagrams and functions relating to its own technological specialty and to the type of diagrams it can generate.

Workspace

The part of the window that displays the information you are working on, such as graphs, tables, and the item list of the application.

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